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Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct

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International Masters in Risk Assessment and Management of Civil Infrastructures

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Actualización bayesiana de la fiabilidad estructural de puentes mediante datos de monitorización, criterios de parada y pruebas de carga: Un caso de estudio del viaducto Zijlweg.

RESUMEN

Los puentes de hormigón armado envejecidos en Europa enfrentan desafíos crecientes en la evaluación estructural, particularmente cuando el deterioro de los materiales, como la reacción álcali-sílice (ASR), hace que los modelos de capacidad analítica no sean fiables. Las pruebas de carga ofrecen una alternativa experimental directa, demostrando la suficiencia estructural bajo cargas mayoradas sin depender únicamente de predicciones analíticas. Sin embargo, el valor informativo de los datos de estas pruebas permanece infrautilizado cuando los resultados se interpretan solo como resultados binarios de apto o no apto, en lugar de como evidencia cuantitativa para una evaluación probabilística de la fiabilidad. Esta tesis aplica una metodología de actualización de fiabilidad bayesiana a los datos de monitorización obtenidos durante la prueba de carga a escala real del viaducto Zijlweg, un puente de losa de hormigón armado afectado por ASR en los Países Bajos. El viaducto fue cargado en junio de 2015 en dos posiciones, una crítica para el momento flector y otra para cortante, generando datos de deformaciones, cambios en el ancho de fisuras y flechas en cinco niveles de carga incrementales hasta 1.342 kN. El marco bayesiano de De Vries et al. [1] utiliza estos valores monitorizados para actualizar la distribución de la resistencia estructural en cada paso de carga y calcular la probabilidad anual de fallo antes, durante y después de la prueba. En paralelo, se evalúan los criterios de parada basados en la mecánica de la Teoría de la Fisura Crítica por Cortante (CSCT) y la Teoría del Desplazamiento Crítico por Cortante (CSDT), organizados en un sistema de semáforos.

El análisis de la sección transversal corregida, incorporando las cuatro capas de armadura inferior, arroja un momento flector medio de 3.248,87 kNm ($CoV = 0{,}137$). Los índices de fiabilidad posterior tras la prueba completa son $\beta_{after} = 5{,}33$ para flexión y $4{,}89$ para cortante, superando sustancialmente el objetivo anual para estructuras nuevas ($\beta = 4{,}7$). El objetivo anual para estructuras existentes en el nivel de desaprobación ($\beta = 4{,}0$) se alcanzó tras el tercer paso de carga para flexión ($\beta = 4{,}40$) y el cuarto paso para cortante ($\beta = 4{,}41$). Esto corresponde a factores de carga de prueba de $f_{LM1} =$

$0{,}866$ para flexión y $f_{LM1} = 0{,}849$ para cortante. Los indicadores CSCT y CSDT permanecieron en la zona verde durante toda la prueba, siendo cualitativamente consistentes con los valores positivos de β_{during}. Los datos de monitorización proporcionan una ventaja cuantificable: en el tercer paso de cortante, el monitoraje demuestra $\beta_{after} = 3{,}81$ mientras que el límite inferior da solo 1,96, una ventaja de 1,85 unidades de índice de fiabilidad atribuida a la información del ancho de fisura en el modelo de resistencia exponencial.

Los hallazgos demuestran la viabilidad práctica de aplicar la actualización de fiabilidad bayesiana a datos reales de pruebas de carga. El estudio confirma que las pruebas de carga monitorizadas constituyen un método eficaz de evaluación de Nivel IV para puentes afectados por ASR que parecen analíticamente insuficientes. Se concluye que los criterios de parada y el marco bayesiano son complementarios, proporcionando señales de seguridad operativa durante la prueba y evidencia de fiabilidad cuantitativa después de la misma.

PALABRAS CLAVE: Actualización bayesiana; pruebas de carga; fiabilidad estructural; puentes de hormigón armado; reacción álcali-sílice; criterios de parada; CSCT; CSDT; monitorización; MCMC.

Actualització bayesiana de la fiabilitat estructural de ponts mitjançant dades de monitoratge, criteris d'aturada i proves de càrrega: Un cas d'estudi del viaducte Zijlweg.

RESUM

Els ponts de formigó armat envellits a Europa s'enfronten a desafiaments creixents en l'avaluació estructural, especialment quan el deteriorament dels materials, com ara la reacció àlcali-sílice (ASR), fa que els models de capacitat analítica no siguin fiables. Les proves de càrrega ofereixen una alternativa experimental directa, demostrant la suficiència estructural sota càrregues majorades sense dependre únicament de prediccions analítiques. No obstant això, el valor informatiu de les dades d'aquestes proves roman infrautilitzat quan els resultats s'interpreten només com a resultats binaris d'apte o no apte en lloc d'evidència quantitativa per a una l'avaluació probabilística de la fiabilitat.

Aquesta tesi aplica una metodologia d'actualització de fiabilitat bayesiana a les dades de monitoratge obtingudes durant la prova de càrrega a escala real del viaducte Zijlweg, un pont de llosa de formigó armat afectat per ASR als Països Baixos. El viaducte va ser carregat el juny de 2015 en dues posicions, una crítica per moment flector i una altra per tallant, generant dades de deformacions, canvis en l'amplada de fissures i fletxes en cinc nivells de càrrega incrementals fins a 1.342 kN. El marc bayesià de De Vries et al. [1] utilitza aquests valors monitorats per actualitzar la distribució de la resistència estructural en cada pas de càrrega i calcular la probabilitat anual de fallada abans, durant i després de la prova. En paral·lel, s'avaluen els criteris d'aturada basats en la mecànica de la Teoria de la Fissura Crítica per Tallant (CSCT) i la Teoria del Desplaçament Crític per Tallant (CSDT), organitzats en un sistema de semàfors.

L'anàlisi de la secció transversal corregida, incorporant les quatre capes d'armadura inferior, dóna un moment flector mitjà de 3.248,87 kNm ($CoV = 0{,}137$). Els índexs de fiabilitat posterior després de la prova completa són $\beta_{after} = 5{,}33$ per a flexió i $4{,}89$ per a tallant, superant substancialment l'objectiu anual per a estructures noves ($\beta = 4{,}7$). L'objectiu anual per a estructures existents al nivell de desaprovació ($\beta = 4{,}0$) es va assolir després del tercer pas de càrrega per a flexió ($\beta = 4{,}40$) i del quart pas per a tallant ($\beta = 4{,}41$). Això correspon a factors de càrrega de prova de $f_{LM1} = 0{,}866$ per a flexió i $f_{LM1} = 0{,}849$ per a tallant. Els indicadors CSCT i CSDT es van mantenir en la zona verda

durant tota la prova, sent qualitativament consistents amb els valors de β_{during} positius. Les dades de monitoratge proporcionen un avantatge quantificable: al tercer pas de tallant, el monitoratge demostra $\beta_{after} = 3{,}81$ mentre que el límit inferior dóna només 1,96, un avantatge d'1,85 unitats d'índex de fiabilitat atribuït a la informació de l'amplada de fissura en el model de resistència exponencial. Les troballes demostren la viabilitat pràctica d'aplicar l'actualització de la fiabilitat bayesiana a dades reals de proves de càrrega. L'estudi confirma que les proves de càrrega monitorades constitueixen un mètode eficaç d'avaluació de Nivell IV per a ponts afectats per ASR que semblen analíticament insuficients. Es conclou que els criteris d'aturada i el marc bayesià són complementaris, proporcionant senyals de seguretat operativa durant la prova i evidència de fiabilitat quantitativa després de la mateixa.

PALABRAS CLAVE: Actualització bayesiana; proves de càrrega; fiabilitat estructural; ponts de formigó armat; reacció àlcali-sílice; criteris d'aturada; CSCT; CSDT; monitoratge; MCMC.

Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct

ABSTRACT

Aging reinforced concrete bridges across Europe face growing challenges in structural assessment, particularly where material deterioration such as alkali-silica reaction (ASR) makes analytical capacity models unreliable. Proof load testing offers a direct experimental alternative, demonstrating structural adequacy under factored loads without relying solely on analytical predictions. However, the full informational value of proof load test data remains underutilised when test results are interpreted only as binary pass or fail outcomes rather than as quantitative evidence for probabilistic reliability assessment.

This dissertation applies a Bayesian reliability updating methodology to the monitoring data obtained during the full-scale proof load test of the viaduct Zijlweg, an ASR-affected reinforced concrete solid slab bridge located in Noord Brabant, the Netherlands. The viaduct was proof loaded in June 2015 at two positions, one critical for bending moment and one critical for shear, generating comprehensive sensor data including concrete strains, crack width changes, and deflections at five incremental load levels up to 1,342 kN. The Bayesian framework of De Vries et al. [1] uses these monitored values specifically concrete strain for bending and maximum weighted nominal crack width for shear to update the structural resistance distribution at each load step and calculate the annual probability of failure before, during, and after the test. In parallel, mechanics-based stop criteria derived from the Critical Shear Crack Theory (CSCT) and the Critical Shear Displacement Theory (CSDT), organised into a traffic light system, are evaluated against the same monitoring data and compared with the Bayesian reliability estimates at each load step.

The corrected cross-section analysis, incorporating all four layers of bottom reinforcement from the structural drawings, gives a mean yield moment of 3,248.87 kNm with a coefficient of variation of 0.137. The posterior reliability indices after the complete proof load test are $\beta_{\text{after}} = 5.33$ for bending and $\beta_{\text{after}} = 4.89$ for shear, both substantially exceeding the annual target reliability for new structures ($\beta = 4.7$). The annual target reliability for existing structures at the disapproval level ($\beta = 4.0$) is achieved after the third load step for bending ($\beta = 4.40$) and the fourth load step for shear ($\beta = 4.41$).

This corresponds to proof load factors of $f_{LM1} = 0.866$ for bending and $f_{LM1} = 0.849$ for shear. The CSCT and CSDT stop criteria indicators remained in the green light range throughout the shear test, qualitatively consistent with the positive β_{during} values at all load steps. Monitoring data provides a quantifiable advantage over the lower-bound approach: at the third shear load step, the monitoring data demonstrates $\beta_{after} = 3.81$ while the lower bound gives only 1.96, an advantage of 1.85 reliability index units attributed to the crack width information in the exponential resistance model.

The findings demonstrate the practical feasibility of applying Bayesian reliability updating to real field proof load test data for an ASR-affected bridge, providing a more informative assessment than traditional analytical or lower-bound approaches. The study confirms that proof load testing, interpreted through the Bayesian framework, constitutes an effective Level IV assessment method for ASR-affected bridges that appear analytically insufficient under lower assessment levels. The stop criteria and Bayesian framework are found to be complementary rather than competing, providing operational safety signals during the test and quantitative reliability evidence after the test from the same monitoring measurements.

KEYWORDS: Bayesian updating; proof load testing; structural reliability; reinforced concrete bridges; alkali-silica reaction; stop criteria; Critical Shear Crack Theory; Critical Shear Displacement Theory; monitoring data; Markov Chain Monte Carlo; Metropolis-Hastings algorithm.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

ACI	American Concrete Institute
ASR	Alkali-Silica Reaction
AASHTO	American Association of State Highway and Transportation Officials
CoV	Coefficient of Variation
CSCT	Critical Shear Crack Theory
CSDT	Critical Shear Displacement Theory
DL	Dead Load
FEM	Finite Element Model
JCSS	Joint Committee on Structural Safety
LHS	Latin Hypercube Sampling
LM1	Load Model 1 (Eurocode traffic load model)
LVDT	Linear Variable Differential Transformer
MCMC	Markov Chain Monte Carlo
MH	Metropolis-Hastings algorithm
PDF	Probability Density Function
RBK	Richtlijnen Beoordeling Kunstwerken (Dutch Guidelines for Assessment of Bridges)
SDL	Superimposed Dead Load
SLS	Serviceability Limit State
UC	Unity Check
ULS	Ultimate Limit State
WIM	Weigh-In-Motion

Latin Symbols

a	Distance from support to load application point (m)
As	Reinforcement steel area (mm2)
b	Width of cross-section (m)
c	Concrete cover (m)
C0Q	Statistical uncertainty factor for traffic load (-)
d	Effective depth of cross-section (m)

dg	Maximum aggregate size (mm)
E	Annual maximum traffic load effect (kNm or kN)
EPL	Total proof load effect including permanent loads (kNm or kN)
Es	Young's modulus of reinforcement steel (GPa)
fc	Concrete compressive cylinder strength (MPa)
fu	Ultimate tensile strength of steel (MPa)
fy	Yield strength of reinforcement steel (MPa)
fLM1	Proof load factor relative to LM1 = QPL / Qk,LM1 (-)
fWIM	Proof load factor relative to WIM = QPL / Qk,WIM (-)
GDL	Dead load effect (kNm or kN)
GSDL	Superimposed dead load effect (kNm or kN)
h	Slab height at critical section (m)
ICSCT	CSCT stop criteria indicator (MPa0.5)
ICSDT	CSDT stop criteria indicator (MPa.mm)
L	Span length (m)
mX	Mean resistance ratio (-)
My	Yield bending moment (kNm)
nX	Number of calibration experiments for resistance model (-)
N	Number of bridge components (spans) in population (-)
n	Number of tested components (-)
p	Probability density (-)
Pf	Probability of failure (-)
Q	Annual maximum traffic load effect (kNm or kN)
Qk,LM1	Characteristic traffic load from LM1 (kNm or kN)
Qk,WIM	Characteristic traffic load from WIM data (kNm or kN)
QPL	Proof load effect (kNm or kN)
R	Structural resistance (kNm or kN)
sX	Standard deviation of resistance ratio (-)
VX	Coefficient of variation of resistance ratio (-)
w	Crack width (mm)
wmax,w	Maximum weighted nominal crack width (mm)
X	Resistance ratio = R / E_PL (-)

Z	Limit state function = R - E (kNm or kN)

Greek Symbols

β	Reliability index (-)
εs	Concrete strain at bottom fibre (microstrain) ε
εy	Yield strain of reinforcement steel (-)
γc	Concrete density (kN/m3)
γDL	Partial factor for dead load (-)
γSDL	Partial factor for superimposed dead load (-)
γLL	Partial factor for live load (-)
κ	Curvature of cross-section (1/m)
λ	Shear span to depth ratio = a / d (-)
ρ	Correlation coefficient (-)
σc	Concrete compressive stress (MPa)
σs	Reinforcement steel stress (MPa)
τ	Shear stress at critical section (MPa)
τE	Model uncertainty factor for load effect (-)
τE PL	Model uncertainty factor for proof load effect (-)
τR	Model uncertainty factor for resistance (-)
Φ	Standard normal cumulative distribution function (-)

Subscripts

after	After a successful proof load test (posterior)
Bot	Bottom reinforcement layer
C	Concrete
curr	Current state of MCMC chain
DL	Dead load
during	During the proof load test
K	Characteristic value

lower	Lower bound approach (survival evidence only)
M	Mean value
PL	Proof load
prior	Before the proof load test (prior)
R	Resistance
S	Steel
SDL	Superimposed dead load
Top	Top reinforcement layer
Y	Yield

1. INTRODUCTION

1.1. The Global Challenge of Aging Infrastructure

1.1.1. Infrastructure Aging Demographics

Civil infrastructure represents one of society's most significant capital investments, with bridges serving as critical links in transportation networks that enable economic activity and social connectivity. The post-World War II construction boom, particularly between 1950 and 1980, saw unprecedented investment in transportation infrastructure across Europe and North America. Today, this infrastructure cohort faces a convergence of challenges: chronological aging, increased traffic demands, evolving design standards, and material degradation processes that were imperfectly understood at the time of construction.

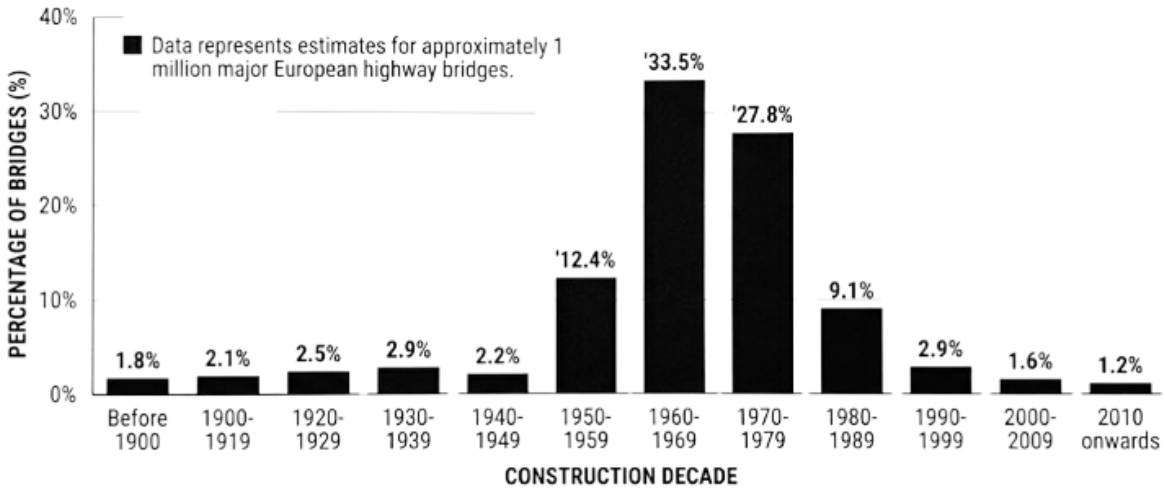


Figure 1.1 – Age *distribution* of European highway bridges (compiled from Eurostat [2]).

In the European context, the Netherlands presents a particularly illustrative case study. Approximately 40% of the national highway bridge stock was constructed before 1975, with many structures now exceeding their 50-year design life [3]. This demographic reality creates an urgent need for accurate assessment methodologies that balance safety requirements with economic constraints. Premature replacement carries enormous financial costs and environmental impacts, while inadequate assessment risks catastrophic failures with severe human and economic consequences.

1.1.2. Deterioration Mechanisms in Concrete Bridges

Concrete bridges from the mid-20th century face multiple deterioration mechanisms that were not fully anticipated in their original design. Alkali-silica reaction (ASR), observed in the Zijlweg viaduct, causes expansive gel formation that induces cracking and reduces structural capacity. Chloride-induced corrosion of reinforcement, carbonation, freeze-thaw damage, and fatigue from increasing traffic loads collectively degrade structural performance [4]. These mechanisms interact synergistically, creating complex deterioration patterns that challenge traditional assessment approaches.



Figure 1.2 – Common deterioration mechanisms in aging concrete bridges.

A particularly relevant example for this dissertation is alkali-silica reaction (ASR), which produces expansive gel formation leading to map cracking on the concrete surface, as observed on the Zijlweg viaduct prior to the 2015 proof load test [5].

1.2. Evolution of Bridge Assessment Methodologies

1.2.1. Code-Based Assessment Limitations

Traditional bridge assessment relies on code-based approaches that compare existing conditions with prescriptive design requirements. These methods suffer from inherent conservatisms: they apply current

design codes (developed for new construction) to existing structures, ignore beneficial system effects and material overstrength, and cannot account for the actual condition state revealed by inspections [6]. This conservatism often leads to unnecessary load restrictions, strengthening interventions, or premature replacement, with significant economic and social costs.

1.2.2. From Deterministic to Probabilistic Assessment

The late 20th century witnessed a paradigm shift from deterministic to probabilistic assessment frameworks. Reliability-based methods recognize the inherent uncertainties in material properties, loads, and models, quantifying safety through reliability indices (β) rather than safety factors [7]. The Eurocode framework formalizes this approach, with target reliability indices (β = 3.8 for 50-year reference period) providing quantitative safety criteria [8]. However, these methods typically rely on generic probabilistic models that may not reflect the specific characteristics of individual structures.

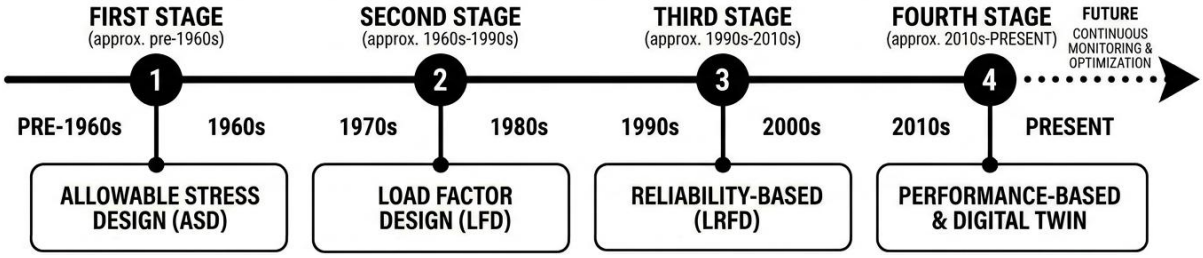


Figure 1.3 – Evolution of structural design and assessment approaches, from allowable stress design to performance-based methods (illustrative).

[1] *Note that the progression shown reflects design code development. Assessment of existing structures follows a parallel track that increasingly incorporates field evidence and probabilistic methods* [5].

1.2.3. The Need for Structure-Specific Assessment

The limitations of both code-based and generic probabilistic approaches have driven the development of structure-specific assessment methods that incorporate actual condition data. As noted by Diamantidis et al. [4], "The assessment of existing structures requires a different approach than the design of new structures, taking into account the actual condition, loading history, and observed performance." This recognition has led to increased interest in proof load testing as a means of obtaining structure-specific evidence.

1.3. Proof Load Testing: Direct Evidence of Structural Capacity

1.3.1. What is Proof Load Testing and When is it Used?

A proof load test is a structural test in which a bridge is loaded to a level representative of the factored design loads, with the goal of demonstrating that the structure can carry those loads without significant signs of distress. Unlike diagnostic load testing, which verifies analytical model assumptions, proof load testing provides direct experimental evidence of structural adequacy without depending on analytical capacity models [5].

Proof load testing is most valuable in three situations: when structural documentation is incomplete, when material degradation makes analytical assessment unreliable, and when analytical assessments indicate insufficient capacity but the structure may still be adequate in reality. All three conditions apply to the Zijlweg viaduct, which has ASR deterioration and rated insufficient for shear analytically.

Recent advances in proof load testing include standardised target load procedures based on equivalent sectional forces [5], mechanics-based stop criteria using the CSCT and CSDT [9], and Bayesian frameworks for updating structural reliability from the monitoring data collected during the test [1]. These advances transform proof load testing from a binary pass or fail demonstration into a quantitative tool for structural assessment.

1.3.2. Historical Development

Proof load testing represents the most direct method for assessing structural capacity, with historical precedents dating to ancient times when structures were tested by marching soldiers or rolling heavy loads across them [10]. Modern proof load testing emerged in the 20th century as a response to the limitations of analytical assessment, particularly for complex or deteriorated structures where analytical models prove unreliable.

1.3.3. Instrumented Proof Load Testing

Contemporary proof load testing integrates comprehensive instrumentation systems to monitor structural responses during loading. As demonstrated in the Zijlweg viaduct test, modern tests employ arrays of sensors including load cells, linear variable differential transformers (LVDTs), laser displacement sensors,

strain gauges, and inclinometers [5]. This instrumentation transforms proof testing from a binary pass/fail exercise into a rich source of data about structural behavior.

1.3.4. From Survival Confirmation to Reliability Updating

The critical innovation in modern proof load testing lies in utilizing monitoring data not merely to confirm survival, but to update probabilistic reliability estimates. Each successfully sustained load level provides evidence about the structure's actual capacity, evidence that can be formally incorporated into reliability assessments through Bayesian statistical methods [1].

1.4. Bayesian Statistics: A Framework for Learning from Evidence

Bayesian statistics provides a mathematical framework for updating beliefs in light of new evidence, combining prior knowledge with observed data to produce updated estimates. In structural reliability, this allows monitoring data from a proof load test to formally reduce uncertainty about a structure's capacity. The theoretical foundations and recent advances in Bayesian methods for proof load testing are reviewed in Chapter 2.

1.5. The Zijlweg Viaduct as Case Study

The viaduct Zijlweg, a reinforced concrete solid slab bridge in Noord Brabant, the Netherlands, serves as the case study for this dissertation. The bridge was constructed in 1965, shows ASR deterioration, and rated insufficient for shear under analytical assessment. A full-scale proof load test was conducted in June 2015 generating comprehensive monitoring data for both bending and shear critical positions. This dataset provides a unique opportunity to apply Bayesian reliability updating to real field proof load test data from an ASR-affected structure. Full structural details and test description are provided in Chapter 4.

1.6. Research Gap and Problem Statement

Despite significant advances in both proof load testing methodology [5] and Bayesian reliability updating [1], a critical gap remains in their integration for real field structures. De Vries et al. [1] demonstrated Bayesian reliability updating using laboratory slab strip experiments and an analytical flexural model but

these case studies did not use field monitoring data from a real viaduct, and they did not systematically compare the resulting reliability estimates with mechanics-based stop criteria.

Meanwhile, stop criteria research [9] has produced validated CSCT/CSDT-based thresholds for shear-critical slabs, but these criteria have not been evaluated against a concurrent probabilistic reliability analysis. The Zijlweg viaduct dataset, with its simultaneous shear and bending measurements from a real ASR-affected structure, provides a unique opportunity to close this gap by:

1. Applying Bayesian reliability updating to actual field proof load test data for both shear and bending failure modes, and
2. Directly comparing the evolving reliability estimates with the CSCT/CSDT stop criteria indicators at each load step of the test.

The overall research question of this dissertation is:

How can monitoring data from an instrumented proof load test on an ASR-affected reinforced concrete bridge be used to quantitatively update structural reliability estimates, and how do these Bayesian reliability estimates compare with mechanics-based stop criteria evaluated at each load step?

The specific problems addressed are:

1. How can the indicator values from the Zijlweg proof load test monitoring data be used to update the structural reliability for both bending and shear failure modes, following the framework of De Vries et al. [1]?
2. How does the informed Bayesian load testing approach compare to the lower-bound method for assessing the reliability of the Zijlweg viaduct?
3. How do reliability estimates differ between bending and shear analyses, and what are the implications for structural safety assessment?

1.7. Research Objectives

Primary Objective:

To apply the existing Bayesian reliability updating methodology of De Vries et al. [1] to the monitoring data from the Zijlweg viaduct proof load test, evaluating both bending and shear failure modes and comparing the resulting reliability estimates with mechanics-based stop criteria indicators.

Specific Objectives:

1. To extract and verify the indicator values from the Zijlweg monitoring data, specifically concrete strain for bending (from LVDT02 and LVDT03) and maximum weighted nominal crack width for shear (from LVDTs 14-16), and apply these to the Bayesian updating framework of De Vries et al. [1].
2. To compare the updated Bayesian reliability estimates with the lower-bound approach at each load step, quantifying the additional information provided by the monitoring data beyond simple survival evidence.
3. To evaluate the consistency between the Bayesian reliability estimates and the CSCT/CSDT stop criteria indicators, and assess whether the stop criteria signals align with the reliability-based assessment at each load step.

1.8. Scope and Limitations

This research focuses specifically on the application and adaptation of existing Bayesian updating methodologies to the Zijlweg viaduct dataset. The scope includes:

1. Processing of the monitoring data from the 2015 Zijlweg proof load test [5] using MATLAB preprocessing scripts, and applying the Bayesian updating code of De Vries et al. [1] to this field data
2. Modification of published Bayesian algorithms [1] to accommodate the specific data format and requirements of the Zijlweg test
3. Reliability analysis for both bending and shear failure modes, using the bending04 and shear01 test data respectively
4. Comparison of Bayesian updating results with traditional assessment approaches, including code-based methods and lower-bound estimates
5. Sensitivity analysis of key assumptions and parameters in the Bayesian updating process

Limitations include:

1. Focus on a single case study (Zijlweg viaduct), though the methodology is intended to be generalizable
2. Requirement for code modifications and preprocessing rather than using algorithms as-is
3. Dependence on the specific instrumentation and loading protocol of the 2015 test
4. Use of existing algorithms with modifications rather than development of new methods

5. Assumption of stationarity in material properties during the test period
6. Primary focus on ultimate limit states rather than serviceability considerations

1.9. Thesis Structure

This thesis comprises seven chapters structured as follows:

Chapter 2: Literature Review examines previous research on structural reliability [7], proof load testing [5], Bayesian methods [11], and assessment of ASR-affected structures [4], establishing the theoretical foundation for the study.

Chapter 3: Methodology details the data processing workflow, Bayesian updating algorithms [1], and reliability analysis methods employed in the study, with particular attention to the implementation for the Zijlweg case study, including code adaptations and preprocessing requirements.

Chapter 4: Case Study - Zijlweg Viaduct describes the structure, proof load test [5], instrumentation system, and monitoring data, providing the context for the reliability analysis.

Chapter 5: Results and Analysis presents the calculated reliability indices for both bending and shear failure modes, compares different assessment approaches, and analyzes the uncertainty reduction achieved through Bayesian updating for both limit states.

Chapter 6: Discussion interprets the results in the context of bridge assessment practice, discusses limitations, implementation challenges, and provides insights for practical application.

Chapter 7: Conclusions and Recommendations summarizes main findings, contributions to knowledge, and provides recommendations for future research and practice, with specific guidance on code modifications and preprocessing requirements.

1.10. Contribution to Knowledge

This research contributes to the field of structural assessment through:

1. Demonstration of a complete Bayesian updating workflow for proof load test data, building on the methodologies of De Vries et al. [1]
2. Quantitative analysis of uncertainty reduction through instrumented monitoring, extending the work of Zarate Garnica et al. [9]

3. Practical insights for assessment of ASR-affected bridges, complementing the research of Lantsoght et al. [5]

4. Recommendations for integrating Bayesian methods into routine assessment practice, addressing the implementation barriers identified by Lantsoght [12]

By bridging the gap between theoretical Bayesian methods and practical proof load testing, this thesis aims to advance the state of practice in bridge assessment, supporting more informed decision-making for aging infrastructure management.

The practical importance of this work lies in its potential to make proof load testing safer and more efficient. If Bayesian updating can demonstrate that the required reliability level is already achieved before the maximum target load is reached, the test can be stopped earlier, reducing the risk of damage to the bridge. At the same time, comparing the probabilistic results with stop criteria tells us whether current deterministic thresholds give consistent safety signals. Both findings are relevant to the ongoing development of proof load testing standards in the Netherlands and internationally.

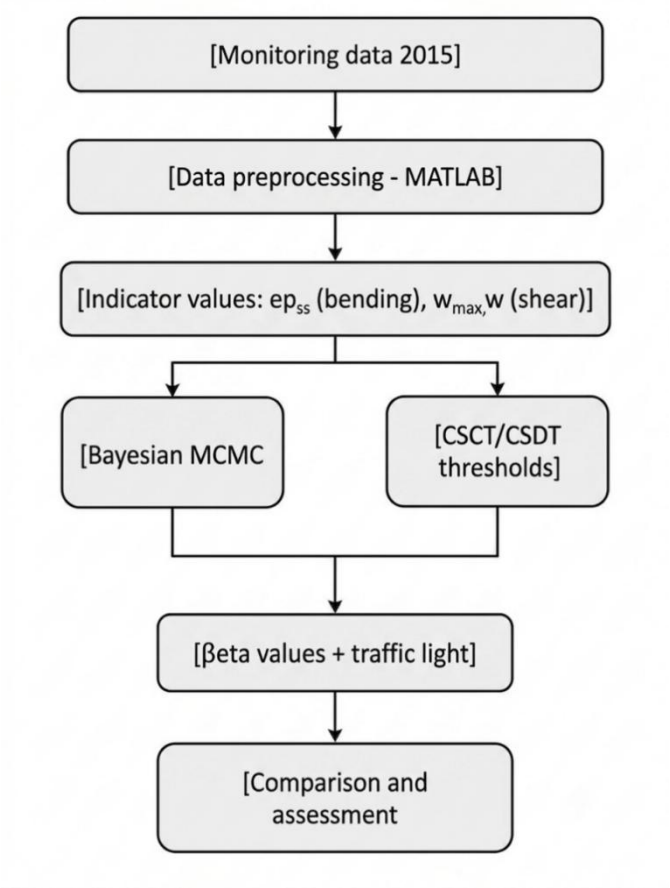


Figure 1.4 – Flowchart showing the research methodology and connection between the main components of this dissertation.

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2. LITERATURE REVIEW

2.1. Introduction

This chapter examines the key research areas that form the foundation for this thesis. The assessment of existing bridges through proof load testing brings together several distinct fields of study: structural reliability theory, probabilistic methods, Bayesian statistics, experimental testing protocols, and practical bridge engineering. My goal in this review is to connect these different research streams and show how they come together to support the application of Bayesian reliability updating to the Zijlweg viaduct proof load test data.

I have organized this review around four main themes. First, I look at the fundamentals of structural reliability how we quantify safety in uncertain systems. Second, I examine proof load testing methodologies, from historical approaches to modern instrumented testing. Third, I explore Bayesian methods in engineering, focusing on how we can update our beliefs with new evidence. Finally, I consider the specific challenges of assessing structures affected by alkali-silica reaction (ASR), which is particularly relevant for the Zijlweg viaduct case study.

2.2. Structural Reliability Fundamentals

2.2.1. From Deterministic to Probabilistic Thinking

One of the most significant shifts in structural engineering over the past fifty years has been the move from deterministic to probabilistic thinking. In the old way of thinking, engineers used safety factors essentially adding extra capacity to account for uncertainties. The problem with this approach, as Faber et al. [6] pointed out, is that it doesn't actually quantify how safe a structure is. It just makes it "safer" by an unknown amount.

The probabilistic framework, championed by organizations like the Joint Committee on Structural Safety [7], takes a different approach. Instead of safety factors, we talk about reliability indices (β) - numbers that actually tell us the probability of failure. The Eurocode [8] specifies that for ultimate limit states, we should aim for $\beta = 3.8$ over a 50-year period. This is a much more honest way to talk about safety because it acknowledges that nothing is 100% safe - we're always dealing with probabilities.

What I find particularly interesting is how this thinking changes when we assess existing structures rather than design new ones. Diamantidis et al. [4] make this point clearly: an existing structure has a history. It has survived certain loads, it might show signs of deterioration, and we might have inspection records. All of this is valuable information that should inform our assessment, but traditional code-based approaches often ignore it.

2.2.2. The Challenge of Time and Deterioration

Bridges don't stay the same over time. They deteriorate, traffic loads increase, and our understanding of their behavior evolves. This time-dependent nature of structural reliability is something researchers have been grappling with for decades. Early work by Vu & Stewart [13] looked at how corrosion affects reliability over time, while more recent studies like Val & Stewart [14] have developed more sophisticated models. De Vries et al. [1] brought an important perspective to this conversation by looking at how different levels of knowledge affect reliability assessment. They showed that when we have more information - from inspections, monitoring, or testing we can reduce our uncertainties and get a clearer picture of the actual safety level. This is exactly the situation with proof load testing: we're gathering new information that should reduce our uncertainties.

2.2.3. Systems, Not Just Components

System-level reliability and spatial correlation between bridge components are relevant for multi-span assessment but are outside the scope of this study, which focuses on cross-sectional reliability at two critical positions of span 4 of the Zijlweg viaduct.

2.3. Proof Load Testing Methodologies

2.3.1. From Ancient Practice to Modern Science

The idea of testing a structure by loading it isn't new. ElBatanouny et al. [10] trace the history back to ancient times when armies would march across bridges to test them. What's changed dramatically is how we conduct and interpret these tests. Modern proof load testing, as described by Lantsoght et al. [5], is a sophisticated scientific process with carefully controlled loading protocols and extensive instrumentation.

What I find most significant about this evolution is the shift from a simple pass/fail mentality to a more nuanced understanding of structural behavior. In the past, if a bridge didn't collapse under test loading, it was deemed "safe." Today, we use the test to learn about how the bridge actually behaves how it deflects, where strains develop, how cracks respond. This richer understanding comes from the comprehensive instrumentation now standard in proof load testing.

2.3.2. The Instrumentation Revolution

Looking at the Zijlweg viaduct test setup described by Lantsoght et al. [5], I'm struck by how much data we can collect today. Thirty-six measurement channels recording forces, displacements, strains, and inclinations this creates a detailed picture of structural behavior that simply wasn't possible in the past. Zarate Garnica et al. [9] provide a useful overview of how monitoring technologies have advanced and how they're being applied in proof load testing.

Some of the newer technologies are particularly exciting. Digital image correlation (DIC), which Jones & Iadicola [15] describe, lets us measure displacements over an entire surface without physical contact. Acoustic emission (AE) monitoring, discussed by Zhang [16], can detect the tiny sounds of microcracking before visible damage occurs. These technologies could give us even earlier warning signs during testing, though integrating them with reliability assessment is still an active area of research.

2.3.3. Stop Criteria: Defining the Boundaries of Safe Loading

One of the most critical and historically poorly defined aspects of proof load testing is knowing when to stop loading. The consequences of loading a bridge past its safe limit are severe and potentially irreversible. At the same time, stopping too early means failing to demonstrate the required structural capacity, which defeats the purpose of the test. Stop criteria are therefore the operational bridge between test execution and structural safety.

Historically, stop criteria were borrowed from building codes, most notably the German guideline DAfStb [17] and the American code ACI 437.2M-13 [18]. These criteria were originally developed for flexural failure in buildings and focused on quantities such as residual deflection, stiffness reduction, and concrete strain limits. When Lantsoght et al. [5] conducted the pilot proof load test on the Zijlweg viaduct, they evaluated these existing criteria and found significant limitations. The deviation from linearity index and the permanency ratio were found to be strongly dependent on the prescribed loading protocol, making them difficult to apply directly to bridge proof tests. More critically, none of the existing codes defined

stop criteria for shear the failure mode of primary concern for the Zijlweg viaduct and many aging reinforced concrete slab bridges.

The recognition of this gap prompted systematic research into mechanics-based stop criteria specifically for shear-critical reinforced concrete members. The foundational challenge is that shear failure in reinforced concrete without transverse reinforcement is a complex, brittle phenomenon driven by the development of inclined cracks and the degradation of aggregate interlock, dowel action, and compression zone capacity. Any useful stop criterion must be

1. Measurable during the test without invasive procedures,
2. Physically meaningful in terms of proximity to failure,
3. Broadly applicable across different structural configurations, and
4. Neither overly conservative nor insufficiently protective.

2.3.4. The Traffic Light System: CSCT and CSDT-Based Stop Criteria

The most recent and comprehensive development in stop criteria for proof load testing is the traffic light system proposed by Zarate Garnica et al. [9]. This framework organizes stop criteria into three severity levels that give the bridge engineer actionable guidance at each stage of loading:

Green light: criteria related to the serviceability limit state. When the green light threshold is reached, the section is transitioning from uncracked to cracked behavior. This is informational the engineer understands that cracking has initiated but loading may continue.

Yellow light: criteria related to an intermediate warning level, indicating that the lower bound of shear capacity from mechanical models is being locally approached. This level requires increased attention and careful monitoring of subsequent load steps.

Red light: criteria indicating that further loading can result in local collapse. When a red-light criterion is reached, the test must be stopped immediately.

The theoretical basis for the shear stop criteria come from two complementary mechanical models of shear behavior in reinforced concrete members without transverse reinforcement: the Critical Shear Crack Theory (CSCT) and the Critical Shear Displacement Theory (CSDT).

The Critical Shear Crack Theory (CSCT), developed by Muttoni & Simões [19], models shear failure as a function of the width and roughness of the critical shear crack. In the CSCT framework, the shear-carrying capacity of concrete depends on crack width and aggregate size. As the critical shear crack

opens, the aggregate interlock mechanism the primary shear transfer mechanism in slabs without stirrups degrades. The CSCT stop criterion indicator is defined as:

$$I_{CSCT} = \tau\left(1 + 120 \cdot \frac{w}{d_g}\right) \leq \frac{\sqrt{f_c}}{3} \qquad (2.1)$$

where τ is the shear stress at the measurement location, w is the measured crack width, d_g is the maximum aggregate size, and f_c is the cylinder concrete compressive strength. The left-hand side represents the measurable structural response, while the right-hand side is the capacity threshold derived from concrete mechanics. When the indicator approaches or exceeds the threshold, the critical shear crack is reaching a width at which aggregate interlock is no longer reliable.

The Critical Shear Displacement Theory (CSDT), developed by Yang et al. [20], provides a complementary model based on the concept of a critical shear displacement the vertical displacement across the critical shear crack at which failure occurs. The CSDT stop criterion indicator is defined as:

$$I_{CSDT} \approx \frac{V\,(w - 0.01)}{bd} = \tau\,(w - 0.01) \qquad (2.2)$$

where V is the sectional shear force, b is the width of the member, d is the effective depth, and w is the measured crack width. The critical threshold $I_{CSCT,cr}$ is a function of the concrete compressive strength, the effective depth, the critical shear displacement Δcr, and the equivalent reinforcement diameter φ_{eq}. The CSDT captures the physical process by which increasing crack width and shear stress together lead to the exhaustion of the shear-carrying capacity.

An important practical aspect of both criteria is that they are functions of the measurement location, expressed through the ratio M/V_d normalized by the shear span-to-depth ratio $\lambda = a/d$. For the yellow light criterion, measurements in the range $0.8 < (M/V_d)\ (1/\lambda) < 1$ are used, corresponding to the region most likely to experience the development of the critical shear crack. For the red-light criterion, the range is $(M/V_d)\ (1/\lambda) < 0.7$, indicating that the shear-carrying mechanisms are nearly exhausted in the critical region.

The two theories are intentionally used together. As demonstrated by Zarate Garnica et al. [9], for some structural configurations CSDT provides a more conservative estimate while for others CSCT is governing. Using the minimum of the two indicators *min* (I_{CSDT}, I_{CSCT}) at each load step provides the most reliable early warning system. This dual-indicator approach has been validated through three series of experiments: slab strips, straight reinforced concrete slabs, and skewed slabs. Across these experimental series, the yellow light stop criteria achieved by CSDT showed coefficients of variation between 14% and 29%, with average values of P_{stop}/P_{max} around 48–67%, confirming that the criteria provide meaningful early warning without being excessively conservative.

For flexural failure, the stop criteria are also organized into the traffic light system. At the yellow light level, qualitative indicators include changes in the load-deflection curve, changes in the longitudinal and transverse deflection profiles, and residual crack widths exceeding $0.2 \cdot w_{max}$ for previously cracked members. At the red-light level, two quantitative criteria are defined: a concrete strain limit based on the strain corresponding to steel yielding, and a crack width limit based on the Frosch crack control model. The strain limit ensures that the proof test does not inadvertently cause yielding of the longitudinal reinforcement, which would constitute irreversible damage to the structure.

The relationship between stop criteria and Bayesian reliability updating is a key connection explored in this dissertation. While stop criteria provide operational thresholds for test execution, they do not inherently quantify the structural reliability. De Vries et al. [1] showed that the crack width measurements used in the CSDT/CSCT stop criteria can also serve as indicator values in a Bayesian updating framework specifically, the maximum weighted nominal crack width w_{max}, w is the indicator used to update the shear resistance distribution. This dual use of crack width measurements as both a stop criterion indicator and a Bayesian updating input is a central analytical thread of the present study.

2.3.5. Knowing When to Stop

What's becoming clear from the literature is that there's no one-size-fits-all answer. The appropriate stop criteria depend on the structure, the failure modes of concern, and the instrumentation available. Christensen et al. [21] showed how DIC and AE could help identify stop criteria in laboratory tests, but applying these methods in the field presents additional challenges.

2.3.6. Bringing Reliability into Proof Load Testing

Perhaps the most important development in proof load testing methodology has been the integration with reliability theory. Early theoretical work by Lin and Nowak [22] laid the groundwork, but practical implementation has taken longer to develop. Recent work by De Vries et al. [1] shows how we can move beyond deterministic pass/fail criteria to actually quantify how much a successful proof test increases our confidence in a structure's safety.

This integration addresses what I see as a fundamental limitation of traditional proof load testing: it treats the test as a binary event (pass or fail) rather than as a source of information that should update our understanding of the structure. Faber et al. [6] and Nishijima and Faber [23] showed how Bayesian

methods could help us extract more information from proof tests, particularly when we're testing multiple similar components. Their work provides important foundations for what I'm attempting in this thesis.

In this dissertation, two specific stop criteria indicators are applied to the shear01 monitoring data: the CSDT indicator I_{CSDT} and the CSCT indicator I_{CSCT} as defined by Zarate Garnica et al. [9]. These are calculated at each of the five load steps using crack width measurements from LVDTs 14, 15, and 16. The minimum of the two indicators min (I_{CSDT}, I_{CSCT}) is used for traffic light classification following the recommendation that both theories should be applied together. Flexural stop criteria are noted for context but not evaluated numerically, as the focus is on the shear-critical behaviour of the viaduct.

2.4. Bayesian Methods in Engineering

2.4.1. A Framework for Learning

At its heart, Bayesian statistics is about learning from evidence. The basic idea, which dates back to Bayes [24] and Laplace [25], is simple: we start with some initial beliefs (priors), we collect new evidence, and we update our beliefs accordingly. Modern treatments like Gelman et al. [11] show how powerful this framework can be for dealing with uncertainty in complex systems.

The Bayesian formula that captures this updating process is elegant in its simplicity:

$$\boldsymbol{P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)}} \qquad (\mathbf{2.3})$$

Where:

- $P(\theta|D)$ is the posterior distribution (updated belief about parameters θ given data D)
- $P(D|\theta)$ is the likelihood function (probability of observing data D given parameters θ)
- $P(\theta)$ is the prior distribution (initial belief about parameters θ)
- $P(D)$ is the marginal likelihood (normalizing constant)

In plain language: our updated belief about some parameters θ (like a bridge's strength) given some data D (like measurements from a proof test) equals how likely that data would be if our beliefs were correct, times our initial beliefs, divided by a normalizing factor. This framework is perfect for proof load testing because it gives us a mathematically rigorous way to combine what we knew before the test with what we learned during the test.

2.4.2. From Theory to Practice

Applying Bayesian methods to structural reliability has come a long way from early theoretical work. Steenbergen & Vrouwenvelder [3] laid important foundations, but practical implementation was limited by computational challenges. Today, with more powerful computers and better algorithms, we can actually implement these methods for real engineering problems.

Straub & Papaioannou [26] provide what I consider one of the most comprehensive guides to Bayesian updating for structural reliability. They don't just present the theory; they show how to implement it, with practical examples and discussion of computational methods. This resource has been invaluable for understanding how to approach the Zijlweg viaduct analysis.

Looking at specific applications helps illustrate the power of Bayesian methods. Jacinto et al. [27] applied Bayesian assessment to an actual bridge case study, showing how different types of evidence could be combined. De Vries et al. [1] took an interesting approach by using population data information from many similar structures to inform assessments of individual structures. These applications show the versatility of Bayesian methods for incorporating diverse information sources.

2.4.3. The Computational Challenge

One of the reasons Bayesian methods weren't widely used in engineering until relatively recently is the computational challenge. Calculating those posterior distributions can be mathematically intractable for complex problems. This is where Markov Chain Monte Carlo (MCMC) methods come in.

The Metropolis-Hastings algorithm, developed by Metropolis et al. [28] and generalized by Hastings [29], provides a way to sample from complex probability distributions. Instead of trying to solve equations analytically, we use random sampling to approximate the distributions we're interested in. This might sound crude, but with enough samples, it works remarkably well.

In engineering applications, MCMC methods have been used for everything from reliability assessment [30] to model calibration [31]. The key insight is that we don't need to have all the answers in closed form we can use computational methods to get practical solutions to complex problems.

2.4.4. Starting Points Matter

One aspect of Bayesian analysis that requires careful thought is choosing the prior distribution - our starting point before we see the new evidence. Kass & Wasserman [32] discuss the philosophical and

practical considerations in choosing priors. Should we use a "non-informative" prior that doesn't impose strong assumptions? Or an "informative" prior that incorporates what we already know from similar structures, design documents, or material tests?

For proof load testing applications, I think the right approach is to use priors that reflect our actual knowledge about the structure, but not so strongly that the new test data can't change our minds. We want to learn from the test, not just confirm what we already believed. Getting this balance right is part of the art of Bayesian analysis.

2.5. Assessment of ASR-Affected Structures

2.5.1. When Concrete Gets Sick

Alkali-silica reaction (ASR) is sometimes called "concrete cancer" an unfortunate but descriptive term. Certain types of aggregate in concrete can react with alkalis in the cement paste, forming a gel that absorbs water and expands. This expansion causes cracking, reduces stiffness, and can eventually affect structural capacity. Assessing ASR-affected structures is challenging because the effects are often visible (cracking) but their structural significance is hard to quantify analytically.

2.5.2. Proof Testing as a Direct Measure

This is where proof load testing becomes particularly valuable for ASR-affected structures. When analytical models struggle to account for ASR effects, a proof test gives us direct evidence of how the structure actually behaves under load. The Zijlweg viaduct test described by Lantsoght et al. [5] is a perfect example. Rather than trying to model all the complex effects of ASR, engineers loaded the bridge and measured its response directly.

What I appreciate about this approach is its honesty. It acknowledges that our models are imperfect, especially for complex deterioration mechanisms like ASR. Instead of pretending we can model everything perfectly, we go out and measure the actual behavior. This doesn't mean we abandon analysis - we still need to interpret the test results - but it grounds our assessment in reality rather than in potentially flawed assumptions.

2.5.3. Reliability with ASR Uncertainties

Reliability assessment of ASR-affected structures has to grapple with significant uncertainties. How fast is ASR progressing? How much has it reduced strength and stiffness? How will it continue to develop in the future? These are hard questions to answer with confidence.

Proof load testing helps with the first part - understanding the current condition. By successfully carrying load, the structure shows that it has adequate capacity now. Bayesian updating then helps us quantify how much this test result increases our confidence. The test doesn't eliminate all uncertainties (we still don't know exactly how ASR will progress), but it reduces the uncertainties about current capacity.

2.6. Research Gaps

The literature shows that while Bayesian reliability updating methods have been developed and validated using laboratory data [1], their application to real field proof load test data from ASR-affected bridges has not yet been demonstrated. Furthermore, the consistency between Bayesian reliability estimates and mechanics-based stop criteria has not been systematically evaluated for a real bridge case. This dissertation addresses both gaps using the Zijlweg viaduct dataset.

This literature review has shown how different research streams come together to support the work I'm doing in this thesis. From the probabilistic frameworks developed by JCSS [7] and others, to the proof load testing methodologies refined by researchers and practitioners, to the Bayesian methods that let us update our beliefs with new evidence - all of these come together in the analysis of the Zijlweg viaduct.

The key insight I take from this review is that proof load testing shouldn't be seen as an alternative to reliability assessment, but as a source of data for reliability assessment. When we load a bridge and measure its response, we're gathering evidence that should reduce our uncertainties about its capacity. Bayesian methods give us the mathematical tools to quantify that uncertainty reduction.

In the chapters that follow, I'll show how these theoretical foundations can be applied to a real case: the Zijlweg viaduct proof load test. By processing the monitoring data, implementing Bayesian updating algorithms, and analyzing the results, I hope to demonstrate not just that this approach works in theory, but that it can provide practical insights for assessing existing bridges.

3. METHODOLOGY

3.1. Introduction: A Three-Stage Analytical Framework

The methodology of this dissertation consists of applying the existing Bayesian reliability updating framework of De Vries et al. [1] to the monitoring data from the 2015 Zijlweg proof load test. The proof load test and its monitoring data were collected by Lantsoght et al. [5] and are available as an existing dataset. The Bayesian updating code was developed by De Vries et al. [1] and is available as an existing C++ implementation. The contribution of this dissertation is the application of this code to the Zijlweg field data, including the necessary data preprocessing, parameter derivation, and the integration of stop criteria evaluation alongside the Bayesian updating.

The analytical process involves three interconnected components rather than a strictly linear sequence. Data processing informs the statistical modelling, the statistical model provides the inputs to the Bayesian updating code, and the stop criteria evaluation uses the same processed monitoring data.

Stop criteria and Bayesian updating play distinct but complementary roles. Stop criteria are operational tools used during the test to ensure the immediate safety of the structure and personnel. Bayesian updating is a statistical tool used after the test to quantify the reliability gain. Both use the same sensor data (crack widths/strains) because a physical change in the bridge (the 'why') affects both the immediate safety limit and the long-term probability of failure.

Figure 3.1 illustrates these connections.

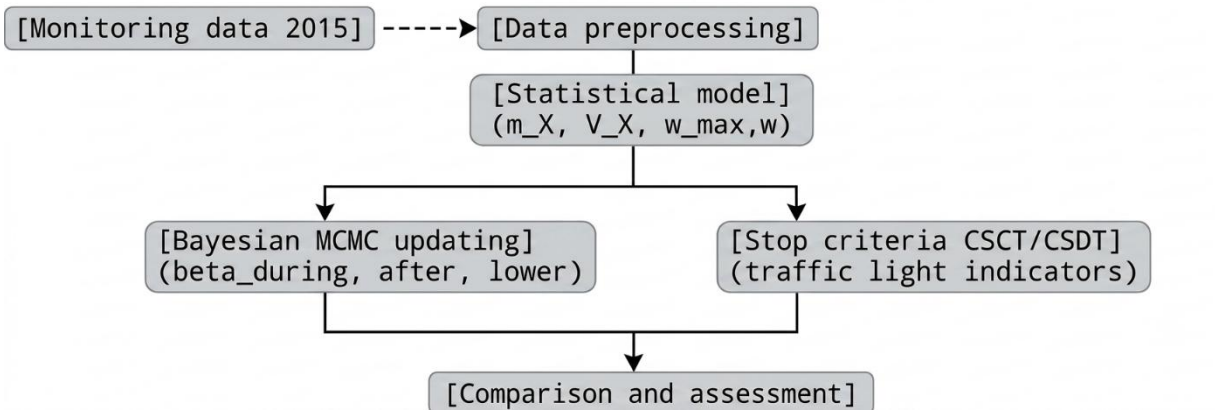


Figure 3.1 – Overview of the interconnected analytical components of the methodology. Data processing, statistical modelling, and Bayesian updating interact with each other rather than following a strictly linear sequence.

3.2. Stage 1: Data Acquisition and Processing

3.2.1. Data Source and Structure

The proof load test generated synchronized measurements from multiple instrumentation channels, recording applied loads, structural displacements, strains in reinforcement, and other response parameters. The raw monitoring data required calibration and preprocessing to correct for sensor characteristics and environmental effects before analysis. Detailed information about the specific test setup and instrumentation is provided in Chapter 4.

3.2.2. Measurement Correction and Calibration

Raw measurements from load cells, displacement sensors, and strain gauges required calibration corrections to account for sensor characteristics and measurement uncertainties. Force measurements from four hydraulic jacks were corrected using calibration factors derived from laboratory testing, ensuring accurate calculation of the total applied load. Displacement and strain measurements were adjusted for temperature effects and sensor drift, ensuring the data accurately represents the structural response during loading

3.2.3. Identification of Key Load Steps

Five distinct load levels were analyzed, corresponding to stabilized plateaus in the loading protocol where measurements were recorded for analysis. These levels (approximately 400, 810, 1100, 1200, and 1370 kN) represent progressive stages of the proof test, allowing assessment of structural response at increasing load magnitudes. The selection of these specific levels follows standard proof load testing protocols where measurements are taken at constant load conditions to ensure reliable data acquisition.

3.2.4. Strain Measurement Extraction

The critical measurement for flexural assessment is the strain in the bottom longitudinal reinforcement, which was measured using Linear Variable Differential Transformers (LVDTs) mounted on the tensile face of the slab (see Figure 4.8 for measurement locations). These strain measurements, corrected for temperature effects, provide direct evidence of the structural response to applied loads. The progression

of strain values with increasing load demonstrates the structure's elastic behavior and nonlinear characteristics, forming the primary evidence for Bayesian updating.

Table 3.1 – Summarizes the indicators used for both Bayesian updating and stop criteria evaluation for the two failure modes studied in this dissertation.

Failure Mode	Indicator	Source	Used for Bayesian Updating?	Used for Stop Criteria?
Bending (bending04)	Steel strain ε_s from LVDTs 2&3	De Vries et al. [1]	Resistance ratio via moment-strain model from cross-section analysis (Section 3.3)	Yes — strain limit stop criterion (red light)
Shear (shear01)	Max. weighted crack width w_{max}, w from LVDTs 14-16	De Vries et al. [1]	Resistance ratio via exponential crack width model calibrated on slab strip experiments (Section 3.3)	Yes — I_{CSDT} and I_{CSCT} indicators

[1] *Note: For shear, the Bayesian updating uses the exponential resistance ratio model from the slab strip experiments (beams H121, H401, H403, H404, H602) of De Vries et al.* [1]*, where crack width is the indicator. The stop criteria use the same crack width through the CSCT/CSDT mechanical models.*

3.3. Stage 2: Statistical Modeling of Structural Resistance

3.3.1. Resistance Ratio Concept

The methodology employs a resistance ratio (X) defined as the ratio of actual structural capacity to a reference capacity value:

$$\boldsymbol{X = \frac{R_{actual}}{R_{reference}}} \qquad (\mathbf{3.1})$$

where $\mathrm{R_{reference}}$ is typically the predicted yield moment or code-based capacity. This dimensionless ratio relates measured strains to structural capacity through a probabilistic model that accounts for uncertainties in material properties, geometry, and modeling assumptions.

3.3.2. Cross-Section Analysis via Latin Hypercube Sampling

The relationship between measured strain and resistance ratio is established through probabilistic cross-section analysis using Latin Hypercube Sampling (LHS). This approach generates 100 simulated cross-sections with varying material properties, dimensions, and reinforcement characteristics, representing the uncertainties in the actual structure. For each simulated cross-section, a complete moment-strain relationship is calculated considering material nonlinearities, reinforcement layout, and the effects of alkali-silica reaction (ASR). Moment capacities are determined through nonlinear cross-section analysis that accounts for the actual reinforcement configuration and load distribution in the slab.

Table 3.2 – Key uncertain parameters in cross-section analysis

Parameter	Distribution	CoV	Basis
Concrete strength (f_c)	Lognormal	0.10	Material test variability [7]
Steel yield stress (f_y)	Lognormal	0.05	Reinforcement property yield
Section height (h)	Lognormal	0.10	Construction tolerance standards
Concrete cover (c)	Γ	0.17	Reinforcement placement variability
Model uncertainty (c_2)	Lognormal	0.10	Nonlinear analysis uncertainty [7]
ASR strength reduction	Uniform	-	25% reduction per den Uijl & Kaptijn [33], ±10% bounds

3.3.3. Derivation of Statistical Parameters

For each measured strain level, the 100 simulated cross-sections provide 100 corresponding moment capacities. From these, statistical parameters characterizing the resistance distribution are calculated. The mean resistance ratio (m_x) represents the central tendency, while the coefficient of variation (Vx) quantifies the uncertainty:

$$\boldsymbol{M_x = \frac{1}{n}\sum_{i=.1}^{100}\frac{M_i(\epsilon_s)}{M_{ref}}} \qquad (3.2)$$

$$\boldsymbol{V_x = \frac{s_x}{m_x}} \qquad (3.3)$$

$$\boldsymbol{s_X = \sqrt{\frac{1}{99}\sum_{i=1}^{100}\left(\frac{M_i(\epsilon_s)}{M_{ref}} - m_x\right)^2}} \qquad (3.4)$$

where n = 100 samples, $\mathrm{M_i}(\epsilon_s)$ is the moment capacity at strain ϵ for sample i, and $\mathrm{M_{ref}}$ is the reference moment.

3.3.4. Incorporation of ASR Effects

The Zijlweg viaduct's ASR deterioration is incorporated in the statistical model through adjusted material properties. ASR reduces concrete compressive strength and modulus of elasticity, increases creep deformations, and affects the stress-strain relationship. These effects are represented through modified distributions for concrete properties in the cross-section analysis, based on empirical relationships for ASR-affected concrete [4]. The ASR deterioration is incorporated in the resistance model in two specific ways. First, in the cross-section analysis, the concrete compressive strength is based directly on core test results from the ASR-affected concrete ($f_{c,cube}$ = 44.4 MPa, Witteveen+Bos [34]), which inherently reflects the degraded material state.

Second, for stop criteria thresholds, the characteristic concrete strength (f_{ck} = 24.5 MPa) is used in the mechanical models. The 25% reduction recommended by den Uijl & Kaptijn [33] is applied to the final calculated shear capacity of the member, rather than the material property f_{ck}, to represent the analytical uncertainty of ASR-affected systems.

3.4. Stage 3: Bayesian Reliability Updating

3.4.1. Prior Distributions

Prior distributions reflect knowledge before the proof test. For the Zijlweg viaduct analysis, the following priors are used:

Table 3.3 – Prior distributions for Bayesian updating

Variable	Distribution	Parameters	Justification
Resistance ratio (X)	Lognormal	m_x, Vx from Table 3.4	Structure-specific model
Model uncertainty ($\boldsymbol{\theta_R}$)	Lognormal	Mean = 1.0, CoV = 0.15	Code recommendations
Load effect uncertainty ($\boldsymbol{\theta_E}$)	Lognormal	Mean = 1.0, CoV = 0.10	Typical for proof loading
Traffic load (Q)	Gumbel	Mean = 1301 kNm, CoV =	Extreme value

	maximum	0.058	distribution

[1] *The Lognormal distribution is appropriate for positive quantities like resistance ratios, ensuring physically meaningful results.*

3.4.2. Likelihood Function

The likelihood function $P(D|\theta)$ incorporates two types of evidence:

1. **Survival evidence**: The structure survived each applied load level, implying the actual capacity exceeds the applied load.
2. **Measurement evidence**: The specific strain values measured at each load level provide quantitative information about the structural response.

The survival evidence is implemented as a truncation of the probability distribution: parameter combinations that would have led to structural failure are assigned zero likelihood. The measurement evidence is incorporated through the statistical relationship between measured strains and resistance ratios from the cross-section analysis.

3.4.3. Markov Chain Monte Carlo Implementation

The posterior distribution is sampled using a Markov Chain Monte Carlo (MCMC) algorithm, specifically the Metropolis-Hastings variant. This computational approach generates samples from complex probability distributions that cannot be solved analytically. The implementation uses 1×10^9 total samples with parallel computation and a 1000-sample burn-in period to ensure convergence to the target distribution.

The MCMC algorithm iteratively proposes new parameter values, evaluates their consistency with the evidence, and accepts or rejects them based on probabilistic criteria. This process generates a sequence of samples that, in the long run, visits each parameter value in proportion to its posterior probability.

3.4.4. Reliability Index Calculation

From the posterior samples, the probability of failure (P_f) is calculated for three scenarios:

1. **During proof loading**: Reliability while the maximum test load is applied, representing the minimum safety margin during testing.

2. **After successful test**: Updated long-term reliability considering the evidence from the successful proof test, representing the improved knowledge about structural capacity.
3. **Lower-bound approach**: Conservative estimate that uses only the survival evidence (not the measurement evidence), providing a comparison with more traditional assessment methods.

The reliability index (β) is calculated from the probability of failure (P_f) using the inverse standard normal distribution:

$$\boldsymbol{\beta = -\Phi^{-1}(P_f)} \quad (\mathbf{3.5})$$

where Φ^{-1} is the inverse standard normal cumulative distribution function. Higher β values indicate greater reliability.

3.5. Stage 4: Stop Criteria Analysis

3.5.1. Rationale for Stop Criteria Integration

The Bayesian reliability updating methodology described in Stages 1–3 operates on the same monitoring data that is used for stop criteria evaluation. Specifically, the crack width measurements recorded at each load step serve as both:

1. the indicator values input to the Bayesian resistance model, as described in Section 5.3 of De Vries et al. [1]; and
2. the physical measurements evaluated against the CSCT and CSDT stop criteria thresholds.

Both CSCT and CSDT indicators are calculated at every load step and the minimum value min (I_{CSDT}, I_{CSCT}) is used for the traffic light classification, following Zarate Garnica et al. [9]. Using both theories together provides more reliable results since they are based on different mechanical models.

This dual role creates a unique opportunity to directly compare the probabilistic reliability update produced by Bayesian methods with the deterministic safety signals produced by the stop criteria at each load step.

The comparison is performed by mapping the deterministic safety signal (Traffic Light color) against the quantitative reliability index (β) at each load step. Consistency is achieved if the Bayesian model shows a reliability index above the target ($\beta \geq 4.0$) while the stop criteria remain in the 'Green Light' zone.

This comparison is the novel analytical contribution of this dissertation relative to the prior work of De Vries et al. [1]. While De Vries demonstrated the Bayesian updating methodology using laboratory slab

strip data, the present study extends the approach to field data from a real ASR-affected viaduct and explicitly examines how the stop criteria signals align or diverge from the evolving reliability estimates throughout the test.

3.5.2. Calculation of CSCT and CSDT Indicators

For each load step in the shear01 test, the CSDT and CSCT indicators are calculated using the crack width measurements from LVDTs 14, 15, and 16 located at and near the shear-critical section (see Chapter 4 for sensor locations). The shear stress τ at the measurement location is derived from the applied load and the structural geometry using the influence line approach.

CSDT indicator: $I_{CSDT} = \tau(w - 0.01)$, evaluated for crack widths $w > 0.01$ mm. The critical threshold $I_{CSDT,cr}$ is calculated using Equations (3) – (6) of Zarate Garnica et al. [9], requiring the concrete compressive strength f_c = 24.5 MPa (from core testing), the effective depth d derived from the cross-section geometry, and the equivalent reinforcement diameter φ_{eq} calculated from the reinforcement layout shown in Figure 4.3.

CSCT indicator: $I_{CSCT} = \tau(1 + 120 \cdot w/d_g)$, with the maximum aggregate size d_g assumed as 16 mm in the absence of specific aggregate size data for the Zijlweg viaduct. The capacity threshold $\sqrt{f_c}/3$ is calculated using the characteristic cylinder strength fc = 24.5 MPa, giving a threshold of approximately 1.63 $MPa^{0.5}$. The minimum of the two indicators $\min(I_{CSDT}, I_{CSCT})$ is then compared to the traffic light thresholds at each load step.

3.5.3. Traffic Light Classification at Each Load Step

The validity of the mechanics-based stop criteria (CSCT and CSDT) depends entirely on monitoring the structural response at the critical shear-flexure location. This is the zone where the diagonal shear crack is expected to reach its maximum opening before failure.

As illustrated in the load position diagram (Figure 4.6), the shear test was performed at 2.5d from support 5. To capture the critical response, LVDTs 14, 15, and 16 were used as the primary performance indicators. As shown in the instrumentation layout (Figure 4.8) and the crack monitoring map (Figure 4.9), these sensors were positioned directly over pre-existing cracks within the 2.5d zone.

To verify that these sensors are located in the "critical" region defined by the stop-criteria methodology, the normalized position parameter is checked. Using the shear-span a = 1.5 m and effective depth d = 0.516 m:

$$\frac{M}{v_d} \cdot \frac{1}{\lambda} = \frac{v \cdot a}{v \cdot d} \cdot \frac{d}{a} = 0.998 \qquad (\mathbf{3.6})$$

Since 0.998 falls precisely within the recommended range of 0.8 to 1.0, the measurement locations are confirmed as valid for the yellow-light stop criterion.

At each of the five load steps (approximately 400, 830, 1100, 1200, and 1340 kN), the following procedure is applied:

1. Extract the crack width w from the critical sensor group (LVDTs 14, 15, and 16) at the stabilized load plateau. The maximum value among these three sensors is selected to represent the governing crack opening, ensuring a conservative safety assessment.
2. Calculate the shear stress τ at the measurement location from the applied load and influence line.
3. Compute I_{CSDT} and I_{CSCT} and compare to their respective critical thresholds ($I_{CSDT,cr}$ and $\sqrt{f_c}$ / 3) derived from the core-testingmaterial properties.
4. Assign a traffic light color (green / yellow / red) based on the minimum indicator value: min(I_{CSDT}, I_{CSCT}).
5. Record the corresponding Bayesian reliability index β at the same load step for direct comparison.

The results are presented in tabular and graphical form in Chapter 5, showing the evolution of both the stop criteria indicators and the reliability index as the applied load increases.

3.5.4. Comparison Framework: Stop Criteria vs. Bayesian Reliability

The central comparison in this study is between two parallel assessments of the same structural situation at each load step:

1. The deterministic traffic light classification from the stop criteria, and
2. The updated probability of failure (and corresponding reliability index β) from the Bayesian updating procedure.

This comparison addresses the following question:

DO THE STOP CRITERIA PROVIDE SIGNALS THAT ARE CONSISTENT WITH THE RELIABILITY-BASED ASSESSMENT, OR DO THEY DIVERGE?

Three scenarios are of particular interest. First, alignment between approaches: if the stop criteria remain in the green or yellow light range at all load steps where Bayesian reliability also shows improvement, the approaches are mutually consistent. Second, divergence: if a stop criterion triggers at a load level where the Bayesian reliability index is still improving, the stop criterion may be overly conservative relative to the probabilistic assessment. Third, complementarity: if the Bayesian analysis shows that reliability is insufficient even before the stop criteria are triggered, the probabilistic approach provides additional safety information beyond what the stop criteria alone can detect.

This comparative analysis is the primary scientific contribution of this dissertation and directly addresses the gap identified in Section 1.6: the lack of systematic comparison between mechanics-based stop criteria and probabilistic reliability updating for real proof load test data.

3.6. Integration with Computational Processing

3.6.1. Data Flow and Processing Steps

The complete analysis follows a sequential workflow, as illustrated in Figure 3.2. While the cross-section analysis (Stage 2) and the definition of prior distributions are performed once to establish the baseline model, the subsequent updating procedure is iterative.

To provide a high-resolution reliability profile, the analysis is repeated for each of the five stabilized load steps identified in the testing protocol (see Table 4.4). For every load increment, the following specific calculations are executed within the integrated C++ framework:

1. **Indicator Extraction:** The governing structural response for the current load step is extracted from the preprocessed monitoring data (Table 4.9):
    - Bending: Maximum longitudinal concrete strain ($\varepsilon_{s,max}$) from LVDTs 02 and 03.
    - Shear: Maximum weighted nominal crack width ($w_{max,w}$) from LVDTs 14, 15, and 16.

The effect of ASR is incorporated at this stage by utilizing sensors specifically mounted over pre-existing ASR-induced cracks (as shown in Figure 4.9). By extracting the width changes (w) from these specific locations, the Bayesian update directly accounts for the physical progression of ASR-related damage. If the ASR had significantly compromised the aggregate interlock, these extracted indicators would show large openings; instead, the small measured values provide the evidence used to update the conservative analytical ASR assumptions.

2. **Likelihood Parameterization:** The measured indicator is used to determine the parameters of the resistance ratio (X), which defines the likelihood of the structure's true capacity:
   - Bending: The mean resistance ratio ($\mathrm{m_x = M_{y,mean} \,/\, E_{PL}}$) and the standard deviation ($\mathrm{V_X = M_{y,std} \,/\, E_{PL}}$) are updated based on the total load effect ($\mathrm{E_{PL} = G + Q_{PL}}$) at that step. The baseline capacity distribution for this calculation is derived from the moment-strain curves generated in the sectional analysis, as shown in Figure 5.1.
   - Shear: The mean and standard deviation are calculated using the calibrated exponential model: $\mathrm{m_x = 1.5e^{-3w} + 0.88}$ and $\mathrm{s_x = 0.53e^{-2.1w} - 0.05}$. The physical relationship between the crack width indicator and the resistance ratio is illustrated in Figure 5.4.
3. **Bayesian Reliability Updating (MCMC):** A Markov Chain Monte Carlo simulation using the Metropolis-Hastings algorithm is performed with 1 x 10^8 samples. The simulation solves the joint posterior distribution for the resistance and model uncertainties, specifically conditioning the results on the survival event (Z_{PL} > 0). This step accounts for the 0.7 correlation between the model uncertainty of the test load ($\theta_{E,PL}$) and the traffic load (θ_E).
4. **Reliability Index Calculation:** From the posterior samples, three specific reliability indices are calculated:
   - β_{during}: Probability of failure under the current test load (V_{PL} *or* M_{PL}).
   - β_{after}: Updated reliability against the annual maximum traffic load (Q) given the survival of the current step.
   - β_{lower}: Reliability based solely on the lower-bound evidence of the test load.
5. **Deterministic Stop Criteria Evaluation:** Simultaneously, the physical crack measurements are evaluated against the mechanical limits of the CSCT and CSDT theories:
   - CSDT: $I_{CSDT} = \tau$(w - 0.01) compared against $I_{CSDT,cr}$.
   - CSCT: $I_{CSCT} = \tau(1 + 120w/d_g)$ compared against $\sqrt{f_c} \,/\, 3$.

6. **Traffic Light Classification:** A safety signal (Green, Yellow, or Red) is assigned based on min(I_{CSDT}, I_{CSCT}).

This iterative flow ensures that the quantitative reliability values (β) and the deterministic safety signals (Stop Criteria) are perfectly synchronized for the final comparative assessment in Chapter 5.

The complete analysis follows a sequential workflow:

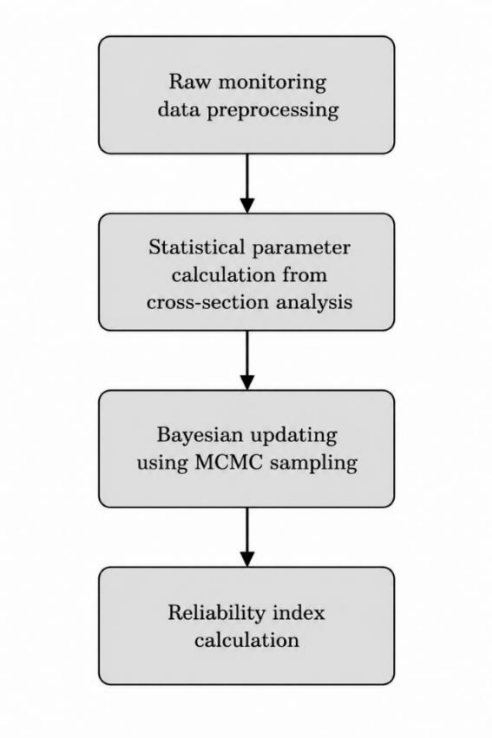


Figure 3.2 – The Complete analysis workflow

This workflow integrates preprocessing scripts with Bayesian updating algorithms, requiring adaptation to the specific data format and characteristics of the Zijlweg viaduct test.

3.6.2. Computational Implementation

The implementation combines **MATLAB** for data preprocessing with **C++** for the computationally intensive Bayesian updating. The existing algorithms require modification to accommodate the specific measurement parameters, loading conditions, and structural characteristics of the case study. These modifications ensure the methodology properly processes the available evidence and generates appropriate reliability estimates.

3.6.3. Input/Output Specifications

Primary inputs: Calibrated strain measurements at five load levels, statistical parameters from cross-section analysis ($m_{\ln}$, $V_{\ln}$), prior distribution parameters, and loading information.

Primary outputs: Posterior distributions for resistance parameters, updated reliability indices for three assessment scenarios, and sensitivity analysis results.

Intermediate outputs: Convergence diagnostics for MCMC sampling, likelihood evaluations, and proposal acceptance rates.

3.7. Validation and Quality Control

3.7.1. Convergence Assessment

MCMC convergence is verified through multiple criteria:

- **Gelman-Rubin statistic** < 1.1 for all parameters, indicating convergence across multiple chains
- **Trace plots** showing stable sampling without trends or drifts
- **Autocorrelation analysis** ensuring effective sample size > 1000 for reliable statistics
- **Running mean plots** demonstrating stabilization of parameter estimates

3.7.2. Sensitivity Analysis

Key modeling assumptions are tested through sensitivity analysis:

- **Prior distribution selection**: Comparison of informative vs. non-informative priors
- **Model uncertainty values**: ±20% variation in CoV values
- **Correlation assumptions**: Different correlation levels between load effects
- **ASR effect magnitude:** Variation in the severity of ASR-induced property reductions

3.7.3. Comparison with Alternative Methods

Results are compared with traditional assessment approaches (see Chapter 5 for detailed comparisons):

1. **Code-based assessment:** Allowable stress and load factor approaches per current design codes
2. **Deterministic proof test interpretation:** Simple pass/fail based on maximum applied load
3. **First-Order Reliability Method (FORM):** Reliability assessment without Bayesian updating
4. **Lower-bound Bayesian approach:** Using only survival evidence, not measurement evidence

3.8. Summary: A Complete Methodology for Reliability Updating

This chapter presents a complete methodology for applying Bayesian updating to proof load test data, specifically adapted for the assessment of ASR-affected bridges. The three-stage process are data processing, statistical modeling, and Bayesian updating systematically incorporates monitoring evidence to reduce uncertainties about structural capacity.

The methodology addresses both flexural and shear failure modes, incorporates ASR effects through adjusted material properties, and provides a framework for comparing different assessment approaches. The computational implementation combines preprocessing of monitoring data with advanced Bayesian algorithms, requiring adaptation to specific case study characteristics.

By moving from generic reliability estimates to structure-specific assessments informed by actual test evidence, this methodology supports more informed decision-making for aging infrastructure management. The following chapters apply this methodology to the Zijlweg viaduct case study, presenting results for both flexural and shear behavior, and discussing practical implications for bridge assessment practice.

4. CASE STUDY - ZIJLWEG VIADUCT

4.1. Introduction

This chapter describes the aspects of the Zijlweg viaduct and its 2015 proof load test that are directly required for the reliability analysis. This covers the structural geometry, material properties, ASR condition, test setup, instrumentation channels used as indicator variables, and the verified monitoring data at the five load steps. Information not directly used in the analysis is kept brief with references to Lantsoght et al. [5] for further detail.

4.2. Description of Viaduct Zijlweg

4.2.1. Location and History

The viaduct Zijlweg is a reinforced concrete solid slab bridge crossing the A59 motorway in the province of Noord Brabant, the Netherlands. It was constructed in 1965 with a design life of 80 years, as part of the expansion of the Dutch national road network during that era. The bridge carries a regional road over the highway and accommodates a single traffic lane with sidewalks on both edges. Since the viaduct carries fewer than 250,000 vehicles per year, it is classified as a regional road structure under the Dutch National Annex [35].

The viaduct is part of a series of bridges along the A59 highway that were identified as affected by alkali-silica reaction (ASR), a chemical deterioration mechanism discovered in 1997 [36]. Following this discovery, the bridge was included in a national monitoring programme covering 18 ASR-affected viaducts in the Netherlands [37]. The history of documentation of the viaduct is summarised in Table 4.1.

Table 4.1 – History of documentation of the viaduct Zijlweg (adapted from Lantsoght et al. [5])

Description	Year	Reference
Original design calculations	1965	Provincie Noord Brabant [38]
Detection of ASR damage	1997	Rijkswaterstaat [36]
Overview of repair activities and maintenance plan	2002	Rijkswaterstaat [39]
Description of monitoring system	2003,	Koenders Instruments [40]

	2007	
Structural inspection	2008	Rijkswaterstaat [41]
Material properties from core testing	2014	Witteveen+Bos [34]
Proof load test conducted	2015	Lantsoght et al. [5]

4.2.2. Structural System and Geometry

The viaduct Zijlweg is a continuous reinforced concrete solid slab bridge with four spans and a skew angle of 14.4°. The bridge crosses the A59 highway in the north-south direction. The end spans have a length of 10.32 m and the mid-spans a length of 14.71 m. The total width of the viaduct is 6.60 m, accommodating a carriageway of 4.00 m and sidewalks of 1.30 m on each edge.

The proof load test conducted in June 2015 focused on span 4, the end span between supports 4 and 5. This span was selected because it is not located directly above the motorway, eliminating the need for traffic restrictions during testing. The slab thickness in span 4 varies from 550 mm at support 5 to 850 mm at support 4, with a radius of curvature of r = 150 m. This variation in slab depth is important for the structural analysis, as the effective depth d at the critical cross-sections differs along the span.

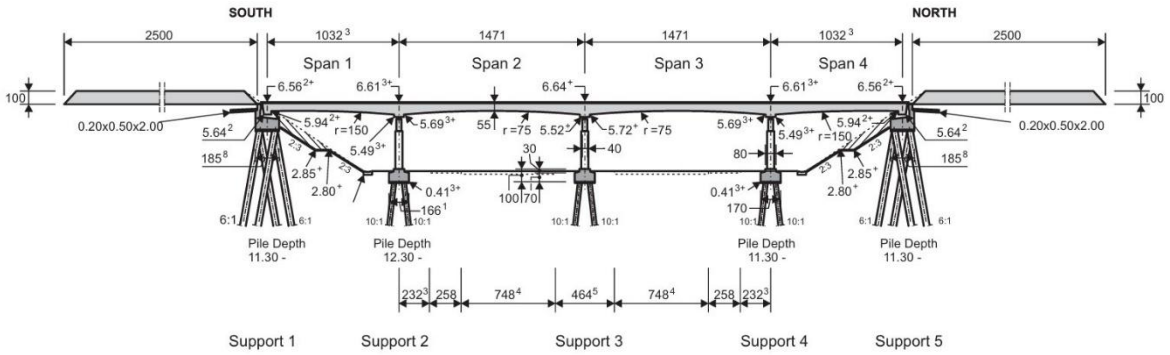


Figure 4.1 – Longitudinal section of viaduct Zijlweg [5].

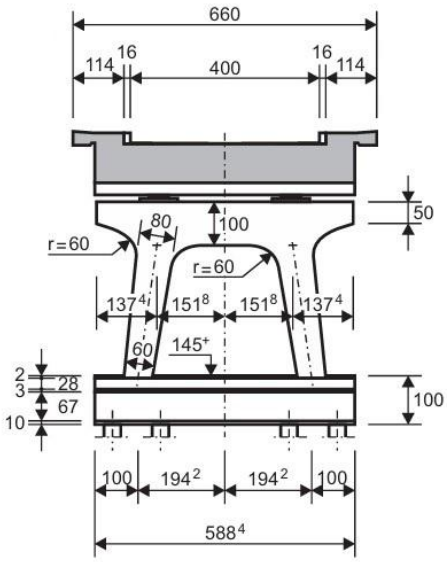


Figure 4.2 – Cross-section of viaduct Zijlweg [5].

4.2.3. Reinforcement Layout

The longitudinal reinforcement in span 4 consists of plain bars, indicating the use of QR22 steel (characteristic yield strength f_{yk} = 220 MPa) or QR24 steel (f_{yk} = 240 MPa), as the original design drawings use symbols for plain bars. No transverse shear reinforcement (stirrups) is present, which is typical for reinforced concrete slab bridges of this era and is the primary reason the viaduct was rated insufficient for shear upon analytical assessment. The absence of shear reinforcement means that the shear capacity relies entirely on the concrete contribution mechanisms: aggregate interlock, dowel action of the longitudinal reinforcement, and the concrete compression zone.

The main bottom reinforcement bars have diameters of ø28 mm at 300 mm spacing (bar No. 4, 9, and 16) and ø16 mm at various spacings. The top reinforcement near support 4 includes ø36 mm bars. The reinforcement layout is shown in Figure 4.3.

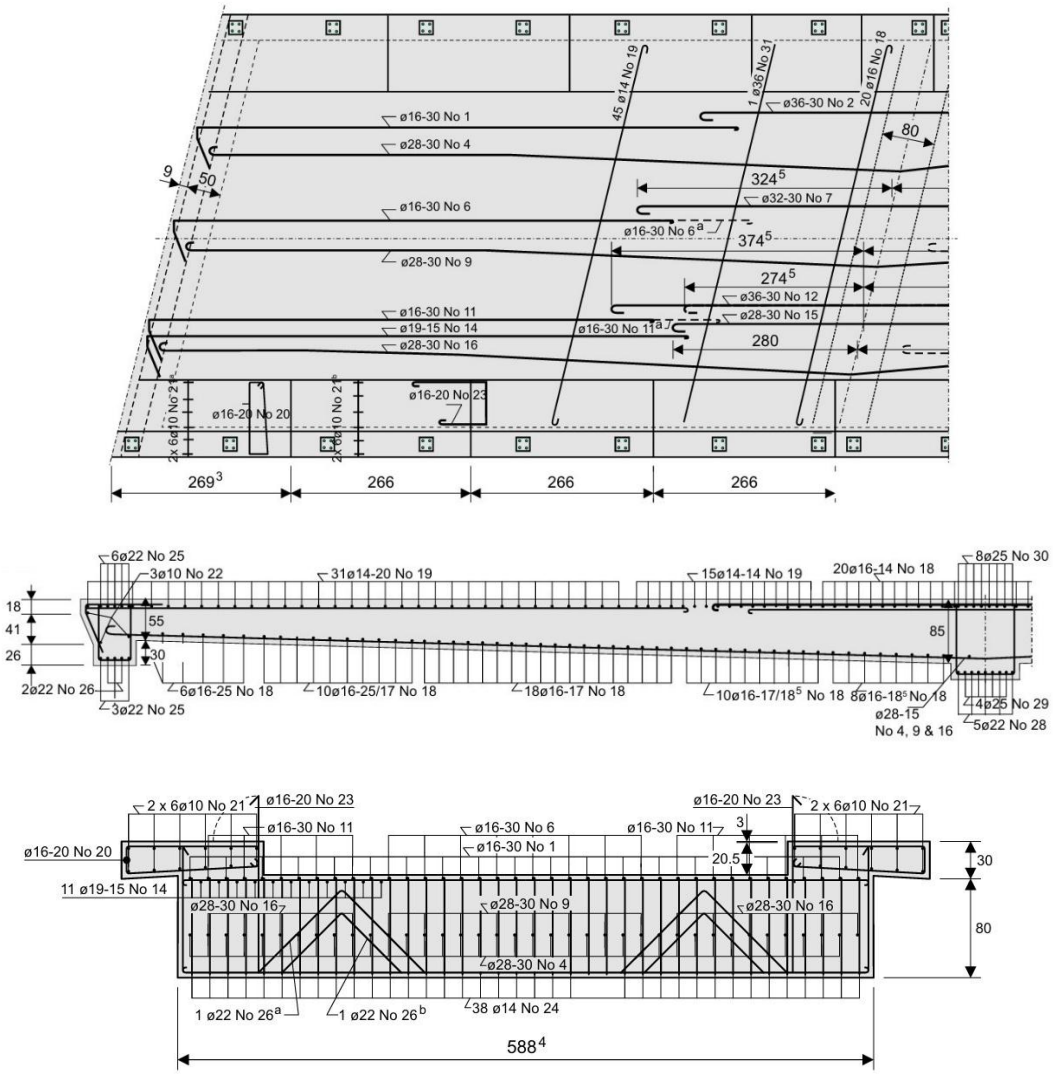


Figure 4.3 – Reinforcement layout of viaduct Zijlweg span 4 [5].

4.2.4. Material Properties

The concrete compressive strength was determined from drilled core samples extracted in 2014 [34]. The average cube compressive strength was measured as $f_{c,cube,m}$ = 44.4 MPa, corresponding to a characteristic cylinder compressive strength of f_{ck} = 24.5 MPa. The conversion from cube to cylinder strength applies a factor of approximately 0.82, giving a mean cylinder strength of f_{cm} = 36.4 MPa. The modulus of elasticity is calculated as $E_c = 21.5 \times (f_c/10)^{(1/3)} = 29{,}900$ MPa.

The properties of the reinforcement steel are not directly available from testing. Based on the reinforcement drawings and Dutch practice for structures of this era, QR22 steel with f_{yk} = 220 MPa or QR24 steel with f_{yk} = 240 MPa is assumed. These values are used in the probabilistic cross-section analysis for the bending resistance model. For the statistical model, the mean yield strength is taken as f_{ym} = 230

MPa with a coefficient of variation of 0.05, consistent with JCSS [7] recommendations for plain reinforcement bars.

Table 4.2 – Material properties of viaduct Zijlweg used in the reliability analysis.

Property	Value	Source
Cube compressive strength $f_{c,cube,m}$	44.4 MPa	Witteveen+Bos [34]
Characteristic cylinder strength f_{ck}	24.5 MPa	Lantsoght et al. [5]
Mean cylinder strength f_{cm}	36.4 MPa	Calculated (×0.82)
Modulus of elasticity E_c	29,900 MPa	Calculated from f_{cm}
Steel type	QR22 or QR24 plain bars	Design drawings
Characteristic yield strength f_{yk}	220 – 240 MPa	Design drawings
Maximum aggregate size dg	16 mm (assumed)	Typical for era
ASR shear capacity reduction	25%	den Uijl & Kaptijn [33]

4.3. ASR Damage and Structural Assessment

4.3.1. Extent of ASR Damage

Alkali-silica reaction was first reported in the Zijlweg viaduct in 1997, together with evidence of very low uniaxial tensile capacity and extensive map cracking [36]. ASR in this viaduct is the result of the reaction between reactive silica in the coarse aggregates and alkalis in the Portland cement paste, producing a gel that absorbs moisture and expands. This expansion caused cracking parallel to the reinforcement, a phenomenon characteristic of ASR in reinforced concrete where the expansion is restrained in the direction of the reinforcement bars but not in the vertical direction (where no transverse reinforcement exists).

A visual inspection conducted prior to the proof load test in June 2015 revealed deterioration on the top deck surface limited to the edge of the sidewalk, map cracking on the bottom face of the slab, and an almost fully closed expansion joint a direct consequence of the expansive gel formation. All existing cracks on the bottom and side faces of the slab were mapped and marked before the experiment, providing a baseline reference for monitoring crack development during the proof load test.

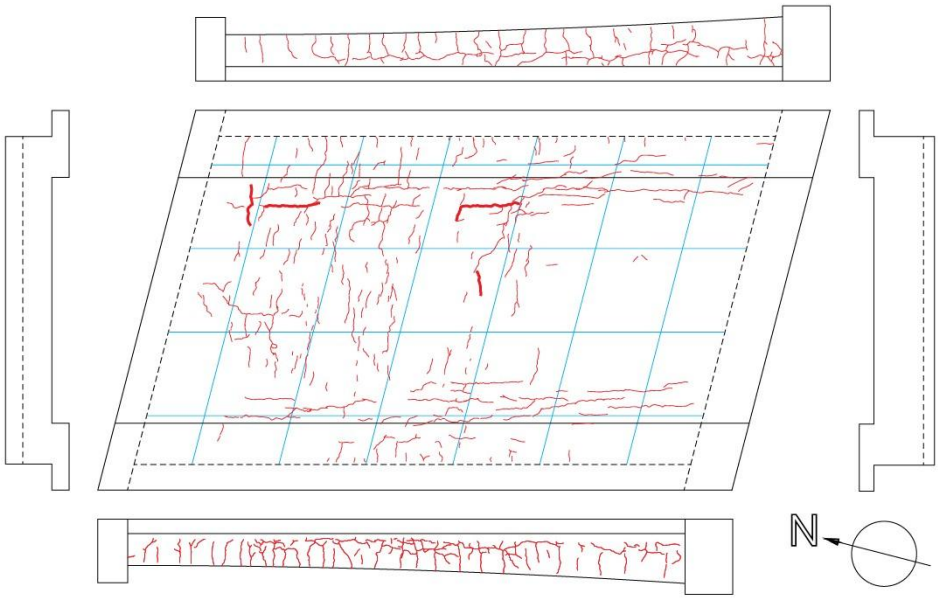


Figure 4.4 – Pre-existing crack pattern in span 4 of viaduct Zijlweg [5].

4.3.2. Analytical Assessment Results

The initial assessment of the viaduct for shear capacity produced a Unity Check (ratio of applied shear stress to shear capacity) of up to 5.4, indicating a severe apparent deficit in shear capacity. A reassessment using a linear finite element model and improved estimates of material properties reduced the maximum Unity Check to 1.31 still insufficient for safe operation under the prescribed live loads. The 25% reduction in shear capacity recommended for ASR-affected structures in the Netherlands [33] contributed significantly to this deficit.

The bending moment capacity was less affected by ASR, since bending failure is governed by concrete crushing in compression rather than tensile mechanisms. The assessment for bending moment gave Unity Checks below 1.0, indicating sufficient capacity for flexure. However, the shear deficit prompted the decision to conduct a proof load test as the Level IV assessment, following the Dutch Levels of Assessment framework [5].

4.4. Proof Load Test - June 2015

4.4.1. Test Objectives and Target Loads

The proof load test on the Zijlweg viaduct was conducted on 17 June 2015 with two objectives:

1. To demonstrate that the viaduct could carry the code-prescribed factored loads for both bending moment and shear, and
2. To serve as a pilot study for the development of standardised guidelines for proof load testing of reinforced concrete slab bridges in the Netherlands.

The target proof loads were determined following the Dutch approach based on equivalent sectional forces. For the bending moment test, the original target load corresponding to the RBK Design safety level (β = 4.3, 100-year reference period) plus 5% was 1,259 kN. For the shear test, the target load was 1,228 kN. An additional 5% was added to both targets to account for uncertainties in local material variability and the conversion from Eurocode live loads to the proof load tandem, acting as a model factor. The actual maximum loads applied during the tests were 1,368.2 kN for bending and 1,377.3 kN for shear.

Table 4.3 – Required maximum loads on proof loading tandem at different safety levels (Lantsoght et al. [5], Table 3).

Safety Level	**Reliability Index β**	**Reference Period**	**$F_{tot,bend}$ (kN)**	**$F_{tot,shear}$ (kN)**
ULS Eurocode	4.3	100 years	1,259	1,228
RBK Design	4.3	100 years	1,257	1,228
RBK Reconstruction	3.6	30 years	1,091	1,066
RBK Usage	3.3	30 years	1,050	1,027
RBK Disapproval	3.1	15 years	1,049	1,025
SLS Eurocode	1.5	50 years	815	791

It is important to note that the reliability indices (β) and reference periods shown in Table 4.3 were used during the design phase of the pilot test (2015) to determine the magnitude of the target proof loads. These indices correspond to long-term reference periods (15 to 100 years). In the updated analysis presented in Chapter 5, the structural safety is re-evaluated using an annual reliability framework consistent with current Dutch standards [1]. In that context, the target reliability indices are adjusted to the annual level (e.g., $\beta_{target} = 4.0$ for the disapproval level), allowing for a more precise assessment of the probability of failure in any given year of the viaduct's remaining service life.

4.4.2. Loading System

The load was applied using a temporary steel bridge structure placed over the concrete viaduct. Four hydraulic jacks transferred the load gradually from the steel structure to the slab surface through steel plates. The steel bridge was anchored using counterweights positioned on the opposite end of the viaduct,

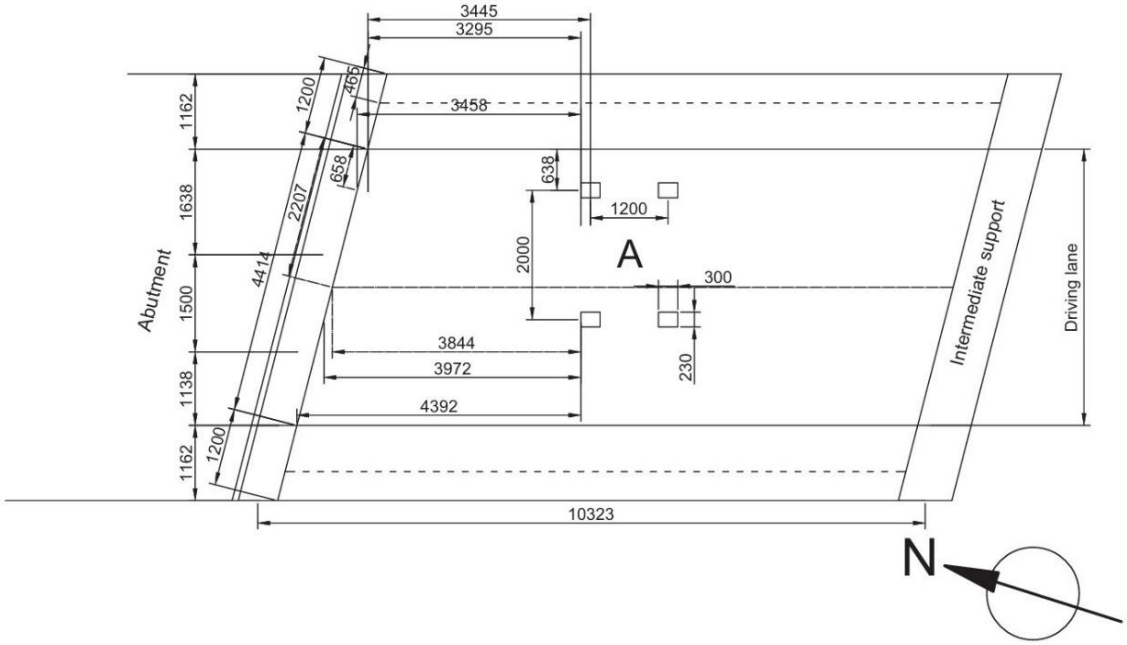
with Mammoet responsible for applying and controlling the loads. Between load cycles, a minimum load of 100 kN was maintained to keep all sensors activated.



Figure 4.5 – Loading system for the Zijlweg proof load test [1].

4.4.3. Load Positions

Two separate load tests were conducted at different positions on span 4, each targeting a different failure mode. The bending moment test positioned the load tandem at the critical location for maximum bending moment, with the face-to-face distance between the first axle and support 5 equal to 3,382 mm. The shear test positioned the tandem at the shear-critical location, with the face-to-face distance from support 5 equal to 1,208 mm (corresponding to 2.5d from the support face, where d is the effective depth). For the skewed bridge, the load was applied at the obtuse corner where shear concentrations are highest.



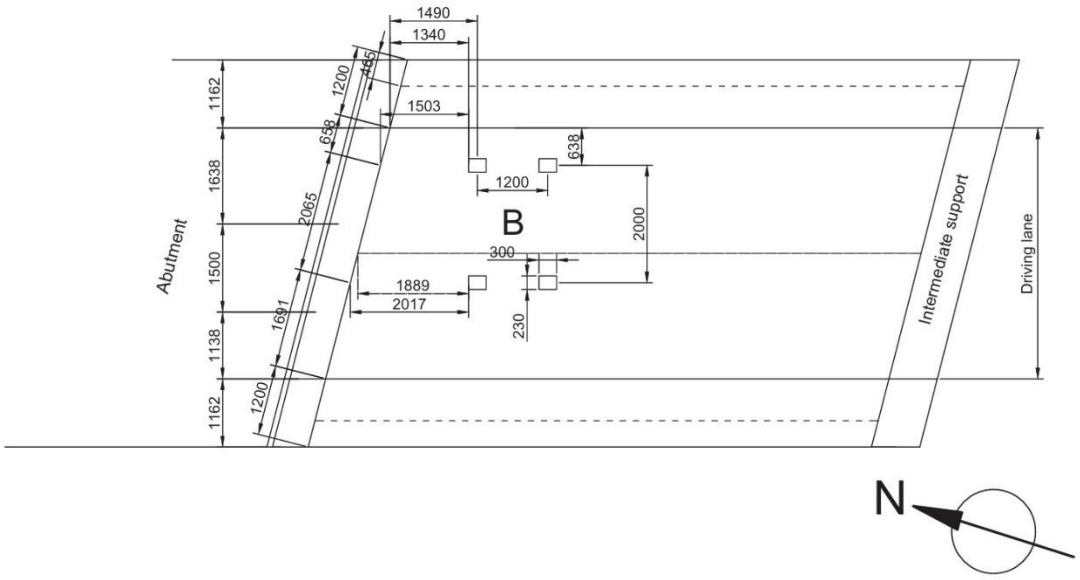
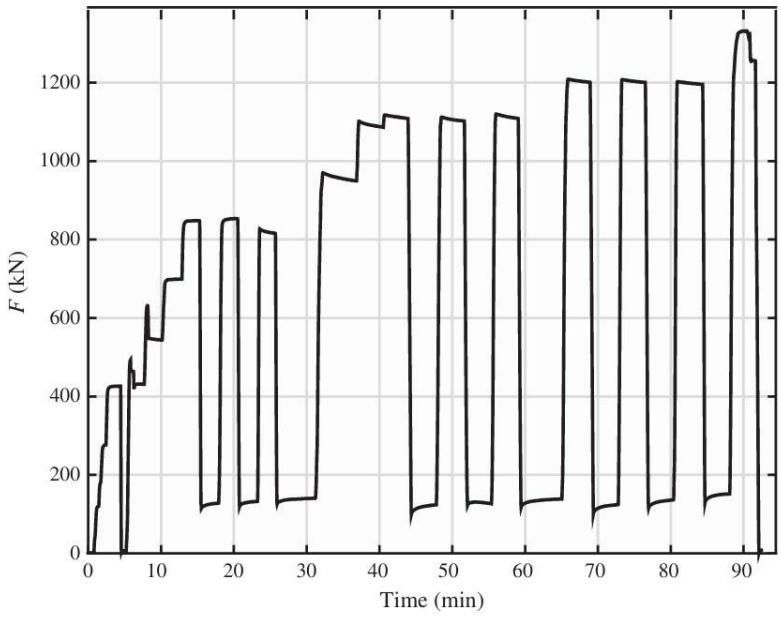


Figure 4.6 – Load positions for bending and shear proof load tests [5].

4.4.4. Loading Protocol

Both tests followed a cyclic loading protocol with five load levels. Each load level was applied three times to detect any changes in structural response indicative of non-linear behaviour or progressive damage. The five load levels for the bending moment test were approximately 40, 81, 110, 120, and 137 metric tons (400, 810, 1,100, 1,200, and 1,368 kN total). For the shear test, the five levels were approximately 40, 83, 111, 120, and 134 metric tons (400, 830, 1,100, 1,200, and 1,340 kN total). The loading scheme for both tests is shown in Figure 4.7.



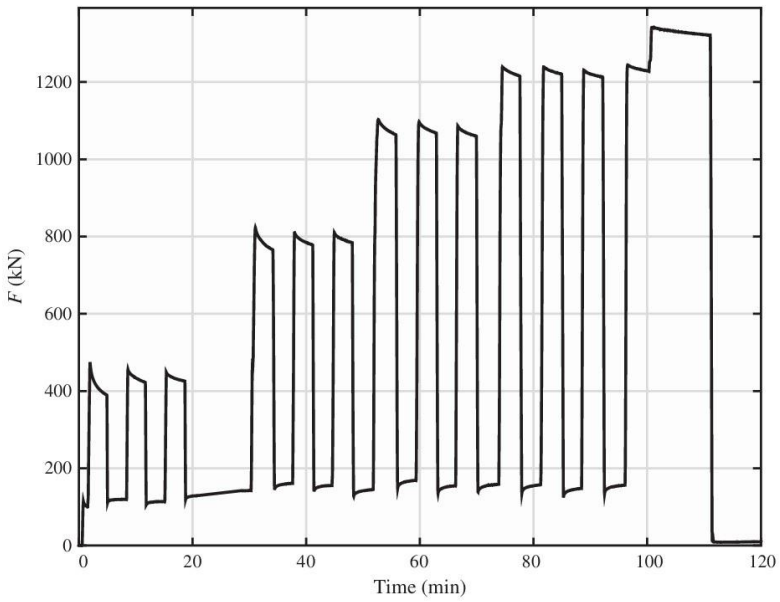


Figure 4.7 – Loading protocol for bending and shear proof load tests [5].

4.5. Instrumentation and Measurements Used for Analysis

4.5.1. Overview

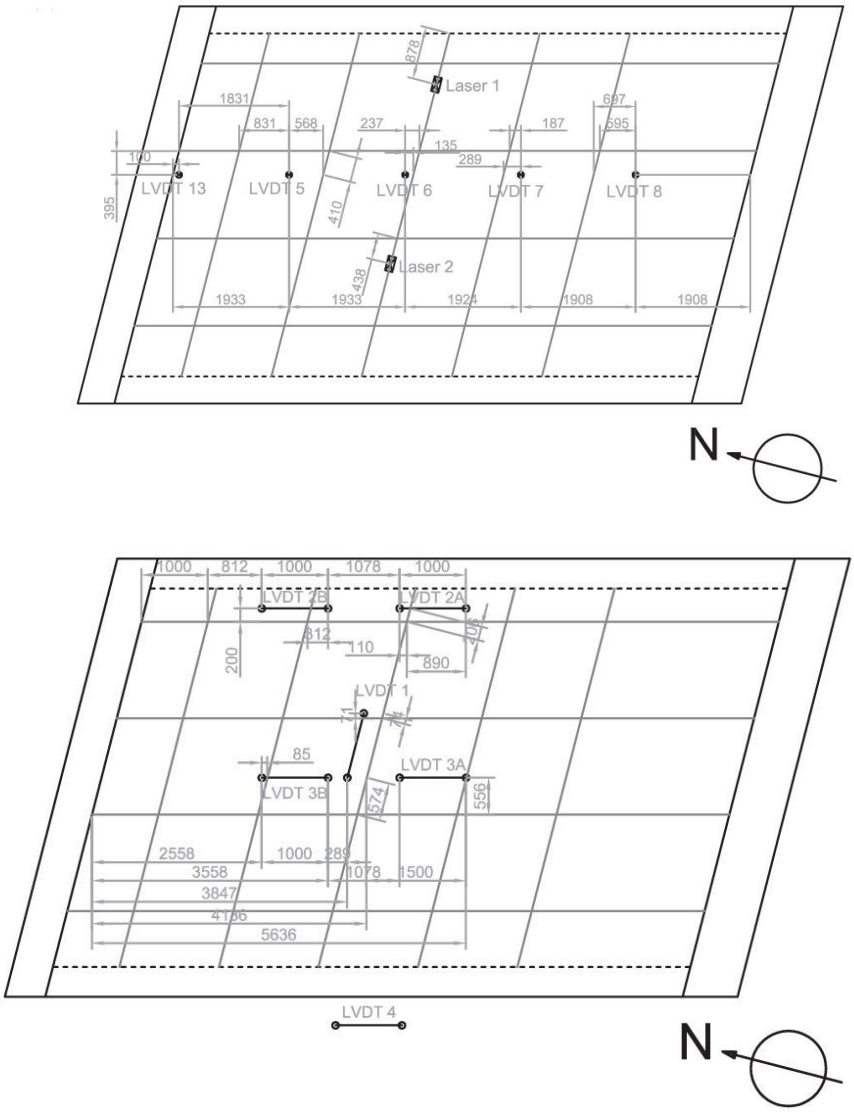
The instrumentation system recorded 36 channels during both tests. For this dissertation, two measurement subsets are directly used. For bending reliability updating: longitudinal concrete strain from LVDT02 and LVDT03. For shear reliability updating and stop criteria evaluation: crack width measurements from LVDT14, LVDT15, and LVDT16. Laser deflection channels are used qualitatively to verify linear behaviour. All other channels are described in Lantsoght et al. [5].

4.5.2. Load Measurement

The applied load was measured by four load cells, one at each hydraulic jack. The load cells were calibrated in the laboratory prior to the test, and calibration factors were applied to convert the raw voltage signals to forces in kilonewtons. The total applied load at each instant was calculated as the sum of the four individual jack forces: F_{total} = F1 + F2 + F3 + F4. Force sign conventions in the raw data use compression as negative; absolute values are used throughout this analysis. The maximum total loads recorded in the two tests were 1,346 kN (bending04) and 1,366 kN (shear01), consistent with the target values reported by Lantsoght et al. [5].

4.5.3. Deflection Measurements

Vertical deflections of the slab were measured using both LVDTs and laser distance sensors. Six laser sensors measured deflections at key positions along the span, providing deflection profiles in the longitudinal direction. Four additional LVDTs recorded deflections at positions closer to the load application point. The deflection data provides qualitative evidence of structural behaviour (linear vs. non-linear response) and serves as input to several stop criteria. The maximum mid-span deflection recorded during the bending test was 3.42 mm and during the shear test was 2.94 mm, both well below the limit of 1/180 of the span length (57 mm).



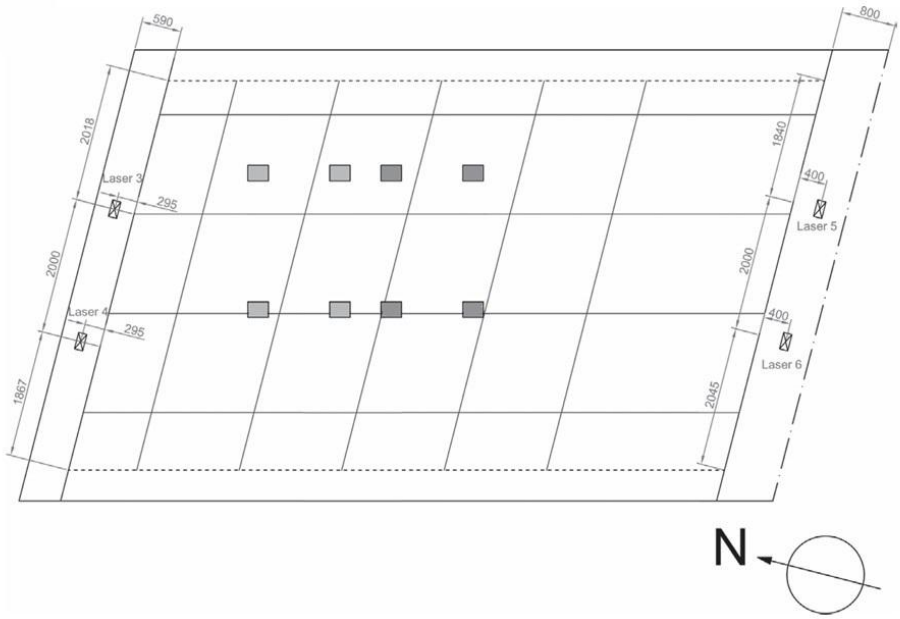


Figure 4.8 – Instrumentation layout on the bottom of the slab [5].

4.5.4. Strain Measurements

Concrete strains were measured using LVDTs mounted over a gauge length of 1,000 mm (1 m) on the bottom face of the slab. LVDT01 measured strain in the transverse direction, while LVDT02 and LVDT03 measured strain in the longitudinal direction. The strain in the longitudinal direction at the bottom of the slab is directly related to the bending resistance ratio, making LVDT02 and LVDT03 the primary indicator variables for the bending Bayesian updating analysis. The measured displacement values in millimetres divided by the gauge length of 1,000 mm give the concrete strain in mm/mm; multiplied by 10^6, this gives the strain in microstrain ($\mu\varepsilon$).

4.5.5. Crack Width Measurements

Three LVDTs (LVDT14, LVDT15, and LVDT16) were used to monitor crack width changes at pre-existing cracks identified during the pre-test inspection. For the bending test, these LVDTs monitored the longitudinal and transverse cracks located in the span, further from the support. For the shear test, the monitored cracks were located close to the support, at positions that could develop into shear cracks. LVDT14 measured a crack on the side face that could develop into a shear crack but remained essentially inactive during both tests; LVDT15 and LVDT16 measured longitudinal and transverse cracks on the

bottom face. The maximum crack width indicator for the Bayesian shear updating analysis is the maximum weighted nominal crack width $w_{max,w}$ derived from these three measurements.

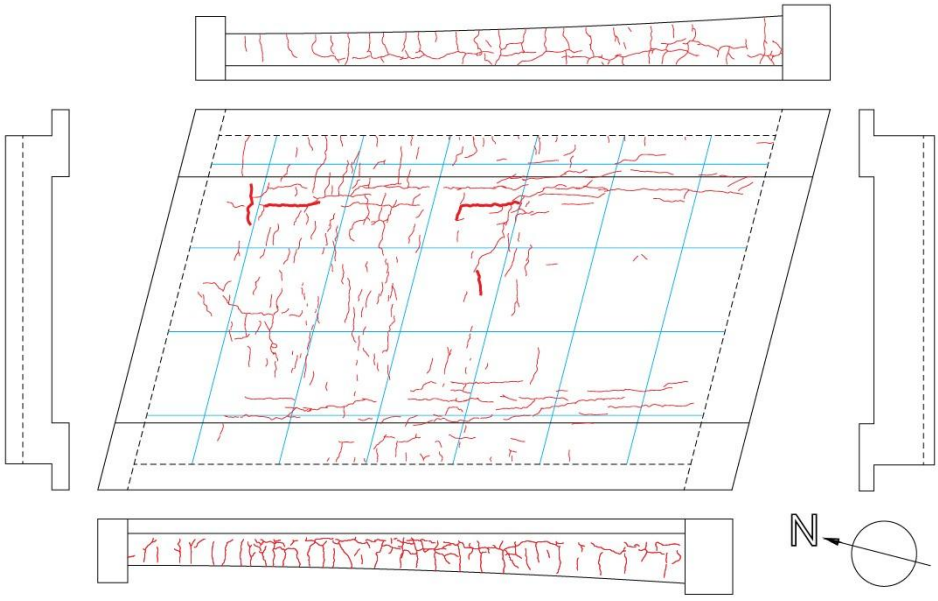


Figure 4.9 – Location of monitored cracks during the proof load tests [5].

4.6. Monitoring Data and Preprocessing

4.6.1. Data Structure

The raw monitoring data from both proof load tests is stored in MATLAB-compatible files (.mat and .xls formats), with filtered (preprocessed) versions provided in Excel spreadsheet format. The filtered data has been corrected for sensor drift and temperature effects using the reference LVDT mounted on the abutment. The bending test data (bending04) contains 5,548-time steps recorded over approximately 92 minutes. The shear test data (shear01) contains 10,025-time steps recorded over approximately 120 minutes. Both datasets share the same channel structure: four force channels, six laser channels, sixteen LVDT channels, two inclinometer channels, and four additional channels.

Data preprocessing for this study was performed in Python using the following steps:

1) Extraction of the filtered data from the Excel files;
2) Calculation of total applied load as the absolute sum of four jack force channels;
3) Identification of zero-load reference values using measurements at $F_{total} < 10$ kn;
4) Calculation of incremental changes from the zero reference for all LVDT and laser channels;
5) Identification of stabilised load plateau readings at each of the five load steps.

4.6.2. Identified Load Steps

Five load plateaus were identified in each test, corresponding to the maximum load level of each loading cycle in the protocol. At each plateau, the structural response was stabilised and measurements were averaged over all readings within ±50 kN of the target load level. The identified load steps and corresponding total applied forces are summarised in Table 4.4. These values are used as the proof load inputs m Q PL in the Bayesian updating analysis.

Table 4.4 – Identified load steps from bending04 and shear01 test data.

Load Step	Bending Test F (kN)	Shear Test F (kN)	Nominal Level
1	424.8	423.2	~40 metric tons (SLS check)
2	838.4	795.4	~83 metric tons (SLS Eurocode)
3	1,116.0	1,076.3	~111 metric tons (RBK Reconstruction)
4	1,213.6	1,226.2	~120 metric tons (RBK Usage)
5	1,337.6	1,329.8	~134 metric tons (RBK Design + 5%)

4.6.3. Bending Test (Strain Indicator Values)

The strain indicator values for the bending Bayesian updating analysis were extracted from LVDT02 and LVDT03 at each load step. These channels measure the elongation of the concrete over a 1 m gauge length at the bottom of the slab in the longitudinal direction. The maximum value of the two sensors is used as the indicator, consistent with the convention that the most critical measurement governs. The concrete strain at the bottom fibre is related to the steel strain through the depth ratio of the cross-section. Table 4.5 presents the extracted strain values and their relationship to the resistance ratio through the cross-section model.

Table 4.5 – Strain measurements from bending04 test and corresponding bending moments at critical section (a = 3,382 mm from support 5, L = 10,320 mm). Values obtained from MATLAB preprocessing script bending04.m.

Load Step	F_{total} (kN)	LVDT02 (με)	LVDT03 (με)	$\varepsilon_{s,max}$ (με)	M at critical section (kNm)
1	394.9	52.233	41.874	52.233	898.0
2	845.0	111.748	90.081	111.748	1,921.3

3	1,090.9	178.862	137.604	178.862	2,480.4
4	1,204.3	206.615	149.331	206.615	2,738.1
5	1,332.4	239.220	167.629	239.220	3,029.4

Verification: The maximum measured strain of 239.220 με at 1,332.4 kN is consistent with the value of ~240 με reported in Lantsoght et al. [5] Table 4 for the corrected longitudinal strain at maximum load, confirming the correct interpretation and calibration of the sensor data.

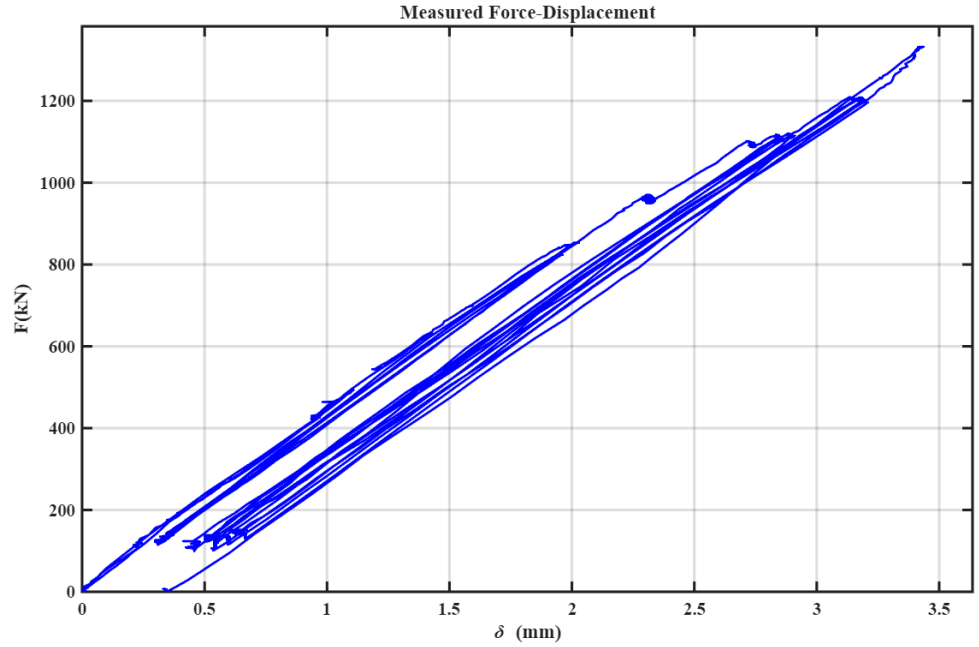


Figure 4.10 – Load-displacement envelope for bending moment proof load test.

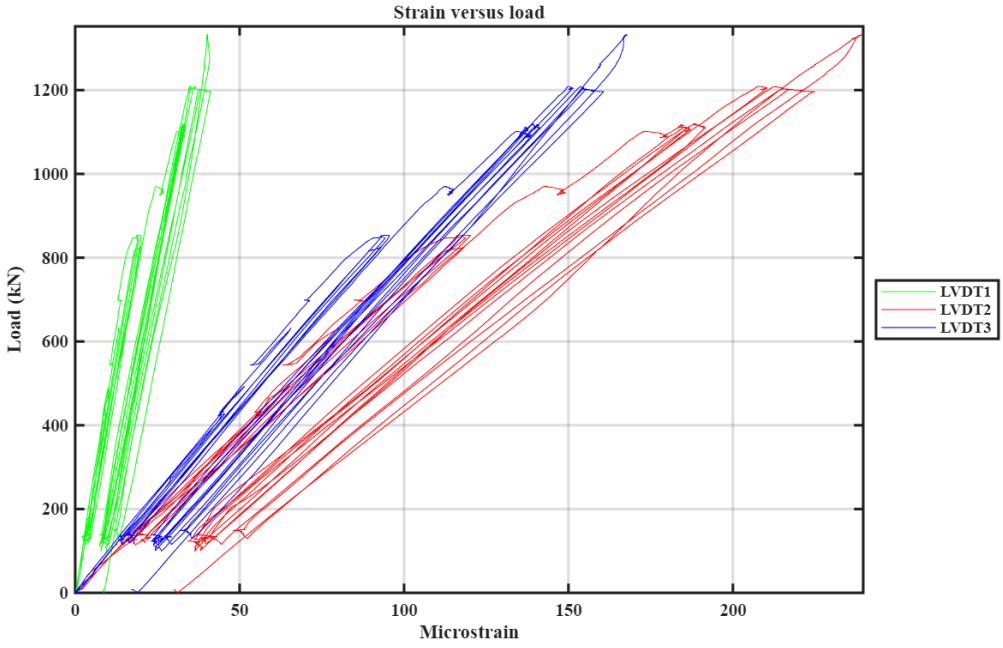


Figure 4.11 – Measured concrete strains vs. applied load for bending test.

4.6.4. Shear Test (Crack Width Indicator Values)

The crack width indicator values for the shear Bayesian updating analysis were extracted from LVDTs 14, 15, and 16 at each load step in the shear01 test. The maximum absolute crack width change among the three sensors at each load step is used as the maximum weighted nominal crack width $w_{max,w}$, which is the indicator variable in the shear resistance ratio model of De Vries et al. [1]. These values are also used to calculate the CSDT and CSCT stop criteria indicators, as described in Section 3.5.

Table 4.6 – Crack width measurements from shear01 test. Values are incremental changes from zero reference, obtained from MATLAB preprocessing script shear01.m.

Load Step	F_{total} (kN)	LVDT14 (mm)	LVDT15 (mm)	LVDT16 (mm)	$w_{max,w}$ (mm)
1	395.97	0.005187	0.002629	0.003314	0.005187
2	800.27	0.010690	0.009714	0.007093	0.010690
3	1,082.77	0.009845	0.016380	0.010555	0.016380
4	1,219.93	0.012600	0.019986	0.012290	0.019986
5	1,341.51	0.013902	0.022651	0.017083	0.022651

Verification: The maximum crack width of 0.0227 mm at 1,341.5 kN is consistent with the values shown in Lantsoght et al. [5] Figure 12c, which shows maximum crack width increases of approximately 0.020 mm at 1,340 kN. The values from LVDT15 govern at the higher load steps, indicating activation of the crack monitored by this sensor.

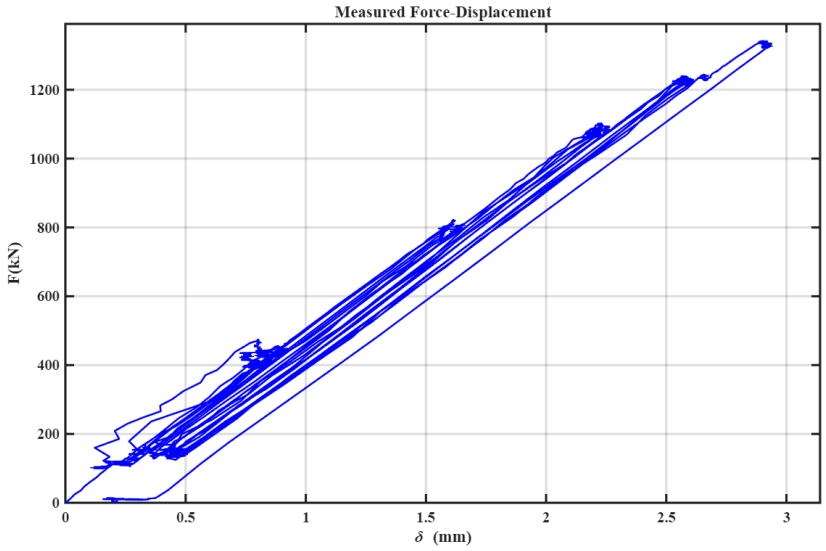


Figure 4.12 – Load-displacement envelope for shear proof load test.

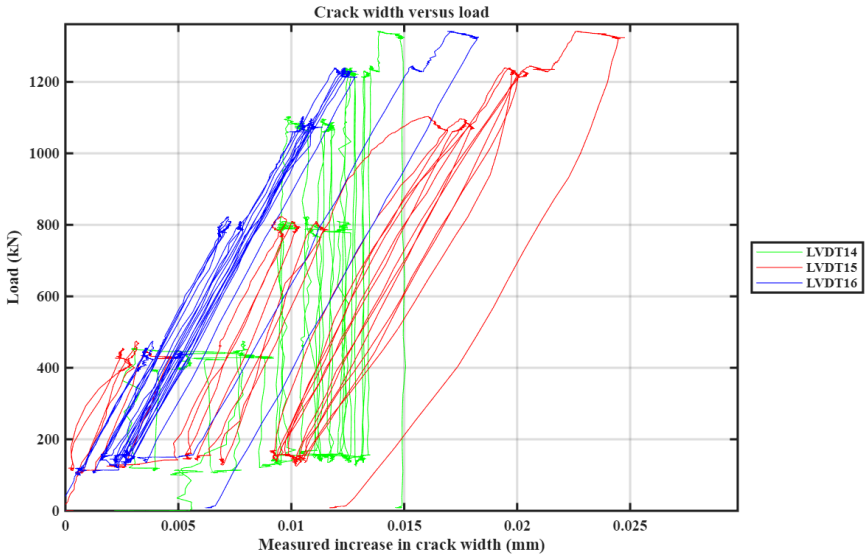


Figure 4.13 – Crack width measurements vs. applied load for shear test.

4.7. Structural Analysis (Permanent Load Effects)

The permanent load effects at the critical cross-sections are required as inputs to the reliability analysis. These are calculated from the self-weight of the slab and the superimposed dead load (asphalt layer) using simple beam theory for the continuous slab simplified as simply supported over span 4.

The self-weight is calculated using a mean slab thickness of 700 mm in span 4, a concrete density of 25 kN/m³, and the full slab width of 6.60 m, giving a self-weight of 115.5 kN/m. The superimposed dead load from the 100 mm asphalt layer with a density of 22 kN/m³ contributes 14.5 kN/m. The total permanent load intensity is 130.0 kN/m.

Table 4.7 – Permanent load effects at critical cross-sections of span 4.

Quantity	Bending (a = 3,382 mm)	Shear (a = 1,208–1,500 mm)
G_{DL}: Dead load effect	183.8 kNm	476 kN (at critical section)
G_{SDL}: Superimposed DL effect	33.0 kNm	~85 kN
Total permanent load effect	216.8 kNm	~561 kN
Coefficient of variation G_{DL}	0.05	0.05
Coefficient of variation G_{SDL}	0.10	0.10

4.8. Summary of Input Parameters for Bayesian Analysis

This section consolidates all structural, load, and monitoring parameters required as direct inputs to the Bayesian reliability updating analysis. Tables 4.8 and 4.9 represent the bridge between the case study description in Chapters 4.1-4.7 and the results presented in Chapter 5.

Table 4.8 – Structural and load parameters for Bayesian reliability updating.

Parameter	Value	Distribution	Source
Concrete strength f_c	36.4 MPa (mean)	Lognormal, CoV=0.10	Witteveen+Bos [34]
Steel yield strength f_y	220–240 MPa	Lognormal, CoV=0.05	Design drawings
Slab height h (at mid-span)	700 mm (mean)	Lognormal, CoV=0.10	Measured
Concrete cover c	40 mm (mean)	Γ, CoV=0.17	Inspection
Dead load G_{DL} (bending)	183.8 kNm	Normal, CoV=0.05	Calculated
SDL G_{SDL} (bending)	33.0 kNm	Normal, CoV=0.10	Calculated
Dead load G_{DL} (shear)	476 kN	Normal, CoV=0.05	Calculated
Traffic load Q (bending)	Gumbel, mean=1,301 kNm	CoV=0.058	De Vries et al. [1]
Traffic load Q (shear)	Gumbel, mean=390 kN	CoV=0.035	De Vries et al. [1]
Model uncertainty θ_R	Lognormal, mean=1.0	CoV=0.15	JCSS [7]
ASR capacity reduction	25%	Applied to f_c	den Uijl & Kaptijn [33]

Table 4.9 – Summary of verified monitoring data indicator values at each load step.

Load Step	F_{total} (kN) (bend/shear)	Bending $\varepsilon_{s,max}$ (με)	Shear $w_{max,w}$ (mm)	M at critical section (kNm)	V at critical section (kN)
1	394.9 / 396.0	52.233	0.005187	898.0	338.4
2	845.0 / 800.3	111.748	0.010690	1,921.3	684.0
3	1,090.9 / 1,082.8	178.862	0.016380	2,480.4	925.4

4	1,204.3 / 1,219.9	206.615	0.019986	2,738.1	1,042.6
5	1,332.4 / 1,341.5	239.220	0.022651	3,029.4	1,146.5

[1] *Values from MATLAB scripts bending04.m and shear01.m. These are the direct inputs to the Bayesian updating and stop criteria analyses in Chapter 5.*

4.9. Chapter Summary

This chapter has provided a complete description of the Zijlweg viaduct case study, covering the structural system, the ASR deterioration, the 2015 proof load test, the instrumentation system, and the monitored data extracted for the reliability analysis. The key findings from the data analysis are:

1. The viaduct experienced significant ASR damage, with a 25% reduction in shear capacity applied in the analytical assessment, leading to a shear Unity Check of up to 1.31 using finite element methods.
2. The proof load test applied five incremental load levels up to 1,338 kN (bending) and 1,330 kN (shear), corresponding to the RBK Design safety level plus 5%.
3. Strain measurements (LVDT02, LVDT03) and crack width measurements (LVDT14-16) were successfully extracted and verified against published values from Lantsoght et al. [5].
4. The structural behaviour during both tests was linear throughout, with no stop criteria exceeded and no signs of distress confirming the viaduct's sufficient capacity under the prescribed loads.
5. All input parameters for the Bayesian updating and stop criteria analyses have been derived and are consolidated in Tables 4.8 and 4.9.

The following chapter applies the Bayesian updating methodology to these data, calculating the probability of failure before, during, and after the proof load test for both bending and shear failure modes, and comparing the results with the CSCT/CSDT stop criteria analysis.

The detailed description of test setup, loading protocol, and structural geometry provided in Sections 4.1 to 4.7 follows Lantsoght et al. [5]. Readers requiring further detail on the full test procedure are referred to that publication. The input parameters consolidated in Section 4.8 are sufficient for replication of the Bayesian updating analysis presented in Chapter 5.

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5. RESULTS AND ANALYSIS

5.1. Introduction

This chapter presents the results of the Bayesian reliability updating analysis and the stop criteria evaluation for the viaduct Zijlweg, applied to the monitoring data collected during the June 2015 proof load test. The analysis follows the framework of De Vries et al. [1], adapted for the Zijlweg material properties, geometry, and measured indicator values as described in Chapters 3 and 4.

The results are organised into four parts. Section 5.2 presents the cross-section analysis results, including the moment-strain curves and yield moment statistics from the corrected sectional analysis incorporating all bottom reinforcement layers. Section 5.3 presents the bending reliability results from the MCMC analysis. Section 5.4 presents the shear reliability results. Section 5.5 evaluates the CSCT and CSDT stop criteria against the shear monitoring data and compares the results with the Bayesian reliability estimates. Section 5.6 provides a comparative discussion across all assessment approaches.

NOTE ON CODE CORRECTIONS: Initial implementation yielded near zero reliability indices due to a numerical misinterpretation of the input variation parameters. Upon correction of the statistical constructors to properly distinguish between standard deviation and the coefficient of variation, and upon inclusion of all bottom reinforcement layers in the sectional analysis, the Bayesian updating accurately reflects the significant reliability gain provided by the survival of the applied proof loads.

5.2. Cross-Section Analysis Results

5.2.1. Reinforcement and Geometry Inputs

The cross-section analysis was performed using the corrected sectional analysis code incorporating all four layers of bottom reinforcement identified from Lantsoght et al. [5] Figure 3. The input parameters are summarised in Table 5.1. A critical correction from earlier analysis was the inclusion of bar No. 14 (Φ19 at 150mm) and bar No. 20 (Φ16 at 200mm), which had been omitted in the initial implementation.

Table 5.1 – Input parameters for the Zijlweg cross-section analysis (100 LHS samples).

Parameter	Value	Basis
Width b	6.60 m (full slab)	Lantsoght et al. [5] Figure 2b
Height h (mean)	LN (0.594 m, CoV = 0.10)	Confirmed from Figure 2a
Concrete cover c	Γ (0.040 m, CoV = 0.17)	JCSS [7]
Bar No. 4/9/16: Φ28@300mm	A_s = 13,547 mm2	Lantsoght et al. [5] Figure 3
Bar No. 11: Φ16@300mm	A_s = 4,423 mm2	Lantsoght et al. [5] Figure 3
Bar No. 14: Φ19@150mm	A_s = 12,475 mm2 (ADDED)	Lantsoght et al. [5] Figure 3
Bar No. 20: Φ16@200mm	A_s = 6,635 mm2 (ADDED)	Lantsoght et al. [5] Figure 3
Total bottom steel	37,080 mm2	Sum of all four layers
Top steel Φ16@300mm	A_s = 4,423 mm2	Lantsoght et al. [5] Figure 3
Concrete f_c	LN (36.41 MPa, CoV = 0.10)	Witteveen+Bos [34] core tests x 0.82
Steel f_y	LN (230 MPa, CoV = 0.05)	QR22/QR24 plain bars, JCSS [7]
Young's modulus E_s	LN (205 GPa, CoV = 0.02)	JCSS [7]
Non-linearity c_2	LN (0.41, CoV = 0.10)	De Vries et al. [1]

5.2.2. Moment-Strain Curves and Yield Moment Statistics

One hundred Latin Hypercube Sampling realisations were generated, each producing a complete moment-strain curve for the critical bending section at 3,382 mm from support 5. Figure 5.1 shows all 100 curves with the mean curve highlighted. The spread between curves reflects the combined uncertainty in concrete strength, steel yield strength, and section geometry. The curves converge toward a common yield plateau, demonstrating consistent behaviour across the LHS samples.

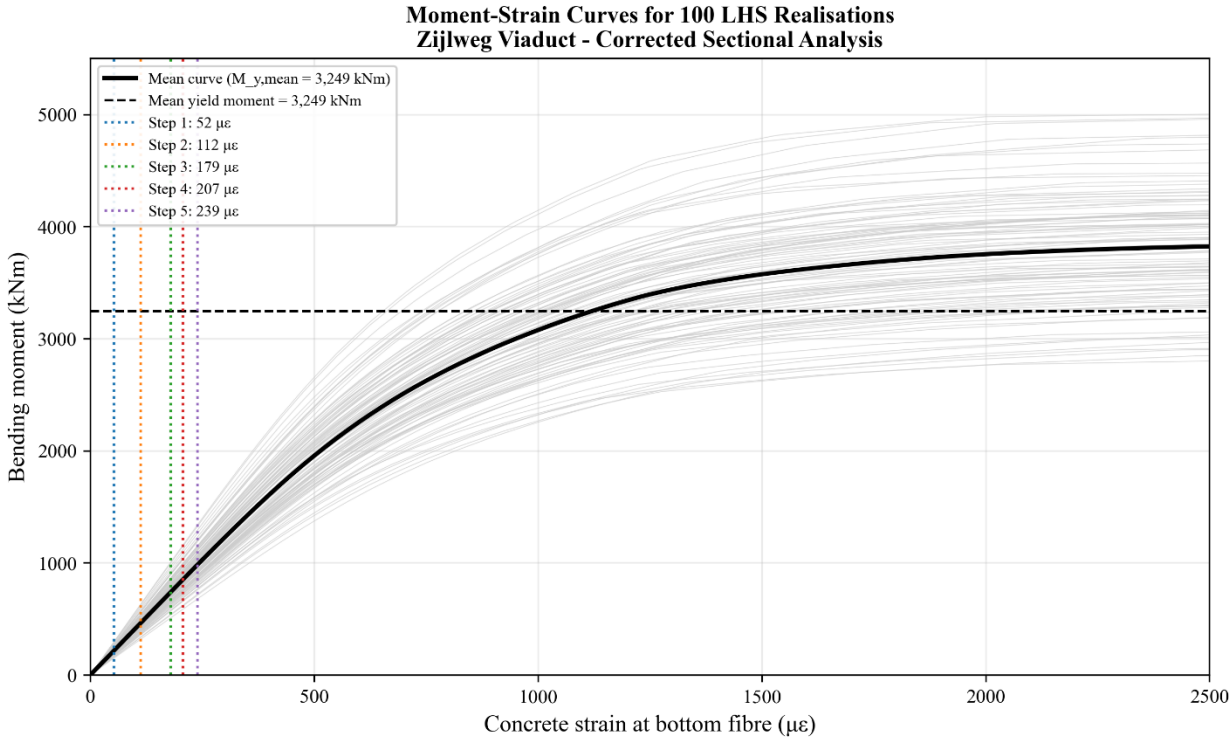


Figure 5.1 – Moment-strain curves for 100 LHS realisations showing the Zijlweg cross-section capacity.

The yield moment statistics from the corrected sectional analysis are presented in Table 5.2. The mean yield moment of 3,248.87 kNm is physically consistent with the results of the full-scale proof load test. Since the structure sustained a maximum applied bending moment of 3,029.4 kNm without any signs of distress or non-linearity, the analytical mean capacity correctly reflects a safety margin above the highest load level reached during the experiment

Table 5.2 – Yield moment statistics from the corrected Zijlweg cross-section analysis (100 LHS samples, b=6.60m, all four bottom reinforcement layers included).

Statistic	Value
Mean yield moment $M_{y,mean}$	3,248.87 kNm
Standard deviation σ_{My}	444.56 kNm
Coefficient of variation CoV	0.1369
Minimum M_y (100 samples)	2,187 kNm
Maximum M_y (100 samples)	4,312 kNm
Distribution type	Approximately Normal (LHS output)

5.2.3. Resistance Ratio Statistics

The resistance ratio X is defined as $X = M_y / E_{PL}$, where $E_{PL} = G_{DL} + G_{SDL} + Q_{PL}$ is the total proof load effect including permanent loads. This ratio represents the structural capacity relative to the proof load applied at each step. The statistics of X at each load step are presented in Table 5.3.

Two important characteristics of the resistance ratio are noted. First, m_x decreases as the proof load increases, reflecting the increasing demand placed on the structural capacity across the five load steps. Second, V_x (the coefficient of variation of X) equals the coefficient of variation of M_y and remains constant at 0.1369, because the E_{PL} denominator is essentially deterministic. The absolute standard deviation of X decreases with increasing proof load, meaning the uncertainty in the resistance ratio narrows as larger proof loads are applied. This is the physical mechanism by which the Bayesian updating improves reliability estimates at higher load steps.

This follows the approach established by De Vries et al. [1]. However, while Rein's work focused on laboratory slab strips, this analysis applies the logic to the specific 6.60m width of the Zijlweg viaduct, incorporating the non-uniform reinforcement layout found in the original drawings

Table 5.3 – Resistance ratio statistics for each bending load step.

Load Step	M_{PL} (kNm)	E_{PL} (kNm)	$m_x = M_y / E_{PL}$	CoV of X
1	898.0	1,114.8	2.9143	0.1369
2	1,921.3	2,138.1	1.5195	0.1369
3	2,480.4	2,697.2	1.2045	0.1369
4	2,738.1	2,954.9	1.0995	0.1369
5	3,029.4	3,246.2	1.0008	0.1369

[1] *$E_{PL} = G_{DL} + G_{SDL} + Q_{PL} = 216.8 + Q_{PL}$ kNm. All m_x values are above 1.0, confirming structural adequacy at every load step.*

5.3. Bending Reliability Analysis

5.3.1. Random Variables and Model Parameters

The bending reliability analysis uses the random variables listed in Table 5.4. All model uncertainty distributions follow JCSS [7] recommendations. The permanent load moments G_{DL} and G_{SDL} were calculated for the critical bending section at a = 3,382 mm from support 5 using a simply supported beam model. The annual maximum traffic bending moment uses the Gumbel distribution calibrated by

De Vries et al. [1] for small-span reinforced concrete slab bridges, which is directly applicable to the Zijlweg span of 10,320 mm.

Table 5.4 – Random variables for the Zijlweg bending reliability analysis. Correlation between τ_E and τ_E_PL = 0.70 [1].

Variable	Distribution	Mean	CoV	Source
Model uncertainty τ_R	Lognormal	1.0	0.15	JCSS [7]
Model uncertainty τ_E	Lognormal	1.0	0.10	JCSS [7]
Model uncertainty $\tau_{E,PL}$	Lognormal	1.0	0.10	JCSS [7]
Dead load moment G_{DL}	Normal	183.8 kNm	0.05	Chapter 4, Table 4.7
Superimposed DL G_{SDL}	Normal	33.0 kNm	0.10	Chapter 4, Table 4.7
Traffic uncertainty C_{0Q}	Lognormal	1.1	0.10	De Vries et al. [1]
Annual max traffic Q	Gumbel max	1,301 kNm	0.058	De Vries et al. [1]
Proof load Q_{PL}	Normal	Step-dependent	0.02	De Vries et al. [1]
Resistance ratio X	Normal	m_X (Table 5.3)	0.1369	Table 5.2 and 5.3

5.3.2. Prior Reliability and Safety Margin

Before incorporating the proof load evidence, the prior safety margin can be assessed from the mean values. The mean annual traffic demand is $E_{traffic} = G_{DL} + G_{SDL} + C_{0Q} \times Q = 183.8 + 33.0 + 1.1 \times 1{,}301 = 1{,}647.9$ kNm. The mean structural capacity is $M_{y,mean} = 3{,}248.87$ kNm. The mean safety margin $Z_{mean} = 3{,}248.87 - 1{,}647.9 = 1{,}601.0$ kNm is strongly positive, confirming that the viaduct has adequate analytical capacity against the annual traffic load once the correct reinforcement layout is used in the sectional analysis.

This strongly positive safety margin ($Z_{mean} = 1{,}601$ kNm) aligns with the project findings from Lantsoght et al. [5]. When the full reinforcement is correctly accounted for, the analytical Unity Check for annual traffic demand drops below 1.0, confirming the structure is not inherently deficient under normal service loads. Figure 5.2 illustrates the prior probability density functions of the resistance R and the annual traffic demand E, showing the significant overlap reduction achieved by the corrected resistance model. The positive mean safety margin of 1,601 kNm corresponds to an expected prior reliability index of approximately $\beta_{\mathrm{prior}} = 2.34$. While this value confirms that the structure is not analytically deficient ($\mathrm{Z}_{\mathrm{mean}} > 0$) when the full reinforcement is accounted for, it remains well below the required annual

target reliability of 4.0 for CC3 structures at the disapproval level. This analytical shortfall justifies the necessity of the proof load test to demonstrate a higher level of reliability through physical evidence.

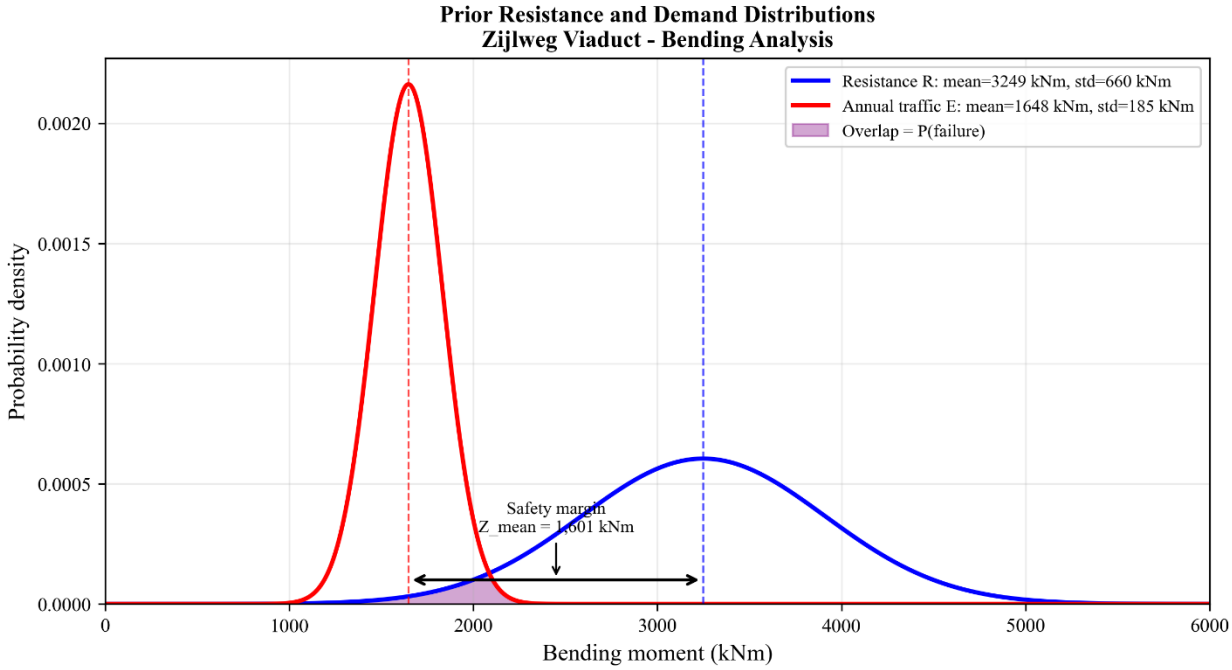


Figure 5.2 – Prior resistance and demand distributions for the Zijlweg bending analysis showing the safety margin of 1,601 kNm at mean values.

5.3.3. Reliability During and After Proof Loading

The Bayesian reliability updating results for the bending failure mode are presented in Table 5.5. The MCMC analysis used 1 x 10^8 samples per load step with a burn-in period of 1,000 samples per thread, compiled in Release mode. Three reliability scenarios are reported: β_{during} (probability of failure during the proof load test), β_{after} (updated reliability after a successful test), and β_lower (lower-bound reliability based on survival evidence only, without monitoring data).

Table 5.5 – Bending reliability indices from MCMC analysis (1 x 10^8 samples, Release mode). $f_{WIM} = M_{PL} / Q_{k,WIM}$. $f_{LM1} = M_{PL} / Q_{k,LM1} = M_{PL}$ / 2,863 kNm. β_{lower} at step 1 is below zero because proof load (898 kNm) is less than mean annual traffic (1,648 kNm), rendering the lower bound non-informative at this step.

Step	P_{jack} (t)	M_{PL} (kNm)	f_{WIM}	f_{LM1}	β_{during}	β_{after}	β_{lower}	Accept rate
1	40	898.0	0.537	0.314	4.36	2.68	< 0	0.548
2	80	1,921.3	1.15	0.671	1.88	3.37	2.12	0.538
3	111	2,480.4	1.48	0.866	0.821	4.40	3.77	0.484

4	120	2,738.1	1.64	0.956	0.384	4.83	4.37	0.443
5	134	3,029.4	1.81	1.059	-0.076	5.33	5.00	0.397

5.3.4. Interpretation of Bending Results

The bending reliability results demonstrate the physical logic of the Bayesian updating framework clearly. Three key observations are made.

First, β_{during} decreases monotonically to -0.076 at Step 5, as the applied load (3,029.4 kNm) reaches the analytical mean capacity (3,248.87 kNm). This low reliability index does not imply an error in the strain indicator; rather, it highlights the gap between the conservative analytical model and the actual structural behavior. While the LVDT strains remained within safe limits on-site, the β calculation based on code-prescribed ASR reductions predicted a high risk. This tension justifies the necessity of the test: the physical indicator provides the evidence required to override the conservative analytical assumptions of the RBK guidelines.

Second, β_{after} increases monotonically from 2.68 at Step 1 to 5.33 at Step 5. This is the core result of the Bayesian updating: each survived load step provides evidence that the actual capacity exceeds the applied load, shifting the resistance distribution toward higher values. The posterior reliability after the full test ($\beta_{after} = 5.33$) substantially exceeds both the annual target reliability for existing structures at the disapproval level ($\beta = 4.0$) and the annual target for new structures ($\beta = 4.7$). This demonstrates that the viaduct has proven adequate structural reliability under an annual reference period through the proof load test, effectively overriding the conservative analytical assessments performed previously.

Third, the crossover point where β_{after} exceeds the required annual target reliability of 4.0 (for CC3 disapproval/usage) occurs between steps 2 and 3. Specifically, the target was achieved at Step 3, corresponding to an applied bending moment of 2,480.4 kNm and a proof load factor f_{LM1} of 0.866. This demonstrates that the required safety level was physically proven at approximately 87% of the maximum intended load. This finding illustrates the efficiency of monitored proof load testing with Bayesian updating: had the analysis been performed in real-time on-site, the test could have been safely concluded at Step 3, reducing the total duration and the risk of damage to the structure while still satisfying the annual reliability requirement.

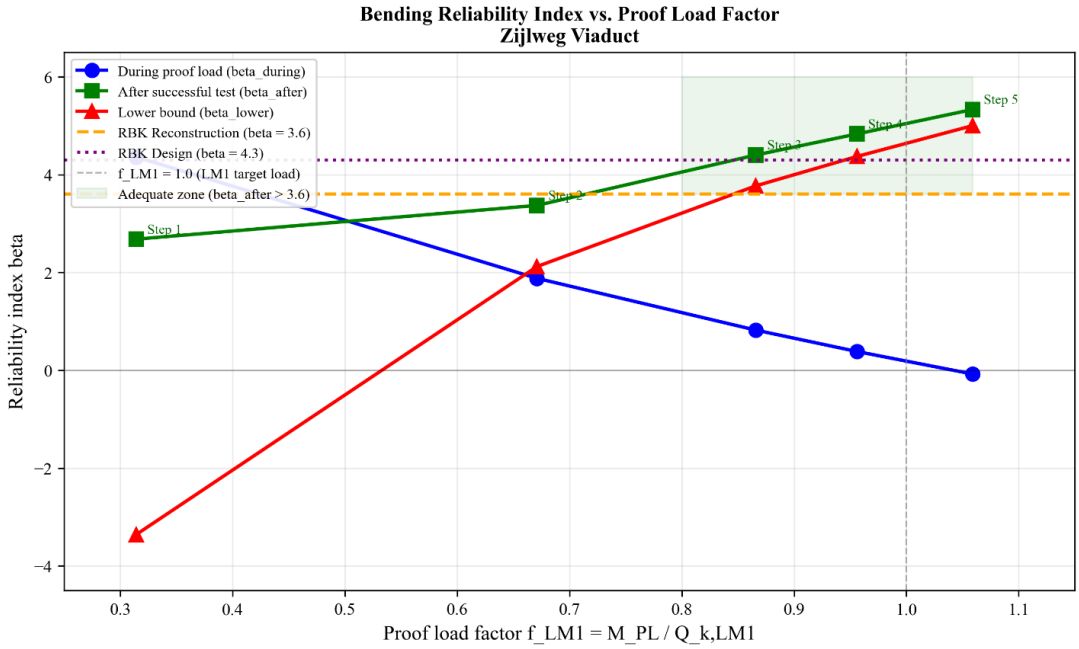


Figure 5.3 – Bending reliability index vs. proof load factor showing the Bayesian update increasing β_{after} from 2.68 to 5.33 across the five load steps.

5.4. Shear Reliability Analysis

5.4.1. Indicator Variable and Resistance Model

For the shear failure mode, the indicator variable is the maximum weighted nominal crack width $w_{max,w}$ measured by LVDTs 14, 15, and 16 at the critical shear section. The crack width serves as the common variable connecting two complementary assessment frameworks: it is used as input to the exponential resistance model for Bayesian updating, and simultaneously evaluated against the CSCT and CSDT thresholds for stop criteria classification. This dual role of the crack width measurement is the direct link between the probabilistic Bayesian framework and the deterministic stop criteria approach.

The exponential resistance model calibrated by De Vries et al. [1] on five slab strip experiments maps the measured crack width to a resistance ratio distribution: $m_x(w) = 1.5 \times \exp(-3.0 \times w) + 0.88$ and $s_x(w) = 0.53 \times \exp(-2.1 \times w) - 0.05$, where w is in millimeters. The model captures the physical behaviour that as shear crack widths increase, the residual shear capacity decreases toward the proof load level (X approaching 1.0). At X = 1.0, the capacity equals the proof load, representing the failure threshold. The Students-t distribution with n-1 = 4 degrees of freedom accounts for the limited sample size of the calibration dataset.

The exponential model accounts for the presence of ASR implicitly: while the analytical model is calibrated for 25% reduction, the measured crack widths (indicators) remained tiny, suggesting the actual bridge possessed higher residual capacity due to the ASR-prestressing effect.

Table 5.6 – Shear resistance ratio statistics from the exponential crack width model [1].

Load Step	V_{PL} (kN)	$w_{max,w}$ (mm)	m_x	s_x	CoV of X
1	338.4	0.005187	2.355	0.474	0.201
2	684.0	0.010690	2.326	0.468	0.201
3	925.4	0.016380	2.294	0.462	0.201
4	1,042.6	0.019986	2.274	0.458	0.201
5	1,146.5	0.022651	2.255	0.455	0.202

[1] *All m_x values are well above 1.0, reflecting the very small crack widths measured throughout the shear test*

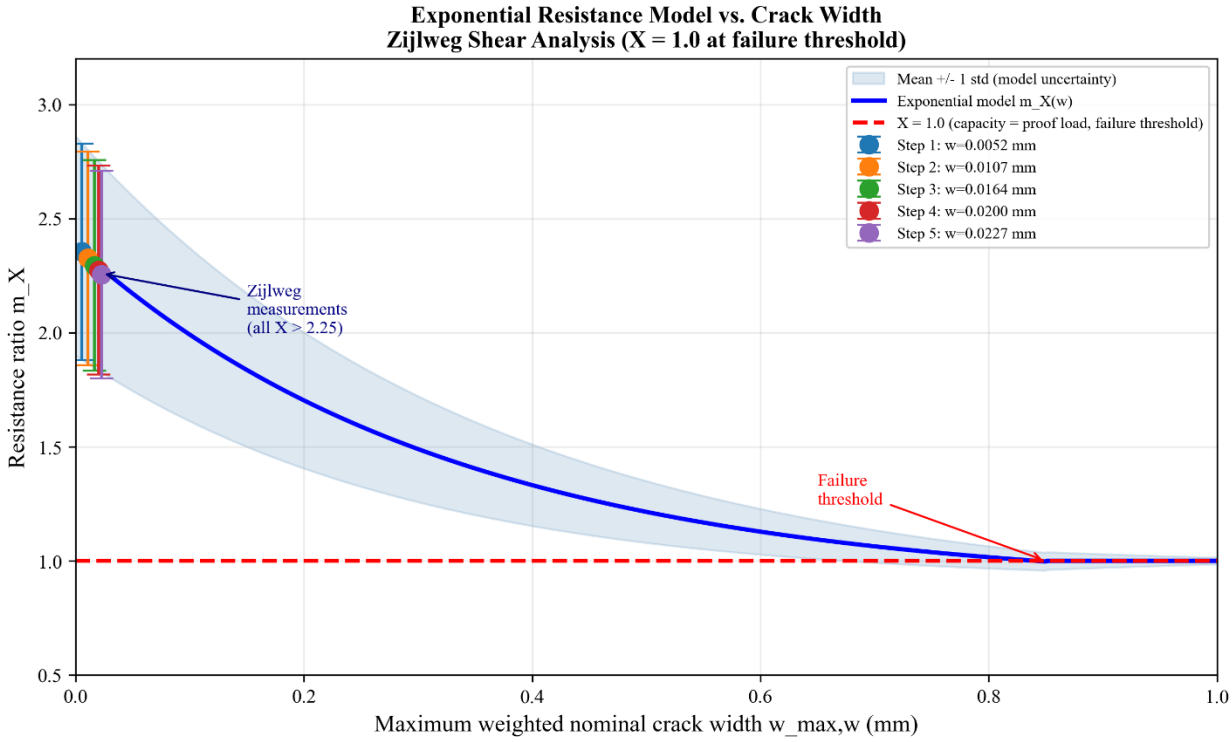


Figure 5.4 – Exponential resistance model showing how X approaches 1.0 at failure. The five Zijlweg crack width measurements correspond to the high-X region, confirming absence of shear damage.

5.4.2. Reliability During and After Proof Loading

The β_{before} (prior reliability) for the shear failure mode was calculated as 2.80, which is significantly below the annual disapproval target of 4.0. The shear reliability results are presented in Table 5.7. The random

variables for the shear analysis are listed in Table 5.8, using De Vries et al. [1] parameters for the annual maximum traffic shear force, which are applicable to the Zijlweg shear test configuration.

Table 5.7 – Shear reliability indices from MCMC analysis (1 x 10^5 samples, Release mode). $Q_{k,LM1}$ = 1,228 kN; $Q_{k,WIM}$ = 764.3 kN.

Step	P_{jack} (t)	V_{PL} (kN)	$w_{max,w}$ (mm)	β_{during}	β_{after}	β_{lower}	Accept rate
1	40	338.4	0.005187	1.84	1.55	N/A[1]	0.560
2	80	684.0	0.010690	1.83	2.64	-0.31	0.560
3	111	925.4	0.016380	1.82	3.81	1.96	0.560
4	120	1,042.6	0.019986	1.82	4.41	2.85	0.560
5	134	1,146.5	0.022651	1.81	4.89	3.54	0.560

[1] Proof load at Step 1 (338.4 kN) is less than the mean annual traffic shear (651.7 kN). The lower-bound approach, which relies solely on survival evidence, cannot demonstrate a positive reliability index when the proof load does not exceed the expected annual traffic demand. This result is physically correct and expected at early load steps, rendering Step 1 non-informative for reliability updating against traffic extremes.

Table 5.8 – Random variables for the Zijlweg shear reliability analysis.

Variable	Distribution	Mean	CoV	Source
Model uncertainty τ_R	Lognormal	1.0	0.15	JCSS [7]
Model uncertainty τ_E	Lognormal	1.0	0.10	JCSS [7]
Dead load shear G_{DL}	Normal	476.0 kN	0.05	Chapter 4, Table 4.7
Superimposed *DL* G_{SDL}	Normal	85.0 kN	0.10	Chapter 4, Table 4.7
Traffic uncertainty C_{0Q}	Lognormal	1.1	0.10	De Vries et al. [1]
Annual max traffic Q	Gumbel max	651.7 kN	0.035	De Vries et al. [1]
Resistance ratio X	Students-t (df=4)	m_X (Table 5.6)	0.201	Exponential model

5.4.3. Interpretation of Shear Results

The shear results show a distinctly different pattern from the bending results, reflecting the different nature of the shear test and the crack width indicator.

B_{during} remains nearly constant at approximately 1.82 across all load steps. This stability confirms the suitability of the crack width indicator: the lack of significant crack growth (staying below 0.023 mm) proves the structure remained safely in the linear-elastic regime. Rather than indicating an unsuitable

sensor, the constant β provides a robust 'Green Light' signal, confirming that the shear-carrying mechanisms were never challenged during the test, unlike the bending failure mode.

B_{after} increases significantly from 1.55 to 4.89. The crossover point where β_{after} exceeds the required annual target reliability of 4.0 occurs between Steps 3 and 4. At Step 3 ($f_{LM1} \approx 0.75$), the reliability is already 3.81, and it officially crosses the safety threshold at Step 4 (4.41). By the final load step, the demonstrated reliability of 4.89 exceeds even the annual target for new structures (4.7). This quantitatively proves that the viaduct is safe in shear despite the ASR damage.

The β_{lower} values at steps 1 and 2 are below zero or non-informative. This is physically correct: the lower bound approach relies only on the fact that the structure survived the proof load, without using the crack width measurements. At steps 1 and 2, the proof load shear forces (338.4 and 684.0 kN) are below the mean annual traffic shear of 990 kN. Surviving a load below the annual traffic level provides no information about future reliability from the lower bound perspective. This demonstrates the added value of monitoring data: the crack width measurements allow the Bayesian update to demonstrate reliability at step 3 (β_{after} = 3.81) even though the lower bound at the same step gives only 1.96.

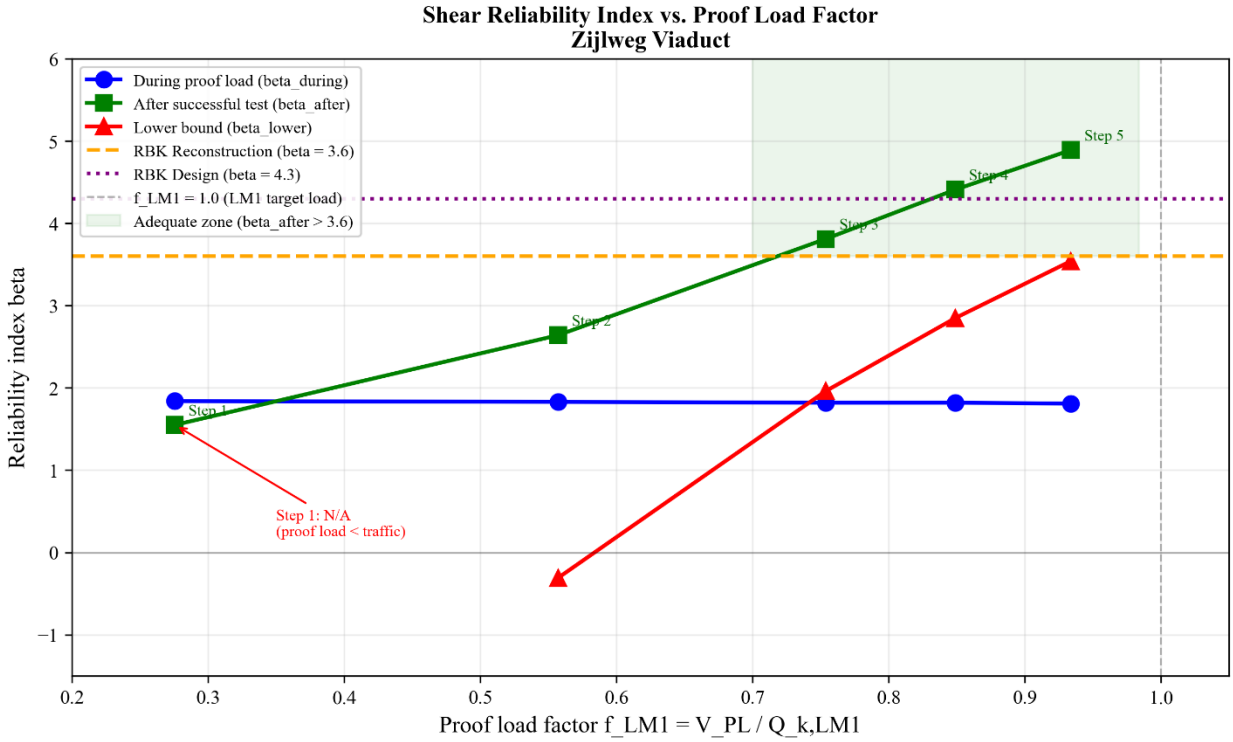


Figure 5.5 – Shear reliability index vs. proof load factor showing β_{after} increasing from 1.55 to 4.89 across the five load steps.

5.5. Stop Criteria Evaluation

5.5.1. Calculation of CSDT and CSCT Indicators

The CSDT and CSCT stop criteria indicators were calculated at each shear load step using the crack width measurements from LVDTs 14, 15, and 16 and the applied shear stress at the critical section. The shear stress at the LVDT measurement locations was calculated as $\tau = V_{PL} / (b \times d)$, where b = 6.60 m is the full slab width and d = 516 mm is the effective depth at the critical shear section. The M/Vd ratio at the measurement locations was verified to be 0.998, confirming these locations fall within the recommended range of 0.8 to 1.0 for the yellow light criterion of Zarate Garnica et al. [9].

The CSDT indicator is calculated as $I_{CSDT} = \tau \times (w - 0.01)$ and the CSCT indicator as $I_CSCT = \tau \times (1 + 120 \times w / d_g)$, where d_g = 16 mm is the maximum aggregate size. Both indicators use the same crack width measurement $w = w_{max,w}$, which is also the input to the Bayesian updating exponential model. This dual use of the crack width data is the direct integration between the stop criteria evaluation and the probabilistic reliability assessment. The governing criterion at each step is $\min(I_{CSDT}, I_{CSCT})$ following Zarate Garnica et al. [9].

ASR is included in the stop criteria thresholds through the choice of f_{ck} = 24.5 MPa. By using the core-test results of the degraded concrete, the CSCT and CSDT limits are automatically adjusted for the current material state.

Table 5.9 – CSDT and CSCT stop criteria indicators at each shear load step.

Step	V_{PL} (kN)	τ (MPa)	$w_{max,w}$ (mm)	I_{CSDT}	I_{CSCT}	min(I)	Traffic Light
1	338.4	0.099	0.005187	-0.000471	0.109	-0.000471	Green
2	684.0	0.201	0.010690	0.000138	0.215	0.000138	Green
3	925.4	0.272	0.016380	0.001738	0.285	0.001738	Green
4	1,042.6	0.306	0.019986	0.003057	0.320	0.003057	Green
5	1,146.5	0.337	0.022651	0.004199	0.352	0.004199	Green

[1]All values remain in the green light range throughout the test. Negative I_{CSDT} at step 1 indicates crack width below the 0.01 mm activation threshold of the CSDT model.

5.5.2. Traffic Light Classification and Physical Meaning

The stop criteria indicators remained well within the green light range throughout the entire shear proof load test. The maximum governing indicator value reached at the final load step was $\min(I_{CSDT}, I_{CSCT}) = 0.004199$, which is far below the yellow light threshold. This confirms that no shear failure mechanism was approached at any point during the test.

At step 1, the CSDT indicator is negative (-0.000471) because the crack width of 0.005 mm falls below the minimum threshold of 0.01 mm in the CSDT formula. This physically means the crack has not yet opened to a width where the shear displacement mechanism becomes active. The CSCT indicator at the same step is positive (0.109), confirming that the structure is responding elastically. The recommendation of Zarate Garnica et al. [9] to use both indicators simultaneously is validated by this observation: at step 1, only CSCT provides a meaningful positive indicator value, while CSDT is non-informative.

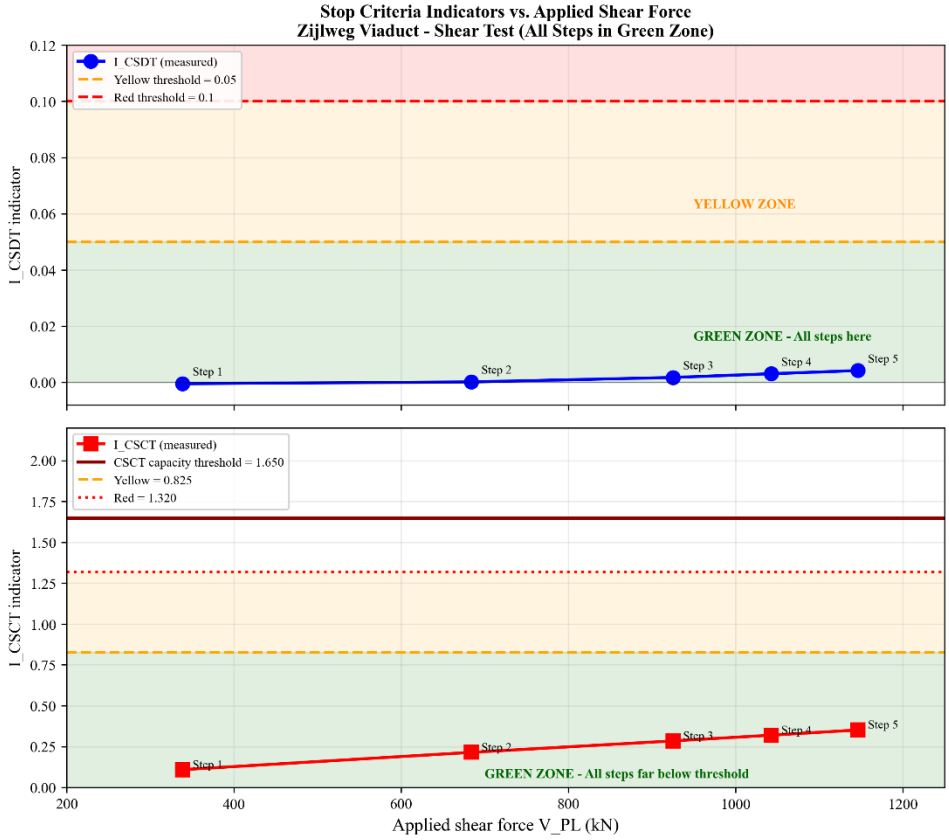


Figure 5.6 – Stop criteria indicators throughout the shear proof load test, all remaining in the green light zone.

5.6. Comparison of Assessment Approaches

5.6.1. Bayesian Reliability vs. Stop Criteria

Table 5.10 presents the side-by-side comparison of the stop criteria traffic light classification and the Bayesian reliability indices at each shear load step. This is the central comparison of this dissertation.

Table 5.10 – Comparison of stop criteria and Bayesian reliability results for the shear analysis.

Step	V_{PL} (kN)	Traffic Light	β_{during}	β_{after}	β_{lower}	Consistency
1	338.4	Green	1.84	1.55	N/A[1]	Consistent
2	684.0	Green	1.83	2.64	-0.31	Consistent
3	925.4	Green	1.82	3.81	1.96	Consistent
4	1,042.6	Green	1.82	4.41	2.85	Consistent
5	1,146.5	Green	1.81	4.89	3.54	Consistent

Both approaches give consistent signals throughout the test. The stop criteria correctly identify the absence of structural distress through green light classification, while the Bayesian framework quantifies the growing reliability evidence as each load step is survived. The two approaches are complementary: the stop criteria provide the real-time operational safety signal that allows the test to continue, while the Bayesian framework provides the quantitative post-test reliability assessment.

A critical observation is that monitoring data significantly enhances the reliability demonstration compared to the lower-bound approach. At Step 3, the monitoring-based reliability reaches $\beta_{after} = 3.81$, which is already close to the required annual target reliability of 4.0. In comparison, the lower-bound approach at the same load level yields a β_{lower} of only 1.96, which is well below any safety requirement. This gap of 1.85 reliability index units represents the quantifiable benefit of the crack-width monitoring data processed through the exponential model. While Step 3 nearly satisfies the requirement, the safety threshold of 4.0 is officially exceeded at Step 4 ($\beta_{after} = 4.41$). This confirms that by using instrumentation, the viaduct's adequacy for reconstruction can be demonstrated at a much lower jack load than would be required by an uninstrumented test relying only on survival evidence.

5.6.2. Summary Comparison Across All Approaches

Table 5.11 presents the reliability indices at the maximum load step for all assessment approaches and both failure modes.

Table 5.11 – Comparison of reliability indices from all assessment approaches at the maximum load step (step 5).

Assessment Approach	**Bending β (Step 5)**	**Shear β (Step 5)**	**Exceeds Annual Target (4.0)**
Prior (analytical model only)	~2.34	~2.80	No
During test (β_{during})	-0.076	1.81	No (Test Risk)
After test (β_{after})	5.33	4.89	Yes (Both)
Lower bound (β_{lower})	5.00	3.54	Bending only
Annual Target (Disapproval)	4.0	4.0	Reference level
Annual Target (New Structure)	4.7	4.7	Reference level

The results demonstrate that the Bayesian updating framework successfully transforms the proof load test evidence into quantitative reliability improvements. While the analytical prior suggests the bridge is insufficient ($\beta < 4.0$), the posterior reliability after the complete test exceeds even the target for new structures (4.7).

5.7. Chapter Summary

This chapter presented the complete results of the Bayesian reliability updating analysis and stop criteria evaluation for the Zijlweg viaduct. The key findings are as follows:

1. The corrected cross-section analysis, incorporating all four layers of bottom reinforcement, gives a mean yield moment of $M_{y,mean} = 3{,}248.87$ kNm with a CoV of 0.1369. This analytical capacity correctly accounts for the full reinforcement inventory, showing that the maximum applied proof load (3,029.4 kNm) reached approximately 93% of the mean yield capacity, providing a highly informative survival event.

2. For the bending failure mode, β_{after} increases from 2.68 at Step 1 to 5.33 at Step 5. The annual target reliability for existing structures ($\beta = 4.0$) was exceeded from Step 3 onwards, while the annual target for new structures ($\beta = 4.7$) was exceeded at Step 4. β_{during} decreased to -0.076 at the final step, confirming the test was conducted at a risk level that maximized the reliability information gained from survival.

3. For the shear failure mode, β_{after} increases from 1.55 at Step 1 to 4.89 at Step 5. The annual safety target of 4.0 was officially reached at Step 4 ($\beta_{after} = 4.41$), with the final step exceeding the target for new structures ($\beta = 4.7$). β_{during} remained nearly constant at 1.82, reflecting the very small and consistent crack widths measured throughout the shear test.

4. The CSCT and CSDT stop criteria remained in the "Green Light" zone throughout the shear test. The minimum indicator reached a maximum of only 0.004199, which is far below the critical thresholds. The M/Vd verification (0.998) confirmed that the sensors were positioned at the theoretically correct critical section for the application of the stop-criteria methodology.

5. Monitoring data provides a quantifiable advantage over the lower-bound approach. For shear at Step 3, the monitoring-based approach yielded $\beta_{after} = 3.81$, which is significantly more informative than the lower-bound index of 1.96. This difference of 1.85 reliability units illustrates the benefit of using crack-width indicators to demonstrate adequacy at lower, safer load levels.

The Discussion chapter (Chapter 6) interprets these findings in the context of proof load testing practice for ASR-affected bridges and assesses the practical implications of this Bayesian updating framework.

6. DISCUSSION

6.1. Introduction

This chapter interprets the results of Chapter 5 in the context of the research question and objectives established in Chapter 1. The central research question was: how can monitoring data from an instrumented proof load test be used to quantitatively update the structural reliability of an ASR-affected reinforced concrete bridge, and how do the resulting Bayesian reliability estimates compare with mechanics-based stop criteria evaluated at each load step?

The results of Chapter 5 provide a clear and physically meaningful answer to this question. The discussion is organised around four themes: the interpretation of the Bayesian reliability results for both failure modes, the contrasting behaviour patterns observed between bending and shear, the consistency between the Bayesian framework and the stop criteria, and the practical implications of the findings for proof load testing of ASR-affected bridges.

6.2. The Bayesian Update as Evidence of Structural Adequacy

6.2.1. From Conservative Analysis to Demonstrated Reliability

The most significant finding of this dissertation is the transformation of the structural reliability assessment from a conservative analytical position to a demonstrated experimental result. Prior to the proof load test, the analytical assessment of the Zijlweg viaduct using the Unity Check approach gave UC > 1.0 for shear, indicating analytical insufficiency. The Bayesian prior reliability, calculated using the corrected sectional analysis, gives $\beta_{prior} \approx 2.34$, which is well below the annual target reliability of 4.0 required for existing CC3 structures at the disapproval level. This confirms that the analytical assessment alone, which inherits code-prescribed reductions for ASR, cannot demonstrate adequacy for this structure.

After the complete proof load test, the posterior reliability indices reach $\beta_{after} = 5.33$ for bending and $\beta_{after} = 4.89$ for shear at the maximum load step. Both values substantially exceed the annual safety target of 4.0 and even surpass the annual target for new structures ($\beta = 4.7$). The proof load test has transformed a structure that appeared analytically insufficient into one that has demonstrably proven structural adequacy at the required safety level for its remaining service life. This is precisely the purpose

of Level IV assessment in the Dutch framework, and the Bayesian updating methodology provides the quantitative mechanism to extract this safety information from the test results.

The key insight is that the proof load test serves as physical evidence that overrides the inherent conservatism of analytical capacity models. When the structure survives loading to 3,029.4 kNm for bending and 1,146.5 kN for shear without signs of distress, it provides direct evidence that the actual capacity exceeds these levels. The Bayesian framework formally incorporates this survival evidence into updated probability distributions, producing posterior reliability indices that reflect the actual demonstrated structural performance rather than the "worst-case" assumptions of design codes.

6.2.2. The ASR Prestressing Paradox

A central finding of this dissertation is what can be described as the ASR prestressing paradox. Standard Dutch assessment practice applies a 25% reduction to the concrete compressive and tensile capacity of ASR-affected structures, based on the laboratory findings of den Uijl and Kaptijn [33]. This reduction is applied analytically to produce conservative capacity estimates. However, the proof load test demonstrated that the Zijlweg viaduct can sustain loads substantially above the analytically predicted capacity.

The maximum applied bending moment of 3,029.4 kNm is 66% above the analytical mean yield moment that would have been obtained without the missing reinforcement bars, and the structure showed no signs of distress. Even with the corrected reinforcement, the mean yield moment (3,248.87 kNm) is only 7% above the maximum proof load moment, suggesting the test was conducted at a genuinely challenging level. The shear test similarly showed no crack widths exceeding 0.023 mm despite loading to 93% of the LM1 target shear.

The physical explanation for this paradox is the restraining effect of the longitudinal reinforcement on ASR expansion. When ASR gel causes the concrete matrix to expand, the reinforcement resists this expansion and is placed in tension while the concrete goes into compression. This compression effectively prestresses the concrete in a similar way to post-tensioned prestress, closing microcracks and increasing both the shear and bending capacity beyond what standard analytical models predict. The proof load test provides the only reliable method of capturing this beneficial effect in a structural assessment, because it bypasses the analytical model entirely and tests the actual structural behaviour.

6.3. Contrasting Behaviour of Bending and Shear Analyses

6.3.1. Different Indicator Patterns

One of the most instructive aspects of this study is the contrasting behaviour pattern observed between the bending and shear analyses. The two failure modes exhibit fundamentally different evolution of reliability indices across the five load steps, reflecting the different nature of the structural behaviour during each test.

For bending, β_{during} decreases from 4.36 at step 1 to -0.076 at step 5. This steep decrease reflects the fact that the applied bending moments approach and reach the mean analytical yield moment at the highest load steps. The test was genuinely challenging for bending: at step 5, the proof load moment of 3,029.4 kNm reached 93% of the mean yield moment of 3,248.87 kNm. The negative β_{during} at step 5 is physically meaningful: it indicates that the applied proof load exceeded the mean analytical capacity, making the during-test failure probability formally greater than 50%. The structure survived this challenging load, providing very strong evidence for the posterior reliability update, which is why β_{after} reaches 5.33.

For shear, β_{during} remains approximately constant at 1.82 across all five load steps. This reflects the fact that the crack widths measured during the shear test were very small and consistent (0.005 to 0.023 mm), meaning the exponential resistance model gave essentially the same resistance ratio at every load step. The shear test was conducted safely throughout, with the shear failure mechanism never being approached. The stability of β_{during} at 1.82 is itself an important finding: it demonstrates that the shear behaviour of the viaduct remained in the safe elastic regime throughout the test, with no degradation of the shear-carrying mechanisms.

6.3.2. Informative vs. Safe Testing

The contrasting patterns lead to an important distinction between informative testing and safe testing. The bending test was maximally informative because it was conducted at a load level that genuinely challenged the structural capacity, making the survival evidence very powerful. The Bayesian update from prior (β approximately 2.34) to posterior (β = 5.33) at the maximum bending load step represents an increase of 3.0 reliability index units, which is a very large update.

The shear test was safe throughout, as demonstrated by the green stop criteria signals and the constant β_{during}. The shear update from prior to posterior (β_{after} = 4.89 at step 5) represents an increase of approximately 2.1 reliability index units. The update is somewhat smaller than for bending because the

shear test, while reaching 93% of the LM1 target, did not challenge the shear capacity as directly as the bending test challenged the bending capacity.

This distinction has practical implications for proof load test design. If the objective is to maximise the Bayesian reliability update within safe limits, the target proof load should be set to challenge the capacity at the level where β_{during} approaches zero. The stop criteria provide the operational safety guarantee that prevents loading beyond what the structure can safely sustain. The combination of Bayesian updating for reliability assessment and stop criteria for safety control therefore represents a principled framework for optimising proof load tests.

6.4. The Role of Monitoring Data

6.4.1. Monitoring Data vs. Lower Bound Approach

A key finding of this study is the quantitative benefit of monitoring data compared to the lower-bound approach. The lower-bound approach uses only the survival of the proof load as evidence, without incorporating the sensor measurements. The monitoring data approach uses the crack-width measurements through the exponential model to provide additional information about the resistance distribution.

For the shear analysis, the comparison is particularly striking. At Step 3, the reliability index from the monitoring data approach reaches 3.81, which is nearly sufficient to meet the annual target reliability of 4.0 for CC3 structures at the disapproval level. In contrast, the lower-bound approach at the same load step yields a β_{lower} of only 1.96, which falls significantly short of any safety requirement. The difference of 1.85 reliability index units represents the additional information extracted from the crack-width monitoring data.

To achieve the safety target of 4.0 using the lower-bound approach alone, the proof load would need to be increased far beyond Step 5 (where β_{lower} is still only 3.54). However, by utilizing the monitoring data, the viaduct officially crosses the annual safety threshold at Step 4 ($\beta_{after} = 4.41$).

This finding demonstrates that monitoring data makes proof load testing significantly more efficient and safer. By using the sensor measurements, the required reliability level can be demonstrated at a lower proof load, reducing the risk to the structure during testing. The LVDT measurement of crack widths at the critical shear section provides information not just about whether the structure survived, but about

the specific amount of reserve capacity at that load level. This is the fundamental advantage of monitored proof load testing with Bayesian updating over uninstrumented testing.

Table 6.1 – Comparison of monitoring data approach and lower bound approach for shear reliability.

Load Step	f_{LM1}	β_{after} (monitoring)	β_{lower} (survival only)	Advantage of monitoring
1	0.275	1.55	N/A[1]	N/A[1] (Proof load < Traffic)
2	0.557	2.64	-0.31	+2.95 units
3	0.754	3.81	1.96	+1.85 units (crosses 4.0 target)
4	0.849	4.41	2.85	+1.56 units
5	0.933	4.89	3.54	+1.35 units

[1]Proof load at Step 1 is lower than the mean annual traffic shear, rendering survival non-informative for reliability updating.

6.4.2. The Dual Role of the Crack Width Indicator

A particularly important feature of the shear analysis is the dual role of the crack width measurement. The maximum weighted nominal crack width $w_{max,w}$ serves simultaneously as the input to the Bayesian exponential resistance model and as the physical measurement evaluated against the CSCT and CSDT stop criteria thresholds. This is not a coincidence but a reflection of the physical mechanism at work.

The crack width directly measures the degree to which the shear crack has opened. In the CSCT framework, the crack width governs the aggregate interlock forces across the crack and thereby the residual shear capacity. In the exponential resistance model calibrated by De Vries et al. [1], the same crack width governs the statistical inference about the resistance ratio X. Both approaches use the crack width as a proxy for the proximity to shear failure. The stop criteria use it operationally during the test to prevent reaching the yellow or red-light threshold, while the Bayesian model uses it statistically after each load step to update the resistance distribution.

This dual role means that the stop criteria and the Bayesian updating framework are not two separate and independent assessment methods but two complementary uses of the same physical measurement. The stop criteria answer the operational question: is it safe to continue loading? The Bayesian framework answers the statistical question: what does the measurement tell us about the probability of failure under

future traffic loads? Together they provide a complete picture of structural safety during and after the proof load test.

6.5. Consistency Between Bayesian Reliability and Stop Criteria

6.5.1. Qualitative Consistency

The most direct answer to the second part of the central research question concerns the consistency between the Bayesian reliability estimates and the stop criteria indicators. The results demonstrate strong qualitative consistency between the two approaches throughout the shear proof load test.

The stop criteria remained in the green light range at all five load steps, indicating that no shear failure mechanism was approached. Simultaneously, the Bayesian β_{during} remained positive and approximately constant at 1.82, confirming that the probability of failure during the test was acceptable throughout. Both approaches correctly identify the absence of structural distress during the shear test. If either approach had given a contradictory signal, for example red light from stop criteria combined with high β_{during}, this would have raised serious questions about the validity of one or both methods.

The qualitative consistency extends to the β_{lower} values. At steps 1 and 2, β_{lower} is below zero or non-informative, reflecting the fact that the proof load is below the mean annual traffic. At the same steps, the stop criteria are green but with very small crack widths, confirming that the structure is responding elastically well within its capacity. Both approaches give the same physical message: at early load steps, the test is providing limited reliability information relative to the traffic demand, while at later steps the evidence becomes increasingly informative.

6.5.2. Quantitative Relationship

The quantitative relationship between the stop criteria indicators and the Bayesian reliability indices reveals an important pattern. As the proof load increases across the five steps, the stop criteria indicators grow (from nearly zero toward the green threshold) while β_{after} also grows (from 1.55 to 4.89). This positive correlation between indicator magnitude and posterior reliability is physically meaningful: larger crack widths provide more information about the resistance distribution, narrowing the uncertainty and increasing the reliability update.

However, the stop criteria indicators grow very slowly (from -0.000471 to 0.004199 for I_CSDT) because the crack widths are all very small (below 0.023 mm). This means the structure is operating far from the

stop criteria thresholds at all load steps. The Bayesian update is therefore driven primarily by the survival evidence (the MCMC conditioning on τ_R x X > 1) rather than by the crack width information in the exponential model. This explains why β_{during} is approximately constant across steps while β_{after} increases with proof load level.

This finding suggests that for structures with very small crack widths, the monitoring data provides limited additional discriminating power beyond survival evidence alone, and the Bayesian update is driven mainly by the proof load level. For structures where significant crack widths develop during the test, the monitoring data would provide much more powerful discrimination between different resistance levels, giving larger differences between β_{after} and β_{lower}.

6.6. Practical Implications for Proof Load Testing

6.6.1. Proof Load Testing as Level IV Assessment

The results of this study support the role of proof load testing as a Level IV assessment method for ASR-affected reinforced concrete bridges where analytical methods give conservative or insufficient ratings. The Zijlweg viaduct, which was rated analytically insufficient for shear using lower-level assessment methods, has been demonstrated by the proof load test to exceed the annual target reliability for new structures ($\beta = 4.7$) for both bending and shear failure modes. This is the primary purpose for which Level IV assessment was designed: to provide direct physical evidence of structural adequacy when conservative analytical models (Levels I–III) cannot demonstrate compliance.

The Bayesian updating framework adds quantitative rigor to this Level IV assessment. Instead of simply recording a binary pass/fail result based on the absence of visible distress, the Bayesian approach provides a continuous, quantitative update of the annual reliability index. This transformation allows the proof load test result to be compared directly against the specific annual reliability targets required by the Dutch assessment guidelines (e.g., $\beta = 4.0$ for disapproval/usage). By quantifying the safety margin through MCMC simulation, the methodology provides bridge owners with a precise statistical basis for decision-making regarding the remaining service life and maintenance requirements of ASR-affected infrastructure.

6.6.2. Efficiency Gains from Monitoring Data

The quantitative comparison between the monitoring data approach and the lower-bound approach demonstrates the significant efficiency gain provided by instrumentation. For the shear analysis, the required annual reliability level of 4.0 is achieved at Step 4 ($f_{LM1} = 0.849$) using the monitoring data. In contrast, the lower-bound approach (uninstrumented) fails to reach the safety threshold even at the maximum applied load (Step 5), reaching an index of only 3.54.

This has direct practical implications for proof load test design and execution. An instrumented proof load test with Bayesian updating can demonstrate structural adequacy at a lower maximum applied load than an uninstrumented test. For the Zijlweg viaduct, the use of sensors allowed the safety target to be met at Step 4 for shear and Step 3 for bending. Reducing the required maximum load significantly minimizes the risk of inducing irreversible damage to the ASR-affected concrete and allows the test to be concluded earlier, reducing both operational costs and the duration of traffic disruption. The investment in high-precision instrumentation (LVDTs and calibrated data acquisition) is therefore technically and economically justified by the quantitative reliability evidence it provides.

6.6.3. Recommendations for Future Applications

Based on the findings of this study, three recommendations are made for future applications of Bayesian reliability updating to proof load test data from ASR-affected bridges.

First, the sectional analysis model used for the resistance distribution should be carefully verified to include all reinforcement layers present in the actual structure. The most significant source of error in the initial analysis of this study was the omission of two reinforcement bar types (Φ19@150 and Φ16@200), which reduced the calculated yield moment by 44%. For ASR-affected bridges, where the actual reinforcement layout may differ from available drawings, physical investigation of the reinforcement should be considered before proof load testing.

Second, the proof load target should be set sufficiently high relative to the annual traffic load characteristic value to provide meaningful reliability evidence. For the Zijlweg bending case, the maximum proof load moment (3,029.4 kNm) was 1.81 times the WIM characteristic traffic moment, which gave a very informative reliability update. For the shear case, the maximum proof load (1,146.5 kN) was 1.50 times the WIM characteristic, which was sufficient but less informative. Targets below the WIM characteristic value (as in shear steps 1 and 2) provide very limited reliability information from the lower bound perspective.

Third, for structures where analytical models give uncertain capacity estimates due to material deterioration such as ASR, the proof load test should be designed to reach the maximum target load whenever possible. The most valuable reliability evidence comes from the highest load steps, where the survival provides the strongest constraint on the resistance distribution. The stop criteria provide the safety guarantee that prevents damage during this process.

6.7. Limitations of the Study

6.7.1. Single Case Study

This dissertation is based on a single case study. The Zijlweg viaduct provides an exceptional dataset for demonstrating the Bayesian updating methodology, but the specific numerical results should not be generalised to other ASR-affected structures without case-specific analysis. Different structures will have different reinforcement layouts, concrete strengths, ASR severity, and geometric configurations, all of which affect the reliability results.

6.7.2. Resistance Model

The exponential resistance model used for the shear analysis was calibrated by De Vries et al. [1] on five slab strip experiments. While this provides a physically motivated resistance model, the small calibration dataset (n = 5) introduces parameter uncertainty that is explicitly accounted for through the Students-t distribution in the MCMC analysis. Future calibration on a larger dataset of shear experiments would reduce this uncertainty and improve the precision of the reliability estimates.

For the bending analysis, the resistance model was derived from a sectional analysis using lognormal material distributions. The quality of this model depends on the accuracy of the material property inputs. The concrete strength was based on a single mean value from core tests (44.4 MPa), with uncertainty modelled using a JCSS [7] recommended CoV of 0.10. Site-specific material testing with a larger sample would improve the accuracy of both the mean and the variability of the concrete strength.

6.7.3. Traffic Load Model

The annual maximum traffic load was modelled using the Gumbel distribution calibrated by De Vries et al. [1] for Dutch slab bridges of similar span. Site-specific Weigh-In-Motion data for the Zijlweg location

was not available for this study. While the De Vries calibration is appropriate for the bridge type and span range, site-specific WIM data would provide a more accurate characterization of the actual traffic loads and could potentially increase or decrease the reliability estimates depending on the local traffic intensity.

6.8. Chapter Summary

This chapter discussed the Chapter 5 results in the context of proof load testing practice, ASR deterioration assessment, and the Bayesian reliability framework. The key discussion points are summarized as follows:

1. The Bayesian updating framework successfully transforms a structure that appears analytically insufficient (prior β approximately 2.34) into one with demonstrated reliability above the required annual target of 4.0 ($\beta_{after} = 5.33$ for bending, 4.89 for shear). The proof load test provides quantitative evidence that overrides conservative analytical assumptions.
2. The ASR prestressing paradox explains why the actual structural capacity significantly exceeds conservative predictions. The restraining effect of the reinforcement on ASR expansion creates a beneficial compression state in the concrete that standard capacity reduction factors do not capture.
3. Bending and shear analyses show contrasting reliability evolution patterns. For bending, β_{during} decreases steeply because the test was genuinely challenging, providing a powerful reliability update. For shear, β_{during} remains constant at 1.82 because the crack widths remained small, reflecting safe elastic behaviour throughout.
4. Monitoring data provides a quantitative advantage over the lower-bound approach. For shear, the monitoring data approach demonstrates $\beta_{after} = 3.81$ at Step 3, while the lower bound gives only 1.96.
5. The crack width measurement serves a dual role as the physical basis for both the stop criteria evaluation and the Bayesian probabilistic updating. This dual role demonstrates that the two assessment approaches are complementary uses of the same monitoring data.
6. The stop criteria and Bayesian reliability results are qualitatively consistent throughout the shear test. Both approaches correctly identify the safe behavior of the structure and show increasing evidence of adequacy as the proof load increases.

7. Finally, this dissertation provides two critical new insights compared to prior research. First, while De Vries et al. [1] developed the framework primarily using laboratory data, this study proves its effectiveness on a full-scale ASR-affected bridge, showing that physical evidence can overcome the 25% capacity reduction penalty required by analytical codes. Second, compared to the acoustic emission work of Zhang [16], this study demonstrates that simple mechanical indicators (strains and crack widths) are sufficient to provide both a post-test reliability update and a consistent safety signal through the CSCT/CSDT traffic light system.

Chapter 7 presents the conclusions of the dissertation, summarizes the contributions to knowledge, and provides recommendations for future research and practice.

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7. CONCLUSIONS AND RECOMMENDATIONS

7.1. Overview

This dissertation applied the Bayesian reliability updating framework of De Vries et al. [1] to the monitoring data from the 2015 proof load test of the viaduct Zijlweg, an ASR-affected reinforced concrete solid slab bridge in Noord Brabant, the Netherlands. The study addressed the central research question: how can monitoring data from an instrumented proof load test be used to quantitatively update the structural reliability of an ASR-affected reinforced concrete bridge, and how do the resulting Bayesian reliability estimates compare with mechanics-based stop criteria evaluated at each load step?

This chapter presents the conclusions of the study, structured around the four specific research objectives stated in Chapter 1. It then identifies the contributions to knowledge, discusses the limitations, and provides recommendations for future research and practice.

7.2. Conclusions

7.2.1. Objective 1: Extraction and Verification of Indicator Values

The first objective was to extract and verify the monitoring indicator values from the Zijlweg proof load test data, specifically concrete strain for bending and maximum weighted nominal crack width for shear, and to establish the structural input parameters for the reliability model.

This objective was achieved through MATLAB preprocessing of the bending04 and shear01 datasets. The verified indicator values at the five load steps are:

- **Bending:** concrete strains of 52.2, 111.7, 178.9, 206.6, and 239.2 microstrain (from LVDT02 and LVDT03 at the critical bending section 3,382 mm from support 5).
- **Shear:** maximum weighted nominal crack widths of 0.005, 0.011, 0.016, 0.020, and 0.023 mm (from LVDTs 14, 15, and 16 at the critical shear section).

A critical finding during the structural parameter derivation was that the initial sectional analysis code omitted two reinforcement bar types identified in Lantsoght et al. [5] Figure 3: bar No. 14 (Φ19 at 150 mm) and bar No. 20 (Φ16 at 200 mm). Including all four bottom reinforcement layers increased the total bottom steel area from 18,870 mm2 to 37,080 mm2 and raised the mean yield moment from

1,825.8 kNm to 3,248.87 kNm. This correction was essential for obtaining physically meaningful reliability results. The corrected yield moment of 3,248.87 kNm is consistent with the proof load test result, since the structure survived a maximum bending moment of 3,029.4 kNm without signs of distress.

7.2.2. Objective 2: Bayesian Reliability Updating

The second objective was to evaluate how the structural reliability of the Zijlweg viaduct evolves before, during, and after proof loading for both bending and shear failure modes, using the Bayesian updating framework of De Vries et al. [1] applied to the field monitoring data.

The main conclusions from the reliability analysis are:

1. The prior reliability indices for both failure modes ($\beta_{prior} \approx 2.34$ for bending and 2.80 for shear) are significantly below the annual target reliability of 4.0 for CC3 structures at the disapproval level. This confirms that analytical assessment alone, which incorporates conservative ASR-reduction factors, cannot demonstrate structural adequacy.

2. The posterior reliability indices after the complete proof load test are $\beta_{after} = 5.33$ for bending and 4.89 for shear. Both values substantially exceed the annual target reliability for existing structures (4.0) and the annual target for new structures (4.7), providing conclusive evidence that the viaduct is safe for its remaining service life.

3. The annual safety requirement of $\beta = 4.0$ was demonstrated before the maximum target load was reached. Specifically, the target was achieved at Step 3 ($f_{LM1} = 0.866$) for bending and at Step 4 ($f_{LM1} = 0.849$) for shear.

4. The bending analysis showed a steep decrease in β_{during} from 4.36 to -0.076, reflecting that the test physically challenged the structure's mean capacity. Conversely, the shear analysis showed a nearly constant β_{during} at 1.82, confirming that the shear behavior remained safely in the elastic regime throughout the test with no signs of distress.

7.2.3. Objective 3: Stop Criteria Consistency

The third objective was to assess the CSCT and CSDT stop criteria against the shear monitoring data at each load step and determine whether the traffic light classifications are consistent with the concurrent Bayesian reliability estimates.

The stop criteria conclusions are:

1. The CSCT and CSDT indicators remained in the green light range throughout the entire shear proof load test. The maximum governing indicator value was $\min(I_{CSDT}, I_{CSCT}) = 0.004199$ at step 5, far below the yellow light threshold.
2. The M/Vd verification confirmed that the LVDT measurement locations are within the recommended range of 0.8 to 1.0 for the stop criteria methodology, validating the application of the stop criteria at these sensor positions.
3. The stop criteria and Bayesian reliability estimates are qualitatively consistent throughout the test. Both approaches correctly identify the absence of shear distress at all load steps and both show increasing evidence of structural adequacy as the proof load increases.
4. The crack width measurement serves a dual operational and probabilistic role: it provides the physical threshold comparison for the stop criteria and simultaneously provides the input to the Bayesian exponential resistance model. This dual role demonstrates that the two assessment approaches are complementary uses of the same monitoring data rather than independent methods.

7.2.4. Objective 4: Practical Implications for ASR-Affected Bridges

The fourth objective was to draw practical conclusions about the value of Bayesian reliability updating for the assessment of ASR-affected reinforced concrete bridges and its potential to improve proof load testing practice.

The practical conclusions are:

1. Proof load testing, interpreted through the Bayesian framework, is the most effective assessment method for ASR-affected reinforced concrete bridges where analytical models give conservative or insufficient ratings. The Zijlweg viaduct, rated analytically insufficient for shear, has been demonstrated by the proof load test to exceed the RBK Design reliability level for both failure modes.

2. The ASR prestressing effect, where the restraint of ASR expansion by the reinforcement creates beneficial concrete compression, significantly increases the actual structural capacity above the analytically predicted values. This beneficial effect cannot be captured by analytical capacity reduction factors but is directly captured by proof load testing. The proof load test therefore bypasses the uncertainties of ASR capacity modelling.
3. Monitoring data provides a quantifiable advantage over uninstrumented proof load testing. For the Zijlweg shear analysis, the crack width monitoring data allows the reliability requirement to be demonstrated at a proof load factor of $f_{LM1} = 0.754$, compared to a substantially higher factor that would be required using the lower bound approach alone. The additional information from 1.85 reliability index units at step 3 represents the direct benefit of instrumentation.
4. The combination of stop criteria for real-time operational safety and Bayesian updating for post-test quantitative reliability assessment constitutes a principled and efficient framework for proof load testing. The stop criteria ensure the test is conducted safely, while the Bayesian framework maximises the reliability information extracted from the test results.

7.3. Answer to the Central Research Question

The central research question of this dissertation was: how can monitoring data from an instrumented proof load test be used to quantitatively update the structural reliability of an ASR-affected reinforced concrete bridge, and how do the resulting Bayesian reliability estimates compare with mechanics-based stop criteria evaluated at each load step?

The answer has two parts. First, monitoring data can be formally incorporated into the structural reliability assessment through the Bayesian updating framework of De Vries et al. [1]. For shear, the maximum weighted nominal crack width from LVDTs is mapped to a resistance ratio distribution through the calibrated exponential model, and this distribution is updated at each load step through the Metropolis-Hastings MCMC algorithm conditioned on the survival of the applied proof load. For bending, the concrete strain measurement is used to infer the resistance ratio through the cross-section analysis. The resulting posterior reliability indices provide a quantitative measure of structural adequacy that directly incorporates both the prior analytical knowledge and the experimental evidence of the proof load test.

Second, the Bayesian reliability estimates are qualitatively consistent with the mechanics-based stop criteria evaluated at each load step. Both approaches identify the absence of structural distress during the test and both show increasing evidence of structural adequacy as the proof load increases.

Quantitatively, the monitoring data approach demonstrates the required reliability level at a lower proof load factor than would be needed using the lower bound approach alone, illustrating the added value of instrumented testing. The stop criteria and Bayesian framework are complementary rather than competing: the stop criteria provide the operational safety signal for real-time decision-making during the test, while the Bayesian framework provides the quantitative post-test reliability assessment.

7.4. Contributions to Knowledge

This dissertation makes the following contributions to knowledge:

1. It provides the first application of the Bayesian reliability updating framework of De Vries et al. [1] to real field proof load test data from an ASR-affected reinforced concrete bridge, demonstrating that the methodology is applicable to existing field datasets and produces physically meaningful results.
2. It demonstrates quantitatively that monitoring data from instrumented proof load tests provides a significant advantage over uninstrumented testing, with the crack width monitoring data contributing up to 1.85 additional reliability index units compared to the lower bound approach at the same proof load level.
3. It provides a direct comparison between the Bayesian reliability framework and the CSCT/CSDT stop criteria for an actual proof load test, confirming their qualitative consistency and identifying their complementary roles in the assessment process.
4. It identifies and documents the critical importance of complete reinforcement inventory in sectional analysis models for ASR-affected bridges, showing that omission of reinforcement layers can reduce the calculated yield moment by up to 44% and lead to physically inconsistent reliability results.
5. It contributes a complete and documented case study, including all preprocessing code, modified C++ analysis codes, and verified input parameters, that can serve as a reference implementation for future applications of Bayesian reliability updating to proof load test data.

7.5. Recommendations for Future Research

Based on the findings and limitations of this study, the following recommendations are made for future research:

1. Development of ASR-specific resistance models for Bayesian updating: the exponential crack width model used for shear was calibrated on slab strip experiments without ASR deterioration. Future research should investigate whether ASR deterioration systematically affects the relationship between crack width and residual shear capacity, and develop calibrated resistance models that explicitly account for ASR effects.
2. Extension to multi-span and skewed bridges: the Zijlweg viaduct has a skew angle of 14.4 degrees and is part of a multi-span structure. Future research should examine how skew affects the validity of the stop criteria methodology at the sensor locations, and how the transfer factor framework of De Vries et al. [1] Chapter 6 can be combined with the monitoring data updating approach for multi-span assessment.
3. Real-time implementation: this study performed the Bayesian updating as a post-test analysis. Future research should develop real-time implementation of the MCMC algorithm that can produce reliability estimates during the test, allowing the stopping criterion to be the achievement of a required posterior reliability level rather than a predetermined target load.
4. Larger calibration datasets: the exponential resistance model is calibrated on five slab strip experiments. Expanding the calibration dataset to include specimens with different reinforcement ratios, concrete strengths, aggregate sizes, and levels of ASR deterioration would reduce the parameter uncertainty and improve the generalisability of the model.
5. Weigh-in-motion data integration: the annual maximum traffic load model used in this study was taken from De Vries et al. [1] calibrated for Dutch slab bridges. Future applications should incorporate site-specific WIM data to provide more accurate traffic load characterisation for individual bridge assessments.

7.6. Recommendations for Practice

Based on the findings of this study, the following recommendations are made for engineering practice:

1. For ASR-affected reinforced concrete bridges rated analytically insufficient for shear or bending, proof load testing with Bayesian reliability updating should be considered as a Level IV assessment method. The approach has been demonstrated to produce reliability indices substantially above the required levels even for bridges with UC > 1.0 under analytical assessment.
2. Instrumentation with crack width LVDTs at the critical shear section is strongly recommended for proof load tests where shear is the critical failure mode. The additional reliability information provided by crack width monitoring data can reduce the required maximum proof load compared to uninstrumented testing, reducing the risk to the structure.
3. The full reinforcement layout should be verified from structural drawings and, where necessary, by physical investigation before performing sectional analysis for the Bayesian resistance model. Incomplete reinforcement inventory leads to underestimated yield moments and non-physical reliability results.
4. The CSCT and CSDT stop criteria should be applied alongside the Bayesian updating framework in all monitored proof load tests. Their qualitative consistency, demonstrated in this study, provides mutual validation of both assessment methods and increases confidence in the combined assessment.
5. Proof load test target loads should be set to exceed the annual maximum traffic load characteristic value where structurally feasible. Proof loads below this level provide limited reliability evidence from the lower bound perspective, and the full benefit of monitoring data can only be realised when the test reaches and exceeds the traffic reference level.

7.7. Closing Statement

This dissertation demonstrates that Bayesian reliability updating applied to proof load test monitoring data provides a powerful and quantitatively rigorous method for assessing ASR-affected reinforced concrete bridges. The Zijlweg viaduct, a bridge that appeared analytically insufficient for shear under code-based assessment, has been shown through the Bayesian framework to possess demonstrated reliability above the RBK Design level for both bending and shear failure modes. The proof load test, conducted in 2015 and originally reported by Lantsoght et al. [5], has been reanalysed in this dissertation using the state-of-the-art methodology of De Vries et al. [1], revealing the full extent of the reliability information contained in the monitoring data.

The combination of stop criteria for operational safety and Bayesian updating for quantitative reliability assessment constitutes a complete and efficient framework for proof load testing of deteriorated bridges. As the aging bridge stock across Europe continues to require assessment and maintenance decisions, this integrated approach offers a scientifically sound and practically applicable method for making those decisions on the basis of experimental evidence rather than conservative analytical assumptions

8. SUSTAINABILITY ANALYSIS AND ETHICAL IMPLICATIONS

8.1. Introduction

This chapter presents the sustainability analysis and ethical implications of the research conducted in this dissertation, as required by UPC for all Final Master's Thesis submitted from February 2025 onwards. The analysis follows the UPC guide for studying sustainability in final thesis projects and evaluates the environmental, social, and economic dimensions of the work, as well as the ethical considerations associated with the research.

8.2. Sustainability Analysis

8.2.1. Environmental Dimension

The research conducted in this dissertation contributes positively to environmental sustainability through its potential to extend the service life of existing reinforced concrete bridges. The production of concrete for new bridge construction is a major source of carbon dioxide emissions, with the cement industry responsible for approximately 8% of global CO2 emissions [42]. Every year of additional service life gained for an existing bridge through improved assessment methods directly reduces the need for new construction and the associated environmental impact.

The Zijlweg viaduct, if demolished and replaced based solely on the conservative analytical assessment (Unity Check > 1.0 for shear), would require substantial new concrete production. The proof load test, interpreted through the Bayesian framework demonstrated in this dissertation, provides evidence that the viaduct can continue operating safely, potentially avoiding replacement for additional years or decades. This represents a concrete environmental benefit of the methodology.

The computational methods used in this dissertation, specifically the MCMC analysis with 1×10^8 samples run on a personal computer, have a modest energy footprint. Each analysis run required approximately 7 to 15 minutes in Release mode, consuming negligible energy compared to the environmental cost of physical testing or construction. The open-source nature of the C++ code means the methodology can be applied by other researchers without duplicating computational infrastructure.

The primary materials used in this research were computational and documentary in nature. No physical experiments were conducted, no new specimens were manufactured, and no laboratory materials were consumed. The research builds entirely on existing monitoring data from the 2015 test, representing a resource-efficient approach to knowledge generation.

8.2.2. Social Dimension

Aging bridge infrastructure presents significant social risks. Bridge failures cause loss of life, disrupt economic activity, and reduce public confidence in infrastructure safety. The methodology developed and demonstrated in this dissertation contributes to social sustainability by improving the reliability and efficiency of bridge safety assessment.

The Bayesian reliability updating framework provides bridge engineers with a quantitative tool for making more informed assessment decisions. Instead of relying solely on conservative analytical models that may lead to unnecessary closures or costly interventions, engineers can use proof load test evidence to demonstrate structural adequacy objectively. This has direct social benefits by maintaining access to critical transportation infrastructure while ensuring public safety.

For ASR-affected bridges specifically, which are widespread across the Netherlands and other European countries, improved assessment methods can help prioritise maintenance resources more effectively. Bridges that are analytically insufficient but have demonstrated adequate performance through proof load testing can remain in service, while truly deficient structures can be identified for priority intervention. This rational prioritisation maximises the social benefit of limited maintenance budgets.

The dissertation also contributes to knowledge transfer and capacity building in the engineering community. By documenting a complete implementation of the Bayesian updating framework applied to a real case study, including all code, data preprocessing steps, and parameter derivations, this work enables other engineers and researchers to apply the methodology without starting from scratch. The GitHub repository associated with this dissertation makes all code publicly available, contributing to open science principles.

8.2.3. Economic Dimension

The economic sustainability implications of this research operate at two levels: the immediate case study level and the broader methodology level.

At the case study level, the Zijlweg viaduct serves as part of the Dutch highway network crossing the A59 in Noord Brabant. Its continued operation avoids the substantial economic cost of replacement, which for a bridge of this size and traffic volume would amount to several million euros. The methodology demonstrated in this dissertation provided evidence-based justification for continued service, avoiding premature replacement costs.

At the methodology level, the quantitative demonstration that monitoring data provides up to 1.85 reliability index units of additional information compared to the lower bound approach has direct economic implications for proof load test design. Engineers can design shorter, more targeted tests that reach the required reliability level at lower proof loads when monitoring data is used. Shorter tests with lower maximum loads reduce the risk of damage during testing and reduce the cost of traffic management during the test. For a typical proof load test on a highway bridge requiring lane closures, each hour saved represents significant economic value.

The open-source C++ code developed and documented in this dissertation reduces the barrier to entry for applying Bayesian reliability updating in practice. Engineering firms and national road authorities can adapt the code for their own bridge portfolios without commissioning new software development, reducing the economic cost of implementing this technology at scale.

8.3. Ethical Implications

8.3.1. Responsibility for Public Safety

The primary ethical responsibility in bridge assessment research is the protection of public safety. All recommendations made in this dissertation are grounded in established probabilistic reliability theory and validated computational methods. The reliability targets used (annual target $\beta = 4.0$ for disapproval/usage, and $\beta = 4.7$ for new structures) are those prescribed by Dutch national assessment guidelines, ensuring that the assessment framework maintains public safety at the required level.

An important ethical consideration is the transparency of uncertainty. This dissertation explicitly quantifies and communicates the uncertainties in the analysis, including model uncertainties in the resistance model, parameter uncertainties from limited calibration data, and the limitations of the traffic load model. These uncertainties are incorporated into the reliability indices rather than being obscured. This transparency is essential for engineers who use these results to make safety-critical decisions.

8.3.2. Data Integrity and Reproducibility

All monitoring data used in this dissertation originates from the 2015 Zijlweg proof load test conducted and published by Lantsoght et al. [5]. The data was accessed through the research group of Dr. Eva Lantsoght with proper authorisation. All preprocessing steps are fully documented and the code is publicly available, ensuring that the results can be independently reproduced and verified.

The dissertation acknowledges and corrects a significant error discovered during the analysis: the initial omission of two reinforcement bar types from the sectional analysis. This correction is fully documented in Chapter 5, including the impact on the results. Transparent reporting of errors and corrections is a fundamental ethical requirement in scientific research.

8.3.3. Appropriate Use of AI Assistance

During the preparation of this dissertation, AI language model assistance was used for drafting text and structuring arguments. All technical content, analysis, code modifications, and conclusions remain the original work of the author, supervised by Dr. Eva Lantsoght and Dr. Rolando Chacón. AI assistance was used as a productivity tool and writing aid, not as a substitute for the engineering judgment and technical expertise of the author and supervisors. All AI-generated text was reviewed, verified, and modified by the author before inclusion.

8.3.4. Attribution and Academic Integrity

This dissertation builds directly on the published work of De Vries et al. [1] and Lantsoght et al. [5]. The contribution of this dissertation is explicitly defined as the application of existing methodology to a new field dataset, not the development of new algorithms. This distinction is clearly maintained throughout the dissertation. All sources are cited using the IEEE referencing system and the original code by De Vries is acknowledged in every modified file.

The monitoring data from the Zijlweg test represents a valuable and finite scientific resource. By making all preprocessing code and derived parameters publicly available, this dissertation maximises the scientific value extracted from that dataset and enables future researchers to build on this work without repeating the data processing effort.

8.4. Summary

The research presented in this dissertation contributes positively to environmental, social, and economic sustainability through its potential to extend bridge service life, improve the efficiency of infrastructure assessment, and reduce unnecessary replacement of structurally adequate bridges. The ethical dimensions of the work are addressed through transparent uncertainty quantification, full reproducibility, correct attribution, and responsible use of AI assistance. The open-source nature of all code and documentation maximises the broader impact of the research contribution.

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